

Collision invariants of a pinned chain and macroscopic transport in truncated wave kinetics

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Abstract

Which quantities do collisions preserve, and how do they determine macroscopic transport? We study these questions in two four-wave resonance models. For the one-dimensional nearest-neighbour pinned chain, with any fixed coupling $0 < d < 1/2$, we prove that every almost-everywhere finite measurable collision invariant for the dispersion $\omega_d(x) = \sqrt{1 - 2d \cos x}$ has the form $A + B\omega_d$, where A, B are constants. Local resonance averages first give a smooth representative; periodic compatibility then determines the invariant. At critical resonances, the energy mismatch and the full collision difference share a vanishing factor. Retaining this cancellation yields a densely defined closed collision form, while the invariant classification and compact resonance averages give a positive spectral gap.

For the wave kinetic equation with quadratic dispersion and a sharp cutoff to a momentum cube in three dimensions, we prove a strong-collision limit from arbitrary smooth positive local Rayleigh–Jeans initial data, throughout any prescribed finite interval on which the corresponding five-moment Euler solution remains smooth and positive. If c is the collision strength, the distribution and macroscopic parameter errors are respectively $O(c^{-3/4})$ and $O(c^{-1})$ in the norms specified below. The collision frequency vanishes at the corners, where the solution retains the free-transport trace of its initial data. Actual five-moment matching gives an exact decomposition of relative entropy into microscopic and macroscopic errors, controlled by weighted dissipation and the flux equations. The microscopic entropy balance further yields strong convergence of the resonance current and a first-order transport law for all five moments. The Onsager tensor is symmetric and nonnegative, with a seven-dimensional kernel in the fifteen-dimensional gradient space. Its eight dissipative directions are the traceless symmetric part of the momentum-parameter gradient and the energy-parameter gradient.

Keywords. Four-wave resonance; collision invariants; pinned chain; Rayleigh–Jeans distribution; Euler limit; Onsager tensor; first-order transport.

1 Introduction

1.1 What collisions preserve

A collision redistributes momentum and energy while preserving certain totals. These conserved quantities determine how many parameters are needed to describe equilibrium, and hence how many unknowns appear in a macroscopic equation. The passage from a collision model to a fluid description therefore begins with a concrete question: what information, beyond the known conservation laws, can survive every collision?

*Companion repository: <https://github.com/veridiscoversLab/resonance-kinetic-lean>.

We consider four-wave collisions. Write k_0, k_1 for the two incoming momenta and k_2, k_3 for the outgoing momenta; the four entries are called the *legs* of a collision. For a function φ of momentum, set $\varphi_i = \varphi(k_i)$ and define the full collision difference

$$\Delta\varphi = \varphi_0 + \varphi_1 - \varphi_2 - \varphi_3. \quad (1.1)$$

A function for which this difference vanishes almost everywhere on the allowed resonances is a *collision invariant*. Constants are always invariant, and conservation of energy makes the dispersion relation invariant as well. The question is whether these elementary examples exhaust the possibilities.

We follow the consequences of this question in two models. In a one-dimensional pinned chain, resonance identities first force measurable invariants to be smooth and then restrict them to constants and energy. In a three-dimensional truncated model, five invariants determine the local equilibrium family. Matching the actual five moments then makes microscopic and macroscopic errors add exactly in relative entropy. The same collision difference thus links classification to transport: the reciprocal of an equilibrium lies in its kernel, while its weighted square measures dissipation away from equilibrium.

This relation is part of the programme of mechanical limits in Hilbert's sixth problem. For hard spheres, Deng, Hani and Ma connected the long-time derivation of the Boltzmann equation to fluid limits. Starting from elastic hard-sphere dynamics with random initial configurations on two- and three-dimensional tori, they obtained compressible Euler and incompressible Navier–Stokes–Fourier systems under the corresponding initial-data and joint scaling assumptions [1, Theorems 1–3]. Weakly nonlinear waves provide another kinetic route. Deng and Hani derived the spatially homogeneous wave kinetic equation from weakly nonlinear Schrödinger dynamics on large tori [2], and extended the derivation to any prescribed time for which the kinetic solution exists [3]. Here we start from specified collision equations and study their resonance geometry and macroscopic limits. The dispersion, momentum cutoff and initial data are stated for each model. Deriving these inhomogeneous collision models from microscopic wave dynamics requires a separate analysis.

1.2 Regularity from resonance compatibility

The dimensionless dispersion of the nearest-neighbour pinned chain is

$$\omega_d(x) = \sqrt{1 - 2d \cos x}, \quad x \in \mathbb{T}_{2\pi} := \mathbb{R}/(2\pi\mathbb{Z}), \quad 0 < d < \frac{1}{2}. \quad (1.2)$$

Here d is a fixed coupling parameter and x is the angular coordinate of lattice momentum. The allowed collisions satisfy

$$\begin{aligned} x_0 + x_1 &= x_2 + x_3 \pmod{2\pi}, \\ \omega_d(x_0) + \omega_d(x_1) &= \omega_d(x_2) + \omega_d(x_3). \end{aligned} \quad (1.3)$$

Momentum is conserved modulo the period, including Umklapp processes across the boundary of a fundamental interval. The angular coordinate x itself is therefore not a globally defined periodic collision invariant.

In their 2006 study of heat transport in this chain, Aoki, Lukkarinen and Spohn asked whether constants and ω_d exhaust the collision invariants [4, after (4.10)]. Lukkarinen later discussed almost-everywhere solutions of this functional equation [5, Section 3.2]. The review by Germain, La and Menegaki [6, Section 7.2] still lists the classification for general $0 < d < 1/2$ as open; its Theorem 4 proves that the space of C^1 solutions is finite-dimensional.

Theorem 2.1 gives the classification. For every fixed $0 < d < 1/2$, any almost-everywhere finite measurable collision invariant φ satisfies

$$\varphi = A + B\omega_d \quad \text{almost everywhere,} \quad A, B \in \mathbb{C}. \quad (1.4)$$

The null sets in the collision identity are measured by the coarea measure on the regular resonance surface; those in the conclusion are measured by Lebesgue measure on the circle. Both measures are defined at the beginning of Section 2. No continuity or integrability is assumed.

The first step explains why a measurable invariant cannot remain arbitrarily rough. For a target point x , choose a smooth resonance chart $(x, H(x, z), z, W(x, z))$, where z is a free parameter and $W = x + H - z$. The collision identity relates $\varphi(x)$ to its values at three other points. These points move nondegenerately with z , so the identity both controls the size of the function and permits averaging. Three changes of variables first compare level sets and yield local boundedness. A smooth function χ , supported inside the chart and having integral one, then gives, for almost every x ,

$$\varphi(x) = \int \chi(z) [\varphi(z) + \varphi(W(x, z)) - \varphi(H(x, z))] dz. \quad (1.5)$$

Since $\partial_z H$ and $\partial_z W$ stay away from zero, the right-hand side can be written as an integral of φ against a smooth kernel. Derivatives in x fall on the kernel, not on the measurable input. The right-hand side defines a local smooth representative, and a finite cover gives a smooth representative on the whole circle. Only then do we differentiate the resonance identity. Second- and third-order compatibility, together with periodic matching, completes the classification. Regularity is a consequence of the collision relations between different points.

1.3 A common vanishing factor

After identifying the kernel, one must give the dissipation integral a suitable domain. The energy constraint degenerates at critical resonances, and its induced measure may have infinite mass. The full collision difference, however, vanishes at the same configurations.

The relation is visible in local coordinates. After permuting the legs, write their positions as $(z + u, z + v, z, z + u + v)$. For a C^2 function φ ,

$$\begin{aligned} F &:= \Delta\omega_d = -uvG_{\omega_d}, & \Delta\varphi &= -uvG_\varphi, \\ G_\varphi(z, u, v) &= \int_0^1 \int_0^1 \varphi''(z + su + tv) ds dt. \end{aligned} \quad (1.6)$$

For example, the local test function $\varphi(z) = z^2$ gives $\Delta\varphi = -2uv$. If $u = 0$ or $v = 0$, the legs coincide in pairs and the difference vanishes. This quadratic function illustrates a local calculation; it is not a periodic invariant on the circle.

When regularizing the energy constraint, we therefore first form $|\Delta\varphi|^2 = u^2v^2|G_\varphi|^2$ and then pass to the singular limit. The square becomes small precisely where the degenerate measure becomes large. Section 2 uses this common factor to construct a densely defined closed dissipation form. The invariant classification and compact resonance averages then give a positive spectral gap off its kernel (Theorems 2.7 and 2.10). The gap constant depends on the fixed pinning parameter and positive resonance weight.

1.4 Moment matching and the decomposition of entropy

The second model has a fixed momentum cube $D = [-R, R]^3$, with $R > 0$, a space variable $X \in \mathbb{T}^3 := \mathbb{T}_{2\pi}^3$, and time $t \geq 0$. Its dispersion is $|k|^2$, so the transport velocity is $v(k) = 2k$. We retain only collisions whose four momenta all lie in D . Writing $\mathcal{C}$ for the resulting four-wave collision operator, we study

$$\mathcal{T}f_c = c\mathcal{C}(f_c), \quad \mathcal{T} := \partial_t + 2k \cdot \nabla_X, \quad c \longrightarrow \infty. \quad (1.7)$$

Here $c > 0$ is the collision strength and $f_c(t, X, k)$ is a positive distribution. Collisions act only on momentum: all four legs share the same time and position. The measure, constants and cutoff in $\mathcal{C}$ are specified in Section 3.

The five collision invariants form the column vector

$$\Psi(k) = (1, k^{(1)}, k^{(2)}, k^{(3)}, |k|^2)^\top,$$

where $k^{(j)}$ denotes the j th component of momentum. For a parameter $\theta \in \mathbb{R}^5$ with $\theta \cdot \Psi > 0$ throughout D , define the Rayleigh–Jeans distribution, abbreviated as RJ,

$$N_\theta(k) = \frac{1}{\theta \cdot \Psi(k)}. \quad (1.8)$$

Its reciprocal is a collision invariant, so $\mathcal{C}(N_\theta) = 0$. Allowing θ to depend on (t, X) gives a local equilibrium.

If the five moments of a positive distribution f lie in the image of the equilibrium moment map, there is a unique $N = N_\theta$ such that $\int_D \Psi(f - N) dk = 0$. We call N the equilibrium matched to the actual five moments. For a reference equilibrium $\bar{N} = N_{\bar{\theta}}$, define the relative entropy

$$\mathcal{H}(f | N) = \int_D \left[\frac{f}{N} - 1 - \log \frac{f}{N} \right] dk. \quad (1.9)$$

The positive upper and lower bounds used below make this integral finite. Moment matching gives the exact identity

$$\boxed{\mathcal{H}(f | \bar{N}) = \mathcal{H}(f | N) + \mathcal{H}(N | \bar{N})}. \quad (1.10)$$

Indeed, the cross term in the expansion is

$$\int_D (f - N)(\bar{N}^{-1} - N^{-1}) dk = (\bar{\theta} - \theta) \cdot \int_D \Psi(f - N) dk = 0.$$

The first term measures the microscopic deviation that leaves the conserved moments unchanged; the second measures the error in the macroscopic parameters. They add exactly because the difference of the reciprocal equilibria remains in the span of the five collision invariants. Proposition 4.1 gives the corresponding variational properties, followed by an example with matched moments but positive microscopic entropy.

1.5 Corner memory and the Euler limit

Evaluating the five conserved fluxes at local equilibrium gives the macroscopic system

$$\partial_t \int_D \Psi N_\theta dk + \sum_{j=1}^3 \partial_{X_j} \int_D 2k^{(j)} \Psi N_\theta dk = 0. \quad (1.11)$$

It is symmetric hyperbolic in the positive parameter domain. Let $\theta_0(X)$ be any smooth initial parameter with $\theta_0 \cdot \Psi$ uniformly positive, and suppose the Euler solution $\bar{\theta}$ with this initial datum stays smooth and positive on a finite interval $[0, T]$. We compare the kinetic solution starting from $f_c(0) = N_{\theta_0}$ with $\bar{N} = N_{\bar{\theta}}$. Write $N_c = N_{\theta_c}$ for the equilibrium matched to f_c .

Let $H_X^s L_k^2$ denote the Sobolev space with up to s weak derivatives in X and values in $L^2(D)$, and take L_t^∞ on $[0, T]$. Theorem 4.3 proves that, for sufficiently large c , a positive classical solution exists up to T , with positive upper and lower bounds independent of c , and

$$\|f_c - \bar{N}\|_{L_t^\infty H_X^3 L_k^2} \leq C_T c^{-3/4}, \quad (1.12)$$

$$\|\theta_c - \bar{\theta}\|_{L_t^\infty H_X^2} \leq C_T c^{-1}. \quad (1.13)$$

The constant C_T depends on finitely many derivatives of the reference solution, its positivity margin and the fixed model parameters. The spatial amplitude of the initial parameter is fixed; it need not shrink as c grows.

This limit allows momentum corners to retain a memory of the initial data. If ξ is a corner of the cube, the collision integral vanishes there, and the positive classical solution satisfies

$$f_c(t, X, \xi) = N_{\theta_0(X-2\xi t)}(\xi). \quad (1.14)$$

The right-hand side generally differs from $\bar{N}(t, X, \xi)$. Collisions are also slow near a corner: at distance r from the corner set, their frequency is of order $r^2 \log^2(2R/r)$. The proof must therefore control the contribution of the whole slow region to dissipation and flux. The corners themselves have zero volume; a thin corner layer has small volume, bounded amplitude, and still collides with the rest of the momentum domain. Weighted energy estimates that retain these interactions, together with amplitude bounds along characteristics, allow corner memory to coexist with (1.12).

Bricmont and Kupiainen [7] studied near-equilibrium diffusive behaviour of a phonon Boltzmann equation, using wave numbers in at least two dimensions, a modified dispersion without the square root, and a diffusive scaling. Our one-dimensional classification retains the square-root dispersion (1.2). The three-dimensional macroscopic theorem uses fixed transport and strong collisions, with (1.11) as its limit. The resonance geometry required by each model is proved separately.

1.6 From dissipation to first-order transport

The Euler equations use the equilibrium flux. The first-order correction comes from the deviation from equilibrium. For a spatial direction $i = 1, 2, 3$ and a conserved quantity $a = 0, \dots, 4$, define the residual flux

$$Z_{c,i}^a = \int_D 2k^{(i)} \Psi_a(k) (f_c - N_c) dk. \quad (1.15)$$

Theorem 6.1 proves, on the entire prescribed time interval,

$$cZ_{c,i}^a \longrightarrow \sum_{j=1}^3 \sum_{b=0}^4 K_{ij}^{ab}(\bar{N}) \partial_{X_j} \bar{\theta}_b \quad \text{in } L^2((0, T) \times \mathbb{T}^3). \quad (1.16)$$

Here K is the Onsager transport tensor. Section 5 defines it through the linearized collision equation after removing the five zero modes, and proves that its entries are finite and that it is symmetric and nonnegative. Symmetry of the collision form gives the reciprocity relation $K_{ij}^{ab} = K_{ji}^{ba}$; the general physical background is described in [8].

The degeneracy of this tensor has a direct geometric description. Write $\theta = (\alpha, \beta, \gamma)$, where $\alpha, \gamma \in \mathbb{R}$ and $\beta \in \mathbb{R}^3$ correspond to the constant, momentum and energy invariants. The fifteen spatial derivatives form a gradient array $\xi = (g, B, h)$, where $g = \nabla_X \alpha$, $B_{jl} = \partial_{X_j} \beta_l$ and $h = \nabla_X \gamma$. On every compact positive parameter set,

$$\xi^T K(N_\theta) \xi \asymp \left| \frac{B + B^T}{2} - \frac{\text{tr } B}{3} I \right|^2 + |h|^2. \quad (1.17)$$

The symbol $\asymp$ denotes two-sided comparison with uniform positive constants, and matrices carry the Frobenius norm. The five-dimensional space of traceless symmetric 3×3 matrices and the three energy-gradient directions account for the eight dissipative directions. Mass gradients, skew-symmetric matrices and scalar matrices account for the seven zero directions. At a nonzero spatial Fourier frequency, gradient compatibility leaves only the mass direction. Coupling to Euler transport then gives quantitative decay for the constant-parameter linear system.

Strong convergence follows from the microscopic entropy balance under actual moment matching: the change in microscopic entropy, four-wave dissipation and work of the residual flux satisfy an exact identity. A weak limit determines a candidate resonance current; the balance also determines its norm and rules out a dissipation defect. In addition to (1.16), we prove

$$c\mathcal{C}(f_c) \longrightarrow \mathcal{T}\bar{N} \quad \text{in } L^2((0, T) \times \mathbb{T}^3 \times D). \quad (1.18)$$

The full signed resonance current converges strongly with respect to the resonance measure as well. These limits give the first-order transport law for the actual five-moment parameters, with a remainder $o(c^{-1})$ in $L^2(0, T; H^{-1}(\mathbb{T}^3))$, where H^{-1} is the dual of H^1 . The convergence is integrated over the entire time interval; first-order strong traces at the initial time require an initial-layer analysis.

1.7 Organization and conventions

Section 2 proves the one-dimensional invariant classification and constructs the closed dissipation form. Section 3 turns to the three-dimensional model, establishing the pair-sum variance representation, corner-frequency estimates and weighted coercivity. Section 4 uses the entropy decomposition to organize non-linear errors and prove the Euler limit. Section 5 determines the transport tensor and its degeneracy, and Section 6 uses the entropy balance to identify the strong resonance current and all five moment fluxes.

Inner products are linear in the first argument. We write $\mathbb{T}_{2\pi}$ for the circle and $\mathbb{T}^3 = \mathbb{T}_{2\pi}^3$ for the spatial torus. The measure dX is periodic Lebesgue measure on $[0, 2\pi)^3$, and dk is ordinary Lebesgue measure on the momentum domain. Constants denoted by C may depend on fixed model parameters; their uniformity is specified in each statement. For $s > 0$, we write $\log_+ s = \log^+ s = \max\{\log s, 0\}$. Notation is introduced where it is used and collected in an index at the end.

Contents

1	Introduction	1
1.1	What collisions preserve	1
1.2	Regularity from resonance compatibility	2
1.3	A common vanishing factor	3
1.4	Moment matching and the decomposition of entropy	3
1.5	Corner memory and the Euler limit	4
1.6	From dissipation to first-order transport	5
1.7	Organization and conventions	6
2	Collision invariants and closed dissipation forms for one-dimensional square-root dispersion	8
2.1	Periodic resonance, coarea measure, and the classification theorem	8
2.2	Nondegenerate resonance charts and regularization of measurable functions	10
2.3	Second- and third-order compatibility on the symmetric resonance branch	13
2.4	Resonance averaging and a positive spectral gap	15
2.5	Critical resonances and cancellation in the collision difference	16
2.6	The maximal closed form, self-adjoint realization, and linearization normalization	19
3	Geometry and degenerate dissipation for four-wave collisions with a cube cutoff	21
3.1	The full resonance measure and two integral kernels	22
3.2	The precise degeneracy of the corner frequency	24
3.3	The five collision invariants and weighted coercivity	26
3.4	Absolute row estimates and the continuous weighted inverse	28
3.5	Linearization and spatial products in a thin corner layer	32
4	The macroscopic limit for positive Rayleigh–Jeans data on a smooth Euler interval	36
4.1	Five-moment parameters and the symmetric hyperbolic system	36
4.2	Local construction and continuation of the transport equation	40
4.3	The nonlinear corrector in two norms	41
4.4	Orthogonal decomposition and differentiated equations through order three	43
4.5	Third-order energy under two-sided corner-layer bounds	45
4.6	Positive exchange in corner layers and amplitude bounds outside them	49
4.7	Continuation on the prescribed interval and convergence rates	51
5	The Onsager tensor and its exact degeneracy	53
5.1	The weighted cell problem and transport tensor	53
5.2	Null directions in the gradient space	54
5.3	The spatial Fourier symbol and transport coupling	55
6	Strong resonance currents and first-order transport of all five moments	57
6.1	Actual moment matching and the entropy identity	59
6.2	A common resonance measure and the limiting weak equation	60
6.3	Norm convergence and recovery of the five-dimensional kernel	61
6.4	The physical collision term and the first-order macroscopic equation	62
7	Structural relations and further questions	64
	Notation index	66

2 Collision invariants and closed dissipation forms for one-dimensional square-root dispersion

We consider the four-wave resonance relation associated with a pinned one-dimensional chain. The angular variable and the usual lattice momentum are related by $x = 2\pi k$. Fix

$$\mathbb{T}_{2\pi} = \mathbb{R}/(2\pi\mathbb{Z}), \quad 0 < d < \frac{1}{2}, \quad \omega(x) = \sqrt{1 - 2d \cos x}, \quad m(dx) = \frac{dx}{2\pi}. \quad (2.1)$$

The dispersion parameter d is unrelated to the small parameter in the macroscopic limit considered later. With d fixed, we write $\omega = \omega_d$, restoring the subscript when comparing parameters. This dispersion and its four-wave collision operator appear in [4, 5]; the collision invariant problem is also discussed in [6]. Every result below holds for each fixed $0 < d < 1/2$, with constants that may depend on d .

A collision consists of two incoming and two outgoing momenta, which we also call its four legs. The invariant relation compares the values of one function at these four momenta. We use two features of this relation. On a nondegenerate resonance chart, the value at the target point equals an average of the signed sum of the other three values. A change of integration variable transfers derivatives in the target variable to a smooth integral kernel. This yields regularity of invariants and compactness for approximate invariants. On a critical chart, the energy difference and the collision difference share a double factor; its cancellation gives finite dissipation for smooth functions. We proceed through the construction of resonance charts, the classification of invariants, the positive spectral gap, and the closed-form realization.

2.1 Periodic resonance, coarea measure, and the classification theorem

The four momenta satisfy

$$x_0 + x_1 = x_2 + x_3 \pmod{2\pi}, \quad \omega(x_0) + \omega(x_1) = \omega(x_2) + \omega(x_3). \quad (2.2)$$

The first two positions x_0, x_1 are incoming and the last two x_2, x_3 are outgoing. Periodic momentum conservation includes both normal and Umklapp processes: for chosen real representatives, the two momentum sums may differ by an integer multiple of 2π . Eliminate the fourth variable, use $(x, y, z) = (x_0, x_1, x_2)$ as coordinates, and set

$$\begin{aligned} w &= x + y - z \pmod{2\pi}, \\ F(x, y, z) &= \omega(x) + \omega(y) - \omega(z) - \omega(w), \\ \Sigma &= \{(x, y, z) \in \mathbb{T}_{2\pi}^3 : F = 0, \nabla F \neq 0\}, \quad d\Xi = (2\pi)^{-3} \frac{d\mathcal{H}^2}{|\nabla F|}. \end{aligned} \quad (2.3)$$

Here $\mathcal{H}^2$ is Euclidean two-dimensional area in local real coordinates, and ∇F is the gradient in the same coordinates. The coarea factor $|\nabla F|^{-1}$ restricts the three-dimensional momentum integral to the zero-energy surface; $(2\pi)^{-3}$ comes from the probability measure normalization of the three remaining angular variables. The measure $\Xi = \Xi_d$ includes all regular resonance branches. It is locally finite and σ -finite on Σ , but its total mass need not remain finite near critical points of the energy. For a periodic function φ , define the collision difference by

$$\Delta\varphi(x, y, z) = \varphi(x) + \varphi(y) - \varphi(z) - \varphi(x + y - z). \quad (2.4)$$

A resonance is trivial if the two pairs of momenta differ only by a permutation; its collision difference vanishes identically. Retaining all four terms serves two purposes: it annihilates both constants and frequency on resonance, and it cancels their common lower-order parts near critical points. The regularization and dissipation estimates below concern this collision difference.

Theorem 2.1 (Finite measurable collision invariants). *Let $\varphi : \mathbb{T}_{2\pi} \rightarrow \mathbb{C}$ be measurable and finite almost everywhere, and suppose that*

$$\Delta\varphi = 0 \quad \Xi\text{-almost everywhere.} \quad (2.5)$$

There are unique $A, B \in \mathbb{C}$ such that

$$\varphi = A + B\omega \quad m\text{-almost everywhere.} \quad (2.6)$$

In particular, φ has a unique smooth periodic representative. Conversely, every affine-frequency representative $x \mapsto A + B\omega(x)$ satisfies the invariant identity on every resonance in (2.2). Real-valued invariants have real coefficients A, B .

Regularity in this theorem is a consequence of the resonance identity. For a measurable function that is merely finite almost everywhere, we first compare level sets along the three non-target legs to obtain an L^∞ bound on the circle. Resonance averages then produce a smooth representative. Only at that stage do we differentiate along resonance charts: second-order compatibility for the odd part and third-order compatibility for the even part, together with periodic matching, determine A, B . We first verify that changing the function on a one-dimensional null set leaves its identity on the resonance surface unchanged.

Lemma 2.2 (Absolute continuity of the four coordinate maps). *Let $E \subset \mathbb{T}_{2\pi}$ be a one-dimensional Lebesgue null set. For each of the maps $\pi_i = x, y, z, x + y - z$, one has*

$$\Xi(\pi_i^{-1}(E)) = 0. \quad (2.7)$$

Thus (2.5) is independent of the almost-everywhere representative of φ .

Proof. The surface Σ is a real-analytic two-dimensional manifold. Choose a countable cover by relatively compact connected analytic charts. We first show that x is nonconstant on every nonempty open surface patch. If $x = x_*$ on such a patch, its tangent space is $\{dx = 0\}$, so (y, z) gives local coordinates on the patch. Consequently $F(x_*, y, z) = 0$ on a two-dimensional open set. Taking the mixed derivative in y, z gives

$$\omega''(x_* + y - z) = 0$$

on a real interval. Analyticity forces ω'' to vanish identically, and periodicity then makes ω constant, contrary to $d > 0$.

On each connected analytic chart, therefore, some coordinate derivative of x is not identically zero. The zero set of a nonzero real-analytic function is Lebesgue null, so the set where $dx = 0$ has zero two-dimensional measure. The analytic fact used here follows from local power series and Fubini: choose a nonzero coefficient in an expansion in one variable; outside a null set of parameters, the resulting one-variable analytic function has discrete zeros, and a countable rectangle cover completes the argument. Away from this null set, x is a local submersion. Shrinking the chart and changing a coordinate whose derivative is bounded away from zero proves (2.7) for $\pi_0 = x$.

The other three legs follow by exchanging the incoming legs, the outgoing legs, or the incoming and outgoing pairs. In the three coordinates left after elimination, these are integer linear transformations with determinant of absolute value 1. They preserve F up to sign, and hence preserve regularity. On every relatively compact regular chart, Ξ has a smooth positive density with respect to two-dimensional coordinate measure. A countable cover gives the claim on all of Σ . $\square$

We will also prove a positive spectral gap for the collision difference, finite energy of smooth functions, and a maximal closed form. We begin with resonance coordinates covering every point of the circle.

2.2 Nondegenerate resonance charts and regularization of measurable functions

Write $v = \omega'$ for the group velocity. We use the first leg x as the target variable and z as the averaging variable, determining y, w from the conservation laws. Suppose a resonant quartet (x_*, y_*, z_*, w_*) has pairwise distinct non-target velocities $v(y_*), v(z_*), v(w_*)$. Then $F_y = v(y_*) - v(w_*) \neq 0$, and the implicit function theorem gives a resonance chart

$$(x, H(x, z), z, W(x, z)), \quad W = x + H - z, \quad (2.8)$$

on a real coordinate rectangle $I \times J$. Differentiating the energy identity gives

$$H_z = \frac{v(z) - v(W)}{v(H) - v(W)}, \quad W_z = \frac{v(z) - v(H)}{v(H) - v(W)}. \quad (2.9)$$

The nonzero denominator allows us to solve the energy equation for $y = H(x, z)$; the two nonzero numerators allow both H, W to be used as one-dimensional changes of integration variable. These three nondegeneracy conditions have distinct roles. After shrinking the rectangle, each leg takes values in a real lift interval of length less than 2π , and H_z, W_z and $v(H) - v(W)$ are bounded away from zero on the closure. For fixed x , the maps H, W are therefore monotone in z and injective even as circle-valued maps. The coarea measure on this chart is

$$d\Xi = (2\pi)^{-3} \frac{dx \, dz}{|v(H) - v(W)|}. \quad (2.10)$$

Lemma 2.3 (Resonance charts covering all target points). *For every $x_* \in \mathbb{T}_{2\pi}$, there is a nontrivial resonant quartet whose first leg is x_* and whose other three velocities are pairwise distinct. Consequently the circle admits a finite cover by target intervals satisfying (2.8)–(2.10).*

Proof. An explicit symmetric quartet treats every target except four points. At those points some velocities coincide, so we construct an asymmetric quartet instead. Reflection, translation, and compactness then give the finite cover. First suppose that

$$x_* \notin \mathcal{E} := \{0, \pi/2, \pi, 3\pi/2\}.$$

Take

$$(x, \pi - x, -x, \pi + x). \quad (2.11)$$

The four legs are distinct and conserve periodic momentum; energy conservation follows from the evenness of ω . Write

$$a = \omega'(x) = \frac{d \sin x}{\omega(x)}, \quad b = \omega'(\pi - x) = \frac{d \sin x}{\omega(\pi - x)}. \quad (2.12)$$

The numbers a, b are nonzero, have the same sign, and are unequal. The non-target velocities $b, -a, -b$ are thus pairwise distinct. In particular, at the center of this chart,

$$F_y = 2b, \quad H_z = \frac{b - a}{2b}, \quad W_z = \frac{-b - a}{2b}. \quad (2.13)$$

To treat the four exceptional points, temporarily allow an auxiliary signed parameter $-1/2 < \tilde{d} < 1/2$, $\tilde{d} \neq 0$, and set

$$\begin{aligned} u &= \sqrt{1 - 2\tilde{d}}, & c_0 &= 1 - u, & q &= \sqrt{4 + 4c_0 - 7c_0^2}, \\ A &= \frac{c_0 + q}{2}, & B &= \frac{q - c_0}{2}. \end{aligned} \quad (2.14)$$

Then $0 < u < \sqrt{2}$, $1 - \sqrt{2} < c_0 < 1$, $c_0 \neq 0$, and

$$\begin{aligned} 2\tilde{d} &= c_0(2 - c_0), & A - B &= c_0, & A + B &= q, \\ AB &= u(1 + 2c_0) > 0, & A^2 + B^2 &= 2 + 2c_0 - 3c_0^2. \end{aligned} \quad (2.15)$$

Since $1 + 2c_0 > 3 - 2\sqrt{2} > 0$, we have $q > |c_0|$ and hence $A, B > 0$. Moreover,

$$(1 - A^2)^2 + (1 - B^2)^2 = c_0^2(2 - c_0)^2 = (2\tilde{d})^2, \quad (2.16)$$

$$\begin{aligned} (2 + c_0)^2 - q^2 &= 8c_0^2 > 0, \\ q^2 - (2 - c_0)^2 &= 8c_0(1 - c_0). \end{aligned} \quad (2.17)$$

Thus $B < 1$, while $A > 1$ if $\tilde{d} > 0$ and $A < 1$ if $\tilde{d} < 0$. Equation (2.16) determines a unique angle y such that

$$\cos y = \frac{1 - A^2}{2\tilde{d}}, \quad \sin y = \frac{1 - B^2}{2\tilde{d}}. \quad (2.18)$$

For positive parameter, $y \in (\pi/2, \pi)$; for negative parameter, $y \in (\pi, 3\pi/2)$. With auxiliary dispersion $\omega_{\tilde{d}}(x) = \sqrt{1 - 2\tilde{d}\cos x}$, we have

$$\omega_{\tilde{d}}(0) = u, \quad \omega_{\tilde{d}}(y) = A, \quad \omega_{\tilde{d}}(\pi/2) = 1, \quad \omega_{\tilde{d}}(y - \pi/2) = B.$$

The equality $u + A = 1 + B$ shows that

$$(0, y, \pi/2, y - \pi/2) \quad (2.19)$$

conserves both periodic momentum and energy.

Put $w = y - \pi/2$. The four velocities are $0, v_y, \tilde{d}, v_w$, respectively, where

$$v_y = \frac{1 - B^2}{2A}, \quad v_w = \frac{A^2 - 1}{2B}. \quad (2.20)$$

Factoring with (2.15) gives

$$\begin{aligned} v_y - v_w &= -\frac{c_0 u(A + B)}{2AB}, \\ v_y - \tilde{d} &= -\frac{c_0 u(A + 1)}{2A}, \\ v_w - \tilde{d} &= \frac{c_0 u(1 - B)}{2B}. \end{aligned} \quad (2.21)$$

Every right-hand side is nonzero. Hence

$$\begin{cases} 0 < v_y < \tilde{d} < v_w, & \tilde{d} > 0, \\ v_w < \tilde{d} < 0 < v_y, & \tilde{d} < 0. \end{cases} \quad (2.22)$$

In particular, all four velocities are pairwise distinct.

Return now to the fixed positive parameter d . With $\tilde{d} = d$, the quartet (2.19) gives a chart at target 0. Exchanging the incoming and outgoing pairs writes it as $(\pi/2, w, 0, y)$ and gives a chart at target $\pi/2$. Reflecting all four legs gives a chart at target $3\pi/2$. Finally, construct (2.19) with $\tilde{d} = -d$ and translate all four legs by π . The identities

$$\omega_d(s + \pi) = \omega_{-d}(s), \quad \omega'_d(s + \pi) = \omega'_{-d}(s) \quad (2.23)$$

place the resulting quartet on the resonance surface for the positive parameter d and give a chart at target π . Both momentum sums increase by 2π , so periodic momentum conservation is preserved. This covers all four exceptional points. For each target, first choose $x_* \in I' \subseteq I$ inside its resonance chart $I \times J$, then choose a finite subcover from these inner intervals I' . The averaging kernels below are defined on the outer intervals I , while uniform estimates are taken on $\overline{I'}$; subsequent chart restrictions leave the chosen cover of the circle intact. $\square$

In this atlas, each of the non-target maps $z, H(x, z), W(x, z)$ permits a change of variable in z . For integrable φ , the resonance average in (2.25) can therefore be written as the integral of φ against a smooth kernel, with target derivatives falling on that kernel. The resonance identity makes the function equal to its average. To cover the finite measurable class of Theorem 2.1, the next proof first uses level sets to establish boundedness, and only then integrates function values.

Lemma 2.4 (From finite measurability to smoothness). *Every measurable function that is finite almost everywhere and satisfies (2.5) belongs to $L^\infty(\mathbb{T}_{2\pi})$ and has a smooth periodic representative. If $\varphi \in L^1$ is already known, this representative satisfies, for each integer $r \geq 0$,*

$$\|\varphi\|_{C^r} \leq C_{d,r} \|\varphi\|_{L^1(m)}. \quad (2.24)$$

Proof. Fix inner and outer intervals and a rectangle from Lemma 2.3, denoted by $I' \Subset I$ and $I \times J$, with $|H_z| \geq c_H > 0$ and $|W_z| \geq c_W > 0$. Set $E_M = \{s : |\varphi(s)| > M\}$. Since the circle has finite measure and φ is finite almost everywhere, $|E_M| \rightarrow 0$. For each fixed $x \in I$, one-dimensional changes of variable give

$$\begin{aligned} |\{z \in J : z \in E_M\}| &\leq |E_M|, \\ |\{z \in J : H(x, z) \in E_M\}| &\leq c_H^{-1} |E_M|, \\ |\{z \in J : W(x, z) \in E_M\}| &\leq c_W^{-1} |E_M|. \end{aligned}$$

Choose a finite M so that the sum of these bounds is less than $|J|/2$. For each x , there is then a set of z of measure at least $|J|/2$ on which the three non-target values have absolute value at most M . Integrability of the function has not been used. Apply Fubini only to the indicator of the set where the invariant identity fails. By (2.10), for almost every x the identity holds for almost every z . Thus, for almost every $x \in I$, the positive measure set just constructed contains a z at which the identity holds, and

$$|\varphi(x)| = |\varphi(z) + \varphi(W(x, z)) - \varphi(H(x, z))| \leq 3M.$$

The same bound holds on I' . The finitely many inner intervals cover the circle, so taking the maximum of their thresholds M gives the global L^∞ bound.

We may now use L^1 regularization. Choose $\chi \in C_c^\infty(J)$ with $\chi \geq 0$ and $\int \chi(z) dz = 1$. The invariant relation gives, almost everywhere,

$$\varphi(x) = \int \chi(z) \varphi(z) dz + \int \chi(z) \varphi(W(x, z)) dz - \int \chi(z) \varphi(H(x, z)) dz. \quad (2.25)$$

Change variables along W and H in the last two terms. For example, if $z = Z_H(x, s)$ is the inverse of $s = H(x, z)$, the corresponding kernel is

$$K_H(x, s) = \frac{\chi(Z_H(x, s))}{|H_z(x, Z_H(x, s))|}.$$

The support of χ has positive distance from ∂J , and $|H_z| \geq c_H$, so the distance between $H(x, \text{supp } \chi)$ and $H(x, \partial J)$ is uniformly positive. The inverse function theorem gives uniform bounds on all derivatives of Z_H for $x \in \overline{I'}$. Consequently K_H vanishes near the boundary of the image interval and extends smoothly by zero to a kernel on the circle, with

$$\sup_{x \in \overline{I'}, s \in \mathbb{T}_{2\pi}} |\partial_x^r K_H(x, s)| \leq C_r \quad (r \geq 0).$$

The W term is treated in the same way; the first kernel is χ , independent of x . Thus the right-hand side of (2.25) is smooth, with each derivative bounded by $C_r \|\varphi\|_1$. The finite cover gives (2.24). Local smooth

representatives agree almost everywhere on overlaps and hence, by continuity, everywhere there. They therefore define a unique global smooth periodic representative.

Lemma 2.2 shows that replacing the function by this representative preserves the invariant identity almost everywhere on all of Σ . The collision difference is continuous on the regular surface, and every regular open patch has positive measure. The identity therefore holds pointwise at every regular resonant point. $\square$

2.3 Second- and third-order compatibility on the symmetric resonance branch

We next prove that a C^1 periodic invariant must have the form $A + B\omega$. Away from $\mathcal{E}$, the symmetric charts of Lemma 2.3 and (2.25) already give local smoothness, so the second and third derivatives below require only the initial C^1 assumption. If the identity is initially known only almost everywhere on Σ , continuity first makes it pointwise on these charts. We treat the odd and even parts separately. For the odd part, two differentiations leave a nonzero coefficient times φ'' . For the even part, we subtract a multiple of the third-order energy identity from the third-order invariant identity to eliminate the unknown derivative of the resonance branch. The resulting ordinary differential equation is solved on two open intervals, whose solutions are then matched by endpoint regularity and periodicity.

Reflection of all four legs, $x_i \mapsto -x_i$, preserves resonance. Hence the odd and even parts separately satisfy the invariant relation. First let φ be odd. On (2.11), the invariant relation gives

$$\varphi(\pi - x) = -\varphi(x), \quad \varphi(\pi + x) = \varphi(x). \quad (2.26)$$

Fix $x \notin \mathcal{E}$ and take the following curve in its resonance chart:

$$z(t) = -x + t, \quad y(t) = H(x, z(t)), \quad w(t) = x + y(t) - z(t). \quad (2.27)$$

Write $y'(0) = \alpha$ and $y''(0) = \beta$. Then $w'(0) = \alpha - 1$ and $w''(0) = \beta$. By (2.26) and oddness, the derivatives at the four base points satisfy

$$\begin{aligned} \varphi'(\pi - x) &= \varphi'(-x) = \varphi'(\pi + x) = \varphi'(x), \\ \varphi''(\pi - x) &= \varphi''(-x) = -\varphi''(x), \quad \varphi''(\pi + x) = \varphi''(x). \end{aligned}$$

Differentiate $\varphi(x) + \varphi(y(t)) - \varphi(z(t)) - \varphi(w(t)) = 0$ twice. The two $\beta\varphi'$ terms cancel, leaving

$$0 = [1 - \alpha^2 - (\alpha - 1)^2]\varphi''(x) = \frac{b^2 - a^2}{2b^2}\varphi''(x). \quad (2.28)$$

Since $a^2 \neq b^2$ in (2.12), we obtain $\varphi'' = 0$ on $\mathbb{T}_{2\pi} \setminus \mathcal{E}$. The C^1 assumption matches the constant values of φ' on the intervening intervals. Periodicity forces their common slope to vanish, and oddness then forces the function itself to vanish.

Now let φ be even and fix $x \in (0, \pi) \setminus \{\pi/2\}$. Along (2.27), write

$$y'(0) = \alpha, \quad y''(0) = \beta, \quad y'''(0) = \gamma, \quad c_a = \omega''(x), \quad c_b = \omega''(\pi - x), \quad r = a/b.$$

The first- and second-order energy identities give, respectively,

$$\alpha = \frac{1 - r}{2}, \quad \beta = \frac{c_a + rc_b}{2b}. \quad (2.29)$$

Define $g(x) = \varphi'(x)/\omega'(x)$. The first-order invariant identity is

$$0 = (2\alpha - 1)\varphi'(\pi - x) + \varphi'(x),$$

and therefore

$$g(\pi - x) = g(x), \quad g'(\pi - x) = -g'(x), \quad g''(\pi - x) = g''(x). \quad (2.30)$$

The third-order energy and invariant identities are

$$0 = 2b\gamma + 3c_b\beta + \omega'''(\pi - x)[\alpha^3 + (\alpha - 1)^3] + \omega'''(x), \quad (2.31)$$

$$0 = 2\varphi'(\pi - x)\gamma + 3\varphi''(\pi - x)\beta + \varphi'''(\pi - x)[\alpha^3 + (\alpha - 1)^3] + \varphi'''(x). \quad (2.32)$$

Subtract $g(x)$ times (2.31) from (2.32). This eliminates the third branch derivative γ . Using $\varphi' = \omega'g$ and (2.30), we have

$$\begin{aligned} \varphi''(\pi - x) - gc_b &= -bg', \\ \varphi'''(\pi - x) - g\omega'''(\pi - x) &= -2c_bg' + bg'', \\ \varphi'''(x) - g\omega'''(x) &= 2c_ag' + ag''. \end{aligned} \quad (2.33)$$

With $H_3 = \alpha^3 + (\alpha - 1)^3 = -r(3 + r^2)/4$, the remaining identity reads

$$-3b\beta g' + (-2c_bg' + bg'')H_3 + 2c_ag' + ag'' = 0.$$

Substitution of (2.29) and simplification yield

$$a(1 - r^2)g'' + 2(c_a + r^3c_b)g' = 0. \quad (2.34)$$

Since $b'(x) = -c_b$, direct differentiation gives

$$\frac{d}{dx} \log \left| \frac{a^2b^2}{b^2 - a^2} \right| = \frac{2(c_a + r^3c_b)}{a(1 - r^2)}. \quad (2.35)$$

Thus (2.34) is equivalent to

$$\left(\frac{a^2b^2}{b^2 - a^2} g' \right)' = 0. \quad (2.36)$$

For the dispersion (2.1), the integrating factor simplifies explicitly:

$$a^2 = \frac{d^2 \sin^2 x}{1 - 2d \cos x}, \quad b^2 = \frac{d^2 \sin^2 x}{1 + 2d \cos x}, \quad \frac{a^2b^2}{b^2 - a^2} = -\frac{d \sin^2 x}{4 \cos x}. \quad (2.37)$$

On each of the intervals $(0, \pi/2)$ and $(\pi/2, \pi)$, we obtain

$$g(x) = A_0 + \frac{B_0}{\sin x}, \quad \varphi'(x) = \frac{d(A_0 \sin x + B_0)}{\omega(x)}. \quad (2.38)$$

For now the constants may differ between the two intervals. An even periodic C^1 function satisfies $\varphi'(0) = \varphi'(\pi) = 0$. Since $\omega(0), \omega(\pi) > 0$, the two endpoints force $B_0 = 0$ on their respective intervals. At $\pi/2$ we have $\omega'(\pi/2) = d \neq 0$, so continuity of φ' also matches the two values of A_0 . Continuity of φ matches the constants of integration. Hence $\varphi = A + B\omega$ on $[0, \pi]$, and evenness and periodicity extend the identity to the circle.

This proves the C^1 classification. Together with Lemma 2.4, it proves Theorem 2.1. Since $\omega(0) \neq \omega(\pi)$, the two coefficients are unique. The converse follows directly from (2.2).

Remark 2.5 (The interval equation and endpoint resonances). Eliminating the B_0 term in (2.38) requires the endpoint conditions. For example, the continuous even periodic function

$$\Phi(x) = d \int_0^{\text{dist}(x, 2\pi\mathbb{Z})} \frac{ds}{\omega(s)}$$

gives $g = 1/\sin x$ on $(0, \pi)$ and satisfies (2.30) and (2.34), but has corners at $0, \pi$. For the positive-parameter value of y in (2.19), put $w = y - \pi/2 \in (0, \pi/2)$. Its collision difference is

$$\Delta\Phi = d \int_0^w \left(\frac{1}{\omega(s + \pi/2)} - \frac{1}{\omega(s)} \right) ds < 0.$$

Here $\omega(s + \pi/2) > \omega(s)$. Thus the endpoint regularity provided by charts at every target excludes this branch, which the third-order equation on the open intervals alone would retain.

Corollary 2.6 (Positive affine-frequency denominators). *Suppose $0 < W < \infty$ almost everywhere and $1/W$ satisfies (2.5). Then $W = (A + B\omega)^{-1}$ almost everywhere, with $A, B \in \mathbb{R}$. If also $W \in L^1(m)$, then*

$$\min_x (A + B\omega(x)) > 0. \quad (2.39)$$

Proof. Theorem 2.1 applies to the finite measurable function $1/W$. Since ω is nonconstant, the identity $\text{Im } A + \text{Im } B\omega = 0$ forces A, B to be real. The continuous denominator is positive almost everywhere and therefore nonnegative everywhere. If it vanishes somewhere, a nonzero affine function of frequency can vanish only at a maximum or minimum of ω . Both extrema are nondegenerate and quadratic, so the reciprocal has a nonintegrable quadratic pole. An identically zero denominator contradicts the almost-everywhere finiteness and positivity of W . This proves (2.39). $\square$

The corollary assumes the resonance invariant identity for $1/W$. Starting from a nonlinear stationary state, one must still justify the collision integral and entropy production identity in the relevant function class to obtain that assumption. Condition (2.39) allows B to be positive, zero, or negative.

2.4 Resonance averaging and a positive spectral gap

Let $a(x_0, x_1, x_2, x_3)$ be a smooth positive weight on the compact four-dimensional torus such that

$$0 < a_- \leq a \leq a_+ < \infty. \quad (2.40)$$

In the reduced coordinates it is always evaluated at $a(x, y, z, x+y-z)$. The lower bound in this subsection requires no additional symmetry. When relating it to the linearized collision operator, we will require invariance of a under exchange of the two incoming legs, exchange of the two outgoing legs, and exchange of the two pairs. Define the extended quadratic form

$$\mathcal{D}_a(\varphi) = \int_{\Sigma} a |\Delta\varphi|^2 d\Xi \in [0, \infty], \quad \varphi \in L^2(m). \quad (2.41)$$

Lemma 2.2 makes this well-defined on L^2 equivalence classes. Let P_0 be the orthogonal projection in $L^2(m)$ onto $\text{span}\{1, \omega\}$.

Theorem 2.7 (Positive spectral gap). *There is $\lambda_{d,a} > 0$ such that every $\varphi \in L^2(m)$ satisfies*

$$\mathcal{D}_a(\varphi) \geq \lambda_{d,a} \|(I - P_0)\varphi\|_{L^2(m)}^2. \quad (2.42)$$

The left-hand side may be infinite.

Proof. We construct a smooth averaging operator K for which the dissipation controls $\varphi - K\varphi$. A bounded sequence with vanishing dissipation is then asymptotic to the image of a compact operator and has a strongly convergent subsequence. We pass zero dissipation to its limit and apply the classification of invariants. Choose a nonnegative smooth partition of unity ρ_i subordinate to the finite inner-interval

cover of Lemma 2.3, with $\text{supp } \rho_i \subseteq I'_i \subseteq I_i$. Choose also $\chi_i \in C_c^\infty(J_i)$, $\chi_i \geq 0$, $\int \chi_i = 1$. For an arbitrary $\varphi \in L^2$, define

$$\begin{aligned} K_i \varphi(x) &= \int \chi_i(z) [\varphi(z) + \varphi(W_i(x, z)) - \varphi(H_i(x, z))] dz, \\ K \varphi &= \sum_i \rho_i K_i \varphi. \end{aligned} \quad (2.43)$$

The change-of-variable argument in Lemma 2.4 shows that K has a global smooth kernel and that, for each integer $r \geq 0$,

$$\|K \varphi\|_{C^r} \leq C_{d,r} \|\varphi\|_2. \quad (2.44)$$

In particular, $K : L^2 \rightarrow L^2$ is compact. It need not be a projection or a self-adjoint operator.

By definition and $\int \chi_i = 1$,

$$\varphi(x) - K_i \varphi(x) = \int \chi_i(z) \Delta \varphi(x, H_i(x, z), z) dz.$$

Applying Jensen's inequality twice gives

$$\begin{aligned} \|\varphi - K \varphi\|_2^2 &\leq \sum_i \int \rho_i(x) \chi_i(z) |\Delta \varphi(x, H_i(x, z), z)|^2 m(dx) dz \\ &\leq C_{d,a} \mathcal{D}_a(\varphi). \end{aligned} \quad (2.45)$$

The last step uses (2.10): the Jacobian factors and weights have uniform bounds on the compact supports, and finite overlap contributes only a finite constant. For a general L^2 function, the three compositions on a chart are locally in L^2 , so Fubini justifies this step. The inequality is automatic when the right-hand side is infinite.

If (2.42) failed, there would be $\varphi_n \perp \text{span}\{1, \omega\}$ with $\|\varphi_n\|_2 = 1$ and $\mathcal{D}_a(\varphi_n) \rightarrow 0$. By (2.45), $\varphi_n - K \varphi_n \rightarrow 0$ in L^2 . Compactness gives a subsequence $\varphi_n \rightarrow \varphi$ strongly in L^2 , with $\|\varphi\|_2 = 1$ and $P_0 \varphi = 0$. Pass to a further subsequence converging almost everywhere in the one-dimensional variable. Lemma 2.2 transfers convergence simultaneously along all four legs to all of Σ . Fatou's lemma gives

$$\mathcal{D}_a(\varphi) \leq \liminf_n \mathcal{D}_a(\varphi_n) = 0.$$

Theorem 2.1 forces $\varphi \in \text{span}\{1, \omega\}$, contradicting its unit norm and orthogonality. This limiting step uses the coarea measure on all regular resonances; the finite averaging charts are used only for the compactness estimate (2.45). Finally, $\mathcal{D}_a \geq a_- \mathcal{D}_1$ permits the choice $\lambda_{d,a} = a_- \lambda_{d,1}$, which also yields a uniform lower bound as the weight varies over a compact positive parameter family. $\square$

2.5 Critical resonances and cancellation in the collision difference

Resonance averaging has already given the positive spectral gap. To construct a dense domain for (2.41), we must also prove that smooth test functions have finite energy near critical points of the energy. The key is the common factor in (2.49) below: $F = -uvG_\omega$ and $\Delta \varphi = -uvG_\varphi$. Forming the collision difference before taking the regularized energy δ limit preserves this cancellation within the same integral. The factorization uses second-order calculus for $\varphi \in C^2$; general L^2 functions will subsequently be treated through the maximal closed difference operator. We first show that every critical resonance admits the required coordinates. Direct calculation gives

$$\omega'(x) = \frac{d \sin x}{\omega(x)}, \quad \omega''(x) = \frac{d[\cos x - d(1 + \cos^2 x)]}{\omega(x)^3}. \quad (2.46)$$

The polynomial $h(t) = t - d(1 + t^2)$ is strictly increasing on $[-1, 1]$, since $h'(t) = 1 - 2dt > 0$, and $h(-1) < 0 < h(1)$. Its unique zero $t_* \in (0, 1)$ is simple. Thus ω' first increases strictly and then decreases strictly on $(0, \pi)$, and has a corresponding unique minimum on $(\pi, 2\pi)$. Both zeros of the curvature are simple; for example, $x_* = \arccos t_*$ satisfies

$$\omega'''(x_*) = -\frac{d(1 - 2d \cos x_*) \sin x_*}{\omega(x_*)^3} \neq 0. \quad (2.47)$$

Lemma 2.8 (Critical points and their local coordinates). *Every critical quartet in the resonance set $\{F = 0\}$ is a trivial permutation. Near each critical point, after permuting the legs and taking local real lifts, one can choose coordinates*

$$x = z + u, \quad y = z + v, \quad w = z + u + v \quad (2.48)$$

in which, for every C^2 function φ ,

$$F = -uvG_\omega, \quad \Delta\varphi = -uvG_\varphi, \quad G_\varphi(z, u, v) = \int_0^1 \int_0^1 \varphi''(z + su + tv) \, ds \, dt. \quad (2.49)$$

The neighborhood can be chosen to have one of the following two forms:

1. $|G_\omega|$ is uniformly bounded away from zero, and the center has $u = v = 0$;
2. $|\partial_z G_\omega|$ is uniformly bounded away from zero, so that $q = G_\omega$ can be used as a coordinate.

Proof. The unimodality following from (2.46) shows that each velocity value has at most two preimages on the circle. If the preimages are distinct, they have curvatures of opposite signs and distinct frequencies: positive and negative velocities lie in their respective semicircles, on each of which ω is strictly monotone; zero velocity occurs only at $0, \pi$. On the other hand,

$$\nabla F = 0 \iff \omega'(x) = \omega'(y) = \omega'(z) = \omega'(w). \quad (2.50)$$

The four legs of a critical quartet thus take at most two values p, q . If p occurs j, k times in the incoming and outgoing pairs, respectively, energy conservation gives

$$(j - k)[\omega(p) - \omega(q)] = 0.$$

Distinct points have distinct frequencies, so $j = k$. The two multisets agree, and the quartet is a trivial permutation. The case of four identical legs is included.

For the trivial pairing $x = z, y = w$, choose (2.48), with $u = 0$ at the center. If the other two points are distinct, choose a nonzero local real representative $v_0 \neq 0$; if all four legs coincide, take $u_0 = v_0 = 0$. The change of coordinates has determinant of absolute value 1, and periodic functions are represented by their periodic real lifts. Two applications of the fundamental theorem of calculus give (2.49). Exchanging the two incoming legs reduces the other trivial pairing to this case.

At a critical point with $u = 0, v \neq 0$,

$$\begin{aligned} G_\omega(z, 0, v) &= \frac{\omega'(z + v) - \omega'(z)}{v} = 0, \\ \partial_z G_\omega(z, 0, v) &= \frac{\omega''(z + v) - \omega''(z)}{v} \neq 0. \end{aligned} \quad (2.51)$$

The last inequality follows because the two preimages of the same velocity have curvatures of opposite signs. At $u = v = 0$, $G_\omega = \omega''(z)$. If this vanishes, then $\partial_z G_\omega = \omega'''(z) \neq 0$ by (2.47), giving the second type of chart. If it does not vanish, we obtain the first type. Every quartet with four identical legs is critical, so the first type must also be included. Compactness of the critical set gives a finite cover by such neighborhoods. $\square$

Equation (2.49) is a double-difference identity. For example, taking $\varphi(t) = t^2$ only in a local real lift gives

$$(z+u)^2 + (z+v)^2 - z^2 - (z+u+v)^2 = -2uv, \quad G_\varphi = 2.$$

This local, nonperiodic calculation explains the factor uv : the four terms cancel the constant and linear parts, leaving only the mixed second-order term. Averaging the second derivative in (2.49) gives the same structure for any C^2 function.

Proposition 2.9 (Finite energy of smooth functions and the regularized limit). *For $\varphi \in C^2(\mathbb{T}_{2\pi})$,*

$$\mathcal{D}_a(\varphi) \leq C_{d,a} \|\varphi\|_{C^2}^2, \quad \int_{\Sigma} a |\Delta \varphi| \, d\Xi \leq C_{d,a} \|\varphi\|_{C^2}. \quad (2.52)$$

Let $\rho_\eta \in C_c^\infty(\mathbb{R})$ be any family of nonnegative functions satisfying

$$\int \rho_\eta(s) \, ds = 1, \quad \text{supp } \rho_\eta \subset [-\eta, \eta]. \quad (2.53)$$

Then, for $\varphi, \psi \in C^2$,

$$\begin{aligned} \lim_{\eta \downarrow 0} \int_{\mathbb{T}_{2\pi}^3} a \Delta \varphi \overline{\Delta \psi} \rho_\eta(F) \, m(dx) m(dy) m(dz) \\ = \int_{\Sigma} a \Delta \varphi \overline{\Delta \psi} \, d\Xi. \end{aligned} \quad (2.54)$$

No evenness, fixed profile, or uniform $O(\eta^{-1})$ height bound is required of ρ_η .

Proof. We first estimate the collision difference in the two types of critical chart and then treat the regular region away from the critical set. In the second type, the nontrivial resonance branch is $G_\omega = 0$. The first type has only trivial zero branches, and its regularized integral must separately be shown to vanish. In a chart of the second type, change variables to $q = G_\omega$. On the nontrivial regular branch $q = 0$, $uv \neq 0$, (2.49) gives

$$|\Delta \varphi|^2 \, d\Xi = (2\pi)^{-3} \frac{|uv| |G_\varphi|^2}{|\partial_z G_\omega|} \, du \, dv, \quad (2.55)$$

$$|\Delta \varphi| \, d\Xi = (2\pi)^{-3} \frac{|G_\varphi|}{|\partial_z G_\omega|} \, du \, dv. \quad (2.56)$$

Both expressions are locally integrable, since $|G_\varphi| \leq \|\varphi''\|_\infty$. The collision difference vanishes on trivial regular branches. Charts of the first type contain only trivial zero branches and hence contribute nothing to the form. Regular compact sets away from the critical set have finite coarea measure, so a finite cover proves (2.52).

Choose a smooth partition of unity subordinate to these charts and the remaining regular region. In a chart of the second type, the regularized integral at fixed u, v has the form

$$u^2 v^2 \int A(q, u, v) \rho_\eta(-uvq) \, dq, \quad (2.57)$$

where A includes the weight, cutoff, $G_\varphi \overline{G_\psi}$, and Jacobian factor; it is continuous and uniformly bounded. The absolute value of (2.57) is bounded by $C|uv|$, using $\int \rho_\eta(-uvq) \, dq \leq |uv|^{-1}$. For each fixed $uv \neq 0$, the one-dimensional approximate identity gives $|uv| A(0, u, v)$. Dominated convergence on the bounded u, v region gives the sesquilinear version of (2.55). The axes $uv = 0$ form a two-dimensional null set, and the uniform $C|uv|$ bound also controls their neighborhoods.

A chart of the first type is centered at $u = v = 0$. After shrinking the neighborhood, set $V = vG_\omega(z, u, v)$; its derivative with respect to v is bounded away from zero. Then

$$F = -uV, \quad \Delta\varphi = -uV \frac{G_\varphi}{G_\omega}, \quad |\Delta\varphi\Delta\psi| \leq Cu^2V^2. \quad (2.58)$$

Take a rectangle $[-L, L]^2$ containing the projection of this chart and set

$$J_j(\eta) = \int_{[-L, L]^2} |uV|^j \rho_\eta(-uV) \, du \, dV.$$

For fixed $u \neq 0$, a one-dimensional change of variables bounds the slice of J_1 by L ; the support condition gives the additional bound $\eta/|u|$. Splitting the integral into $|u| \leq \delta$ and $|u| > \delta$ yields $J_1(\eta) \leq 2L\delta + 2L\eta/\delta$. Choosing $\delta = \sqrt{\eta}$, we obtain $J_1(\eta) \leq 4L\sqrt{\eta}$. On the support of the kernel, $|uV| \leq \eta$, so $J_2(\eta) \leq \eta J_1(\eta)$. The coordinate Jacobian is bounded, and the contribution of this chart satisfies

$$|\text{regularized integral}| \leq CJ_2(\eta) \leq C\eta^{3/2} \rightarrow 0. \quad (2.59)$$

Only (2.53) was used, so the argument applies to any family stated in the proposition. In the remaining regular region, the usual coarea change of variables applies; compact sets away from $F = 0$ eventually contribute nothing. Summing finitely many pieces proves (2.54). $\square$

This limit regularizes the critical energy after the collision difference has been formed. It does not assign a finite measure limit to $\rho_\eta(F)$ alone, without cancellation. For smooth collision differences there is no residual critical contribution to the form.

2.6 The maximal closed form, self-adjoint realization, and linearization normalization

Set

$$\mathcal{H} = L^2(\mathbb{T}_{2\pi}, m), \quad \mathcal{G}_a = L^2(\Sigma, a \, d\Xi), \quad \text{Dom } R_a = \{\varphi \in \mathcal{H} : \Delta\varphi \in \mathcal{G}_a\}, \quad R_a\varphi = \Delta\varphi. \quad (2.60)$$

The domain contains every L^2 function whose collision difference is square integrable; this is the maximal difference operator. The four individual pullbacks need not belong to $\mathcal{G}_a$, since the domain allows cancellation between them. Lemma 2.2 ensures that the operator is independent of representatives.

Theorem 2.10 (Closed form and spectral-gap realization). *The operator R_a is closed and densely defined, with $C^2(\mathbb{T}_{2\pi}) \subset \text{Dom } R_a$. For every $\gamma > 0$, the form*

$$\mathfrak{q}_a(\varphi, \psi) = \frac{\gamma}{4} \langle R_a\varphi, R_a\psi \rangle_{\mathcal{G}_a}, \quad \text{Dom } \mathfrak{q}_a = \text{Dom } R_a \quad (2.61)$$

is closed and densely defined. Its associated nonnegative self-adjoint operator is

$$L_a = \frac{\gamma}{4} R_a^* R_a, \quad \text{Dom } L_a = \{\varphi \in \text{Dom } R_a : R_a\varphi \in \text{Dom } R_a^*\}. \quad (2.62)$$

Moreover,

$$\ker L_a = \text{span}\{1, \omega\}, \quad \mathfrak{q}_a(\varphi, \varphi) \geq \frac{\gamma}{4} \lambda_{d,a} \|(I - P_0)\varphi\|_2^2. \quad (2.63)$$

Proof. Proposition 2.9 gives $C^2 \subset \text{Dom } R_a$, hence density of the domain. Suppose $\varphi_n \rightarrow \varphi$ in $\mathcal{H}$ and $R_a\varphi_n \rightarrow Z$ in $\mathcal{G}_a$. First take a subsequence such that $\varphi_n \rightarrow \varphi$ almost everywhere in the one-dimensional variable. Lemma 2.2 gives simultaneous almost-everywhere convergence of its four pullbacks on all of Σ , so $R_a\varphi_n \rightarrow \Delta\varphi$ almost everywhere. Take a further almost-everywhere convergent subsequence from the $\mathcal{G}_a$ -convergent sequence to obtain $Z = \Delta\varphi$. Thus $\varphi \in \text{Dom } R_a$, proving that the operator is closed.

The square of the graph norm is $\|\varphi\|_2^2 + (\gamma/4)\|R_a\varphi\|_{\mathcal{G}_a}^2$. Closedness makes the domain a Hilbert space in this norm. For every $f \in \mathcal{H}$, the Riesz representation theorem gives a unique $u \in \text{Dom } R_a$ satisfying

$$\langle u, v \rangle + \mathfrak{q}_a(u, v) = \langle f, v \rangle \quad (v \in \text{Dom } R_a).$$

By the definition of the adjoint, this is equivalent to $u \in \text{Dom } L_a$ and $(I + L_a)u = f$. To verify density of the operator domain as well, denote the solution operator by $Sf = u$. Taking $v = u$ gives $\|Sf\|_2 \leq \|f\|_2$; testing the solutions for two right-hand sides against one another gives

$$\langle Sf, g \rangle = \langle Sf, Sg \rangle + \mathfrak{q}_a(Sf, Sg) = \langle f, Sg \rangle.$$

Thus S is bounded and self-adjoint. If $Sf = 0$, the variational identity makes f orthogonal to the dense domain $\text{Dom } R_a$, hence $f = 0$. Therefore $\overline{\text{Ran } S} = (\ker S)^\perp = \mathcal{H}$, while $\text{Ran } S = \text{Dom } L_a$. We now know that L_a is densely defined, symmetric, and nonnegative, and that $I + L_a$ is surjective. For $h \in \text{Dom } L_a^*$, take $u = S(h + L_a^*h)$. Then $(I + L_a^*)(h - u) = 0$; this kernel is orthogonal to $\text{Ran}(I + L_a) = \mathcal{H}$, so $h = u \in \text{Dom } L_a$. This proves self-adjointness and identifies the operator associated with (2.61). Its kernel is $\ker R_a$. Theorem 2.1 identifies that kernel, and Theorem 2.7 gives (2.63). $\square$

The realization in Theorem 2.10 is determined by the maximal integral domain (2.60). Proposition 2.9 shows that C^2 is a dense subspace of finite-energy functions, where density refers to the ambient L^2 space. Establishing that C^2 is a form core or an operator core would require the corresponding approximation in graph norm. In particular, $C^2 \subset \text{Dom } R_a$ does not by itself imply $C^2 \subset \text{Dom } L_a$.

We conclude by relating this realization to the weak collision expression. Suppose a has the four-leg symmetries stated above. For $\varphi, \psi \in C^2$, (2.52) ensures that each integral

$$\int_{\Sigma} a |\Delta\varphi| |\psi(x_i)| \, d\Xi$$

is finite. Exchanging the incoming legs leaves the collision difference unchanged, whereas exchanging the incoming and outgoing pairs reverses its sign. These changes in the reduced coordinates have determinant of absolute value 1 and preserve the coarea measure. We may therefore expand the collision difference of the test function term by term to obtain

$$\mathfrak{q}_a(\varphi, \psi) = \gamma \int_{\Sigma} a \Delta\varphi \overline{\psi(x)} \, d\Xi. \quad (2.64)$$

The same critical coordinates prove the regularized limit with a single collision difference. In a $q = G_\omega$ chart, $uv \int A(q, u, v) \rho_\eta(-uvq) \, dq$ is bounded in absolute value by a constant. For each fixed $uv \neq 0$, the inner integral has its one-dimensional approximate-identity limit. The constant bound is integrable on the bounded u, v region, and the axes remain a two-dimensional null set. In a $G_\omega \neq 0$ chart, the contribution is at most $C\sqrt{\eta}$. Consequently,

$$\begin{aligned} \lim_{\eta \downarrow 0} \gamma \int_{\mathbb{T}_{2\pi}^3} a \Delta\varphi \overline{\psi(x)} \rho_\eta(F) \, dm^3 \\ = \gamma \int_{\Sigma} a \Delta\varphi \overline{\psi(x)} \, d\Xi. \end{aligned} \quad (2.65)$$

This proves the limit directly even for a noneven ρ_η ; symmetrization by exchanging the two pairs is not used at finite η .

For fixed $\varphi \in C^2$, the pushforward of the finite complex measure $\gamma a \Delta\varphi \, d\Xi$ along the first leg is absolutely continuous with respect to m by Lemma 2.2. Thus (2.64) defines an $L^1(m)$ weak output. On $C^2 \cap \text{Dom } L_a$, it agrees with the L^2 output of (2.62).

The corresponding normalization in [4, equations (3.21) and (4.1)] is

$$a = \prod_{i=0}^3 \omega(x_i)^{-2}, \quad \gamma = \frac{9\pi}{4}, \quad \mathfrak{q}_{\text{ALS}}(\varphi, \varphi) = \frac{9\pi}{16} \int_{\Sigma} \prod_{i=0}^3 \omega(x_i)^{-2} |\Delta \varphi|^2 d\Xi. \quad (2.66)$$

The angular normalization $m(dx) = dx/(2\pi)$ and the factor $(2\pi)^{-3}$ after momentum elimination are already included in (2.3).

More generally, let $N = (\alpha + \beta\omega)^{-1} > 0$ and set

$$a_N = \frac{\prod_{i=0}^3 N(x_i)}{\prod_{i=0}^3 \omega(x_i)}. \quad (2.67)$$

The form in the physical relative perturbation $W = N + Ng$ is

$$\mathfrak{q}_N(g, h) = \frac{\gamma}{4} \int_{\Sigma} a_N \Delta(g/N) \overline{\Delta(h/N)} d\Xi. \quad (2.68)$$

The positive bounded invertible multiplication $g \mapsto g/N$ makes this form closed and densely defined, with kernel $N \text{span}\{1, \omega\}$. The inequality

$$\text{dist}_{L^2}(g, N \text{span}\{1, \omega\}) \leq \|N\|_{\infty} \text{dist}_{L^2}(g/N, \text{span}\{1, \omega\})$$

gives the corresponding positive spectral gap. If (α, β) ranges over a fixed compact positive parameter set, the lower bound for the weight and the multiplication bounds are uniform, so the gap constant can also be chosen uniformly.

To check the sign, write the factor in the weak four-wave collision expression as

$$\gamma(\prod \omega_i^{-1}) \prod W_i \Delta(1/W).$$

At $W = N + N^2\varphi$, the identity $\Delta(1/N) = 0$ eliminates the first variations of the four multiplicative weights, while $\delta(1/W) = -\varphi$. The linear term is therefore $-\gamma \int a_N \Delta \varphi$. The positive sign of the dissipation operator agrees with (2.64). The conversion between g and φ is specified by (2.68); the positive multiplication is not a unitary transformation. In particular, for $N = (\beta\omega)^{-1}$ one has $a_N = \beta^{-4} \prod \omega_i^{-2}$, so in the $N^2\varphi$ variable the linear term is $-\beta^{-4} L_{\text{ALS}}\varphi$, in agreement with the temperature factor in [4, equation (3.20)]. The conclusions of this section concern collision invariants, the dissipation gap, and its closed-form realization. A nonlinear diffusion limit with spatial transport requires additional dynamical estimates.

3 Geometry and degenerate dissipation for four-wave collisions with a cube cutoff

We now turn to quadratic dispersion in three dimensions. The full four-leg collision difference again connects conservation with dissipation: after grouping the momenta into incoming and outgoing pairs, the entire dissipation becomes a pair-sum variance over the admissible spherical directions. The pair sums at every center and radius come from the same function and therefore jointly determine the five-dimensional null space. Fix $R > 0$ and let $D = [-R, R]^3$ be the momentum domain; all four legs of every collision are restricted to this same cube. At its corners, the set of admissible collisions degenerates and the collision frequency vanishes. We first compute this rate of degeneracy, then use the frequency as a weight to establish coercivity and an inverse for the linear equation. Finally, we prove the corner-layer

positivity and spatial product bounds needed in the nonlinear energy estimates. To distinguish momentum components from collision-leg indices, write $k = (k^{(1)}, k^{(2)}, k^{(3)})$ and set

$$\Psi(k) = (1, k^{(1)}, k^{(2)}, k^{(3)}, |k|^2)^T, \quad \Theta_+ = \{\theta \in \mathbb{R}^5 : \min_{k \in D} \theta \cdot \Psi(k) > 0\}, \quad N_\theta = (\theta \cdot \Psi)^{-1}.$$

Unless stated otherwise, the parameters lie in a fixed compact set $\Theta \Subset \Theta_+$. Thus N_θ and its parameter and momentum derivatives of any fixed order are uniformly bounded, and $0 < N_- \leq N_\theta \leq N_+ < \infty$. Constants may depend on R, Θ , but not on the strong-collision coefficient introduced later.

3.1 The full resonance measure and two integral kernels

Write $\chi = \mathbf{1}_D$. For a momentum quadruple (k_0, k_1, k_2, k_3) , indices 0, 1 denote the incoming legs and 2, 3 the outgoing legs. We abbreviate $\phi_i = \phi(k_i)$ and $f_i = f(k_i)$, and define

$$d\Xi_D = \delta^3(k_0 + k_1 - k_2 - k_3) \delta(|k_0|^2 + |k_1|^2 - |k_2|^2 - |k_3|^2) \prod_{i=0}^3 \chi(k_i) dk_i, \quad (3.1)$$

$$\Delta\phi = \phi_0 + \phi_1 - \phi_2 - \phi_3. \quad (3.2)$$

The δ distributions here are defined by the coarea formulas below. The four indicator functions jointly restrict a collision and remain present after variables are eliminated. If f also depends on time and space, all four values $f_i = f(t, X, k_i)$ are taken at the same (t, X) . Fix $k_0 = k$ and set $k_2 = k + x, k_3 = k + y, k_1 = k + x + y$. Then

$$d\Gamma_k = \frac{1}{2} \delta(x \cdot y) \chi(k) \chi(k+x) \chi(k+y) \chi(k+x+y) dx dy. \quad (3.3)$$

The momentum constraint is automatic in these coordinates, and the energy difference is $2x \cdot y$, which gives the factor $1/2$. A resonant quadruple is therefore a rectangle with vertex k and orthogonal sides x, y . Relative to the output k_0 , we call the diagonally opposite vertex k_1 the opposite leg, and the neighboring vertices k_2, k_3 the adjacent legs. For $x \neq 0$, the measure is $\frac{1}{2|x|} dx d\mathcal{H}^2(y)|_{x^\perp}$, multiplied by all the indicators above. No additional atom is assigned to $x = 0$. The admissible x, y are bounded, and $|x|^{-1}$ is locally integrable in three dimensions. This defines a finite positive measure, with $d\Xi_D = dk d\Gamma_k$. The collision operator is

$$\mathcal{C}(f)_0 = \int (f_1 f_2 f_3 + f_0 f_2 f_3 - f_0 f_1 f_3 - f_0 f_1 f_2) d\Gamma_{k_0}. \quad (3.4)$$

Lemma 3.1 (Boundedness, continuity, and conservation). *Each trilinear term in (3.4) maps $C(D)^3$ continuously into $C(D)$, with norm bounded by C_R ; the same norm bound holds on $L^\infty(D)$. All five moments satisfy $\int_D \Psi \mathcal{C}(f) dk = 0$. If $N = N_\theta$, then $\mathcal{C}(N) = 0$.*

Proof. The coarea formula bounds the absolute value of each term by $C_R \prod_j \|f_j\|_\infty \int_{|x| \leq C_R} |x|^{-1} dx$. To prove continuity of the output on the closed domain, fix $k_n, k \in D$ with $k_n \rightarrow k$. The output indicator is then identically 1. For almost every x , the point $k+x$ lies on no face of the cube, and x is parallel to no coordinate axis. On the plane $x^\perp$, each face equation for $k+y$ or $k+x+y$ is a nonconstant affine equation, so its boundary has zero area. Hence all indicators and continuous factors converge pointwise. Dominated convergence, first on a fixed bounded planar disc and then against $|x|^{-1} dx$, proves continuity. The same argument applies when the output lies on the boundary; the excluded set of x has zero three-dimensional measure.

For positive f , the cubic polynomial equals $f_0 f_1 f_2 f_3 \Delta(1/f)$. The full measure is invariant under exchanges within either pair and under exchange of the two pairs. Thus, for every bounded ϕ ,

$$\int \phi \mathcal{C}(f) = \frac{1}{4} \int \prod_i f_i \Delta(1/f) \Delta \phi \, d\Xi_D.$$

Taking $\phi = \Psi_j$ gives conservation. This is a polynomial identity, so it also holds for bounded real f : one may first add a positive constant to f and then compare coefficients. Finally, $\Delta(N^{-1}) = \theta \cdot \Delta \Psi = 0$. $\square$

Take $N = N_\theta$. Writing a perturbation as $f = N + Nu$, define the collision frequency and the dissipative operator acting on the relative perturbation u by

$$\nu_N(k) = N(k)^{-1} \int N_1 N_2 N_3 \, d\Gamma_k, \quad L_N u = -N^{-1} D\mathcal{C}(N)[Nu]. \quad (3.5)$$

Its quadratic form is

$$\mathfrak{a}_N(u, v) = \frac{1}{4} \int \prod_i N_i \Delta(u/N) \overline{\Delta(v/N)} \, d\Xi_D. \quad (3.6)$$

On unweighted $L^2(D)$, this is the form of a bounded nonnegative self-adjoint operator, and

$$L_N = \nu_N + K_{1,N} - 2K_{2,N}. \quad (3.7)$$

The two kernels correspond to integration over the opposite and adjacent legs. Below, $k, p \in D$, and χN denotes the extension of N by zero outside D . Fix $k_0 = k, k_1 = p$ and put $m = (k+p)/2, r = |k-p|/2$. The energy difference is $2(r^2 - |k_2 - m|^2)$, whose normal derivative on the sphere has magnitude $4r$. Hence

$$K_{1,N}(k, p) = \frac{|k-p|}{8} \int_{\mathbb{S}^2} (\chi N)(m+r\omega) (\chi N)(m-r\omega) \, d\omega. \quad (3.8)$$

Fix instead $k_0 = k, k_2 = p$, and set $x = p-k, k_3 = k+z, k_1 = p+z$. The energy difference is $2x \cdot z$, so

$$K_{2,N}(k, p) = \frac{1}{2|p-k|} \int_{z \perp (p-k)} (\chi N)(p+z) (\chi N)(k+z) \, d\mathcal{H}^2(z). \quad (3.9)$$

Both kernels are nonnegative and symmetric, and

$$K_{1,N} \leq C|k-p|, \quad K_{2,N} \leq C|k-p|^{-1}. \quad (3.10)$$

Their diagonal values do not affect the integral operators. The two adjacent inputs give the same kernel $K_{2,N}$, which accounts for the factor 2 in (3.7). Boundedness follows from the Schur bounds supplied by (3.10); expanding the full squared difference gives (3.6).

Pair coordinates give another representation of the dissipation. For fixed center V and radius r , the pair $V \pm r\sigma$ has total momentum $2V$ and total energy $2(|V|^2 + r^2)$; only the direction σ varies. A collision therefore compares two directions at the same center and radius, while the sharp cube cutoff determines which directions are admissible.

Proposition 3.2 (Variance representation of the full pair sums). *For $V \in \mathbb{R}^3$, $r > 0$, and $\sigma \in \mathbb{S}^2$, set*

$$A_{V,r}(\sigma) = (\chi N)(V+r\sigma) (\chi N)(V-r\sigma), \quad Z_{V,r} = \int_{\mathbb{S}^2} A_{V,r}(\sigma) \, d\sigma.$$

When $A_{V,r}(\sigma) > 0$, define

$$H_u(V, r, \sigma) = \frac{u(V+r\sigma)}{N(V+r\sigma)} + \frac{u(V-r\sigma)}{N(V-r\sigma)}.$$

Set its value to zero in all other directions. If $Z_{V,r} > 0$, write $\overline{H}_u = Z_{V,r}^{-1} \int A_{V,r} H_u d\sigma$; if $Z_{V,r} = 0$, set $\overline{H}_u = 0$. For bounded u, v ,

$$\mathfrak{a}_N(u, v) = \int_{\mathbb{R}^3} \int_0^\infty r^3 Z_{V,r} \int_{\mathbb{S}^2} A_{V,r}(\sigma) (H_u - \overline{H}_u) \overline{(H_v - \overline{H}_v)} d\sigma dr dV. \quad (3.11)$$

The same identity extends continuously to the space H_ν defined below. In particular, vanishing of the form requires that, for almost every fixed (V, r) , the pair sum arising from the same u be constant over all admissible directions.

Proof. First eliminate the momentum constraint and write $k_0 = V + p, k_1 = V - p, k_2 = V + q$, so that $k_3 = V - q$. The absolute Jacobian determinant of this nine-dimensional transformation is 8, and the energy constraint is $2(|p|^2 - |q|^2)$. Passing to polar coordinates $p = r\sigma, q = s\tau$ and using the energy constraint to eliminate s gives

$$d\Xi_D = 2r^3 \prod_{i=0}^3 \chi(k_i) dV dr d\sigma d\tau.$$

The full difference in (3.6) is then $H_u(\sigma) - H_u(\tau)$, and the two angular weights are $A_{V,r}(\sigma)A_{V,r}(\tau)$. Expanding the product of the two differences gives the cross term $(\int A H_u)(\int A H_v)$. Thus

$$\frac{1}{2} \int \int A(\sigma) A(\tau) [H_u(\sigma) - H_u(\tau)] \overline{[H_v(\sigma) - H_v(\tau)]} d\sigma d\tau = Z \int A(H_u - \overline{H}_u) \overline{(H_v - \overline{H}_v)} d\sigma.$$

This proves the coefficient and all cutoff conditions in (3.11). If $Z = 0$, both sides vanish. The single-leg marginal gives $\int \prod_i N_i |u_i/N_i|^2 d\Xi_D = \int \nu_N |u|^2$. Hence both the uncentered pair-sum map and the map obtained by subtracting its weighted mean are bounded on H_ν . Approximation by bounded functions gives the extension; all pair sums and their means are understood as almost-everywhere representatives for the corresponding measures. $\square$

3.2 The precise degeneracy of the corner frequency

Let Z be the set of eight corners and $r(k) = \min_{\xi \in Z} |k - \xi|_\infty$. For $\eta > 0$, write

$$E_\eta(\xi) = \{k \in D : |k - \xi|_\infty \leq \eta\}, \quad E_\eta = \bigcup_{\xi \in Z} E_\eta(\xi) = \{k \in D : r(k) \leq \eta\}.$$

Near a corner, use the three coordinate products $x^{(j)} y^{(j)}$ as variables. Their positive and negative densities can be computed explicitly. The constraint that their sum vanish expresses the frequency as convolutions for six sign patterns. The squared logarithm below comes from two of the positive product densities.

Proposition 3.3 (The squared-logarithmic corner frequency). *There exist $r_0, c_0, C_0 > 0$ such that, whenever $0 < r(k) < r_0$,*

$$c_0 r(k)^2 \log^2 \frac{2R}{r(k)} \leq \nu_N(k) \leq C_0 r(k)^2 \log^2 \frac{2R}{r(k)}. \quad (3.12)$$

The function ν_N is continuous on the closed cube, and its zero set is exactly Z . In particular, for $p \geq 0$, $\nu_N^{-p} \in L^1(D)$ if and only if $p \leq 3/2$. These conclusions are uniform on Θ .

Proof. It suffices to compute the unweighted frequency $M(k) = \int d\Gamma_k$, since $\nu_N \asymp M$. At the corner (R, R, R) , set $L = 2R, d_j = (R - k^{(j)})/L, e = \max_j d_j$, and put $x = LX, y = LY$. The full admissible domain in each coordinate is precisely

$$\mathcal{P}_d = \{X, Y \in [-1 + d, d] : X + Y \in [-1 + d, d]\}.$$

Push forward the area measure of this planar domain under the product $t = XY$. Denote its density on the positive half-line by $P_d(t)$, and write its density on the negative half-line as $N_d(s)$, with $t = -s$. They are, respectively,

$$P_d(t) = T_{1-d}(t) + T_d(t), \quad T_b(t) = \log \frac{(b + \sqrt{b^2 - 4t})^2}{4t} \mathbf{1}_{0 < t < b^2/4}, \quad (3.13)$$

$$N_d(s) = 2 \log \frac{d(1-d)}{s} \mathbf{1}_{0 < s < d(1-d)}, \quad t = -s. \quad (3.14)$$

The formulas on the right are used only on the intervals specified by the indicators and are defined to be zero elsewhere; also $T_0 = N_0 = 0$. The first formula comes from the two triangles on which the coordinates have the same sign: solve $uv = t, u + v \leq b$ and integrate du/u . The mixed-sign quadrant is exactly $0 \leq X \leq d, 0 \leq -Y \leq 1-d$; its endpoints already satisfy the third indicator, giving the second formula. There are no atoms on the axes.

For the positive densities extended by zero, define $(P_{d_i} * P_{d_l})(s) = \int_0^s P_{d_i}(t) P_{d_l}(s-t) dt$. When the three products sum to zero, their signs, outside the null coordinate axes, must be one negative and two positive, or two negative and one positive. Writing $d = (d_1, d_2, d_3)$, we have $M = L^4 m(d)$, and the six sign patterns give

$$\begin{aligned} m(d) &= \frac{1}{2} \sum_{j=1}^3 \int_0^\infty N_{d_j}(s) (P_{d_i} * P_{d_l})(s) ds \\ &\quad + \frac{1}{2} \sum_{j=1}^3 \int_0^\infty \int_0^\infty N_{d_i}(u) N_{d_l}(v) P_{d_j}(u+v) du dv, \quad \{i, l, j\} = \{1, 2, 3\}. \end{aligned} \quad (3.15)$$

This identity can also be obtained directly in each orthant by the change of variables $t_j = X_j Y_j$. The absolute integrability bounds below also determine the value of the convolution at zero energy. For $e \leq 1/64$, $P_{d_j}(t) \leq 2 \log(1/t)$ and $N_{d_j}(s) \leq 2 \log(e/s) \mathbf{1}_{s < e}$. Setting $t = sv$ gives

$$(P_{d_i} * P_{d_l})(s) \leq Cs \int_0^1 [\log(1/s) + |\log v|] [\log(1/s) + |\log(1-v)|] dv \leq Cs[1 + \log(1/s)]^2.$$

The first line of (3.15) is therefore at most $Ce^2 \log^2(1/e)$. In the second line, set $u = eu', v = ev'$ and use $\log_+(1/(u' + v')) \leq \min(|\log u'|, |\log v'|)$ to obtain the bound $Ce^2[1 + \log(1/e)]$. All endpoint logarithms here are integrable.

For the lower bound, choose $d_j = e$ and restrict to $s \in [e/8, e/4]$ and $t \in [s/3, 2s/3]$. Then $N_e(s) \geq c$, while the triangle side lengths in the other two coordinates satisfy $1 - d_i, 1 - d_l \geq 63/64$. Hence $T_{1-d_i}(t), T_{1-d_l}(s-t) \geq c \log(1/e)$. This positive contribution gives $ce^2 \log^2(1/e)$; the boundary distances in the other two coordinates may vanish. Coordinate reflections handle the remaining corners.

Continuity follows from the same coarea dominated-convergence argument as in Lemma 3.1. At a noncorner point, at least one coordinate j lies in the open interval. For small $s > 0$, set $x^{(i)} = y^{(i)} = -\operatorname{sgn}(k^{(i)})s$ for $i \neq j$ (choose either sign when $k^{(i)} = 0$), and set $x^{(j)} = \sqrt{2}s, y^{(j)} = -\sqrt{2}s$. All three moving legs lie strictly inside the cube, and $x \cdot y = 0, x \neq 0$. A neighborhood on their resonance surface has positive measure, so $M(k) > 0$. At a corner, every coordinate product is nonnegative; only the null axes can make their sum zero, and the preceding density formula gives $M = 0$. Compact sets away from the corners consequently have a uniform positive lower bound. Finally, a corner shell has volume $3r^2 dr$, so integrability of the inverse frequency is equivalent to that of

$$\int_0^{r_0} r^{2-2p} \log^{-2p}(2R/r) dr.$$

At the critical exponent $p = 3/2$, this is $\int dr/[r \log^3(2R/r)]$, which still converges; it diverges for larger p . $\square$

From now on, fix $n_* = (1 + |k|^2)^{-1}$ and $\nu_* = \nu_{n_*}$, and write $H_\nu = L^2(D, \nu_* dk)$. On compact parameter families, $\nu_N \asymp \nu_*$. This weight is dictated by the fact that each single-leg squared integral in the four-leg collision difference is exactly $\int_D \nu_N |u|^2 dk$. Dissipation therefore naturally controls the H_ν norm, whereas the vanishing frequency permits unweighted L^2 mass to concentrate at the corners; see Corollary 3.7.

3.3 The five collision invariants and weighted coercivity

Lemma 3.4 (The five-dimensional space of collision invariants). *If $\phi \in L^2_{\text{loc}}(D^\circ)$ and $\Delta\phi = 0 \Xi_D$ -almost everywhere, then $\phi(k) = a + b \cdot k + c|k|^2$ almost everywhere.*

Proof. The pair representation in Proposition 3.2 shows that a vanishing difference requires the pair sum to be constant over admissible directions for almost every fixed (V, r) . These pair sums, for different (V, r) , all come from the same function ϕ . In any strictly smaller concentric cube, translate all four legs together and convolve in the translation parameter. The momentum and quadratic-energy constraints are preserved under a common translation. Thus the smoothed function ϕ_δ still satisfies the collision relation, provided the four test legs remain in a slightly smaller cube. This operation is legitimate for the almost-everywhere relation with respect to the resonance measure: write the legs as $V + r\sigma, V - r\sigma, V + r\tau, V - r\tau$. Their measure is $2r^3 dV dr d\sigma d\tau$, whose density is strictly positive for $r > 0$. Fubini gives the almost-everywhere relation after common-translation convolution, and smoothness then gives it for every admissible V, r, σ, τ . Hence

$$\phi_\delta(V + r\sigma) + \phi_\delta(V - r\sigma) = \phi_\delta(V + r\tau) + \phi_\delta(V - r\tau).$$

Fix any interior points x, y and write $V = (x + y)/2, v = (x - y)/2$. Rotate v in any coordinate plane, keeping $|v|$ fixed, and differentiate the pair identity at zero rotation. This yields

$$[\partial_i \phi_\delta(x) - \partial_i \phi_\delta(y)](x^{(j)} - y^{(j)}) = [\partial_j \phi_\delta(x) - \partial_j \phi_\delta(y)](x^{(i)} - y^{(i)}).$$

These identities constrain the same gradient field simultaneously. Choose the origin and three nearby points on the coordinate axes. First compare increments between the axis points, then compare any x with the origin and each axis point. This gives a single constant λ such that $\nabla \phi_\delta(x) = b + \lambda x$. Integration over the convex smaller cube gives the stated five-dimensional family of quadratic polynomials; only first-order differentiability is used here.

Finally, take normalized smooth convolution kernels whose support radii tend to zero with a fixed ratio of inner to outer radii. The Lebesgue differentiation theorem gives pointwise convergence on a single set of full measure. Equip the function space on this set with the topology of pointwise convergence. The restrictions of the five-dimensional polynomial family still form a finite-dimensional subspace, which is closed in this Hausdorff topological vector space. The limit on every smaller cube therefore remains in that family. If two polynomials agree almost everywhere on a nonempty open cube, continuity makes them equal everywhere there, and all their coefficients agree. Exhaust D° by countably many smaller cubes and use the Lebesgue-null boundary to obtain one set of coefficients on the full domain. The same smoothing and limiting argument applies to locally integrable functions. $\square$

Lemma 3.5 (Weighted compactness). *Define the weighted kernels*

$$\tilde{K}_{i,N}(k, p) = \nu_N(k)^{-1/2} K_{i,N}(k, p) \nu_N(p)^{-1/2}, \quad i = 1, 2.$$

Both are Hilbert–Schmidt kernels, uniformly on Θ .

Proof. By (3.10), it suffices to prove $\iint \nu_*^{-1}(k) \nu_*^{-1}(p) |k - p|^{-2} dk dp < \infty$. Choose $r_0 > 0$ sufficiently small as in Proposition 3.3, and set $r_j = r_0 2^{-j}$. For $j \geq 1$, combine the shells of equal depth at all eight corners:

$$A_j = \{k \in D : r_{j+1} < r(k) \leq r_j\}, \quad A_0 = \{k \in D : r(k) > r_1\}.$$

These sets partition the cube apart from its corners, and the closure of A_0 stays away from the corners. Write $a_j = \log^{-2}(2R/r_j)$. Proposition 3.3 and the corner-layer volume give

$$|A_j| \leq Cr_j^3, \quad \sup_{A_j} \nu_*^{-1} \leq Ca_j r_j^{-2}, \quad a_j \asymp (1+j)^{-2}.$$

After enlarging the constants, the zeroth layer satisfies the same bounds. For any measurable set with $|A| \leq Cr^3$, split the integral into $|k - p| < r$ and its complement. Three-dimensional polar coordinates give directly

$$\sup_k \int_A |k - p|^{-2} dp \leq |\mathbb{S}^2| r + r^{-2} |A| \leq C' r.$$

Thus, when $r_l \leq r_j$, integration first over A_j and then over A_l gives

$$\int_{A_l} \int_{A_j} \frac{dk dp}{\nu_*(k) \nu_*(p) |k - p|^2} \leq Ca_j a_l \frac{r_l}{r_j} \leq Ca_j a_l.$$

Treat the reverse order symmetrically and use $\sum_j a_j < \infty$. This controls every pair of layers, including neighboring layers, layers at different corners, and the interior region. The corners form a null set and do not affect the integrals. $\square$

The null-space classification and weighted compactness together yield a positive lower bound on the complement of the five-dimensional null space. Two notions of orthogonality must be distinguished: dissipation first controls the weighted distance to the null space, whereas the microscopic constraint uses the unweighted L^2 projection. We prove both statements and their relation.

Proposition 3.6 (Coercivity with a degenerate weight). *The form (3.6) is defined for every $u \in H_\nu$, and*

$$\mathfrak{a}_N(u, u) \geq c_\Theta \inf_{\psi \in \text{span}\{N\Psi\}} \|u - \psi\|_{H_\nu}^2. \quad (3.16)$$

Let P_N be the orthogonal projection in unweighted $L^2(D)$ onto $\text{span}\{N\Psi\}$, and let $Q_N = I - P_N$. This projection extends to H_ν , and

$$P_N u = 0 \implies c_\Theta \|u\|_{H_\nu}^2 \leq \mathfrak{a}_N(u, u) \leq C_\Theta \|u\|_{H_\nu}^2. \quad (3.17)$$

Proof. Four-leg symmetry gives each single-leg integral as $\int \prod N_i |u_i/N_i|^2 d\Xi_D = \int \nu_N |u|^2$. Every single-leg marginal is absolutely continuous with respect to volume. Thus the volume almost-everywhere class of u determines its value on each leg, without assigning additional traces on boundaries or diagonals. The form is therefore defined on all of H_ν and is bounded above by $4 \int \nu_N |u|^2$. Put $v = \sqrt{\nu_N} u$. The full difference defines a bounded operator $\mathcal{R}v = \frac{1}{2} \Delta(\nu_N^{-1/2} v/N)$ with values in $L^2(\prod_i N_i d\Xi_D)$, whose squared norm is the form. First expand for bounded u supported away from the corners, then pass to the limit in H_ν . Lemma 3.5 gives

$$\mathcal{R}^* \mathcal{R} = I + \tilde{K}_{1,N} - 2\tilde{K}_{2,N}.$$

By Lemma 3.4, a zero form implies $u \in \text{span}\{N\Psi\}$. Indeed, $u \in H_\nu$ belongs to L^2 on every smaller cube, which suffices for that lemma; no global unweighted L^2 assumption is made. If there were no positive lower bound on the orthogonal complement of $\sqrt{\nu_N} \text{span}\{N\Psi\}$, choose unit vectors v_j whose form tends to zero. The bounded nonnegative operator $T = \mathcal{R}^* \mathcal{R}$ satisfies $\|Tv_j\|^2 \leq \|T\| \langle Tv_j, v_j \rangle \rightarrow 0$.

Compactness of $T - I$ gives a strongly convergent subsequence of $v_j = Tv_j - (T - I)v_j$. Its limit is a nonzero vector both in the null space and orthogonal to it, a contradiction.

For uniformity in the parameter, put $z = n_* u / N$. Then $\Delta(z/n_*) = \Delta(u/N)$, and comparison of the positive weights bounds $\mathfrak{a}_N(u, u)$ below by a constant multiple of $\mathfrak{a}_{n_*}(z, z)$. Multiplication by N/n_* is uniformly bounded in both directions and identifies the two null spaces exactly. The constant in (3.16) is consequently uniform. Write $\kappa_N = N\Psi$ for the column vector of the five natural zero modes. With e_j the j th standard basis vector of $\mathbb{R}^5$, choose the real orthonormal basis

$$M_N = \int_D N^2 \Psi \Psi^T dk, \quad \kappa_j(k) = N(k) \Psi(k)^T M_N^{-1/2} e_j, \quad 1 \leq j \leq 5.$$

The matrix M_N is strictly positive definite, with a positive lower bound for its smallest eigenvalue on compact parameter sets. The basis therefore depends smoothly on the parameter, and it and its parameter derivatives of every fixed order are uniformly bounded. Since $\kappa_j/\sqrt{\nu_*} \in L^2$, the projection P_N is uniformly bounded on H_ν . If $P_N u = 0$ and ψ is the best weighted approximation from the null space, then $\psi = -P_N(u - \psi)$. Hence $\|u\|_{H_\nu} \leq C\|u - \psi\|_{H_\nu}$, proving the final assertion. $\square$

Corollary 3.7 (Absence of an unweighted spectral gap). *The collision operator has no strictly positive unweighted spectral gap on $Q_N L^2(D)$.*

Proof. Take an L^2 -normalized function g_η supported in a radius- η neighborhood of one corner. This neighborhood has volume $O(\eta^3)$, and the null-space basis is bounded, so $\|P_N g_\eta\|_2 \leq C\eta^{3/2}$. Moreover,

$$\mathfrak{a}_N(Q_N g_\eta, Q_N g_\eta) = \mathfrak{a}_N(g_\eta, g_\eta) \leq C \sup_{E_\eta} \nu_* \longrightarrow 0, \quad \|Q_N g_\eta\|_2 \longrightarrow 1.$$

This is the required microscopic unit sequence with vanishing dissipation. $\square$

3.4 Absolute row estimates and the continuous weighted inverse

Write $K_N = K_{1,N} - 2K_{2,N}$ and $K_{a,N} = K_{1,N} + 2K_{2,N}$.

Proposition 3.8 (Row bounds, squared-row bounds, and vanishing at corners). *Uniformly in the parameter,*

$$\int K_{a,N}(k, p) dp \leq C\nu_*(k), \quad (3.18)$$

$$\int K_{a,N}(k, p)^2 dp \leq C\sqrt{\nu_*(k)}, \quad (3.19)$$

$$\sup_k \int \frac{K_{a,N}(k, p)}{\nu_*(p)} dp \leq C. \quad (3.20)$$

The weighted row $k \mapsto K_{i,N}(k, \cdot)/\nu_*(\cdot)$ is continuous in $L^1(D)$, and

$$\sup_{\theta \in \Theta, k \in E_\eta} \int \frac{K_{a,N}(k, p)}{\nu_*(p)} dp =: \psi(\eta) \longrightarrow 0. \quad (3.21)$$

Proof. Each kernel satisfies $\int N(p) K_{i,N}(k, p) dp = N(k) \nu_N(k)$, so (3.18) retains the frequency factor. For the squared-row bound, (3.10) gives $C\rho$ on $|k - p| \leq \rho$; outside this set, one pointwise kernel bound and (3.18) give $C\nu_*(k)(1 + \rho^{-1})$. Take $\rho \asymp \sqrt{\nu_*(k)}$, capped at a fixed radius. At the corners, nonnegativity and zero frequency make the rows exactly zero.

Near a corner ξ , write $r = |p - \xi|$ and $L(r) = \log(C_R/r)$. The key estimate is the uniform Newton-potential tail bound

$$\sup_{k \in D} \int_{|p - \xi| < \eta} \frac{1 + |k - p|^{-1}}{\nu_*(p)} dp \leq \frac{C}{L(\eta)}. \quad (3.22)$$

Set $\rho = |k - \xi|$. If $\rho > 2\eta$, use the lower bound on the distance and $\int_{r < \eta} \nu_*^{-1} \leq C\eta/L(\eta)^2$. For $0 < \rho \leq 2\eta$, split into $r < \rho/2$, $\rho/2 \leq r \leq 2\rho$, and $2\rho < r < \eta$. The first two contributions are each bounded by $C/L(\rho)^2$, and the last by $C \int_{2\rho}^{\eta} dr/[rL(r)^2]$. For $\rho = 0$, integrate directly down to zero. Away from the corners, the inverse frequency is bounded, proving (3.20); the remaining contribution from a small diagonal ball is $C\rho^2$.

The spherical formula for K_1 is continuous off the diagonal, and its linear upper bound gives continuous vanishing on the diagonal. For K_2 , fix an output k_0 and exclude the three lines $p \in k_0 + \mathbb{R}e_j$ and the point $p = k_0$ from the input variable. Near any remaining p , orthonormal coordinates for $(p - k)^\perp$ can be chosen continuously. The preimage of each cutoff face is either a line in the plane or the empty set, and therefore has zero area. Dominated convergence thus gives kernel continuity for each such fixed p . The excluded input set has zero three-dimensional measure. First remove corner inputs using (3.22), then remove a small diagonal ball, and finally use the same bounded majorant on the remaining domain. This proves continuity of the entire row in L^1 , including outputs on faces, edges, and corners. Pointwise continuity of kernel values is used only for inputs outside the specified null set. Compactness of the closed domain and parameter set gives uniform continuity. Since the corner rows themselves vanish, (3.21) follows. $\square$

Use the Banach spaces

$$\mathcal{X} = L^\infty(D), \quad \mathcal{Y} = \{u : \nu_* u \in L^\infty(D)\}, \quad \|u\|_{\mathcal{Y}} = \|\nu_* u\|_\infty.$$

Functions are understood as almost-everywhere equivalence classes; membership in $\mathcal{Y}$ does not presuppose unweighted L^2 integrability. Since $\nu_*^{-1} \in L^1$, $\mathcal{Y} \subset H_\nu \cap L^1(D)$, so P_N remains defined on these functions. The weight allows solutions to grow near the corners; the uniformly controlled quantity in the equation is $\nu_* u$. This is the space used for the cell problem $L_N u = F$ below.

Proposition 3.9 (The normalized weighted inverse). *For $F \in Q_N \mathcal{X}$, there is a unique normalized solution $u \in Q_N \mathcal{Y}$ of $L_N u = F$, and*

$$\|u\|_{\mathcal{Y}} + \mathbf{a}_N(u, u)^{1/2} \leq C\|F\|_\infty. \quad (3.23)$$

If F is continuous, $\nu_N u$ has a continuous representative. The following absolute row bound also holds:

$$\operatorname{ess\,sup}_{k \in D} N(k) \int N_1 N_2 N_3 |\Delta(u/N)| d\Gamma_k \leq C\|F\|_\infty. \quad (3.24)$$

Here u is an almost-everywhere equivalence class in the weighted space. The continuity assertion specifies only a representative of $\nu_N u$, not corner values for u itself.

Proof. Absorb the diagonal multiplication into the unknown to write the equation as the identity plus a compact operator. The five-dimensional conservation null space corresponds to five compatibility conditions. A finite-rank term removes that null space; after inversion, its contribution vanishes for microscopic right-hand sides. Set $v = \nu_N u$ and $J_N v = K_N(\nu_N^{-1} v)$. Proposition 3.8 and the Arzelà–Ascoli theorem show that $J_N : \mathcal{X} \rightarrow C(D)$ is compact. We first identify the integral operator with the weighted form. For $u \in \mathcal{Y}$ and bounded h , the symmetric absolute kernel satisfies

$$\iint K_{a,N}(k, p) |u(p)h(k)| dk dp \leq C\|h\|_\infty \int \nu_* |u| dp < \infty.$$

Truncate u both near the corners and in amplitude. The resulting u_n converge to u in H_ν , and $\int \nu_* |u_n - u| \rightarrow 0$. Continuity of the form, the absolute integral bound above, and dominated convergence for the diagonal term give

$$\mathfrak{a}_N(u, h) = \int_D (\nu_N u + K_N u) \bar{h} \, dk.$$

This passage to the limit requires only convergence of the pairings, not $\nu_N u_n \rightarrow \nu_N u$ in L^∞ . If $(I + J_N)v = 0$, then $u = v/\nu_N \in H_\nu \cap L^1$, and the identity gives $\mathfrak{a}_N(u, h) = 0$ for all bounded tests. Their density in H_ν implies $u \in \text{span}\{N\Psi\}$. Conversely, these null vectors clearly give zero. Thus $\ker(I + J_N) = \nu_N \text{span}\{N\Psi\}$ has dimension five.

Using the basis κ_j defined above, set

$$\mathcal{A}_{N,0}v = (I + J_N)v + \sum_{j=1}^5 \kappa_j \int \kappa_j \frac{v}{\nu_N} \, dk.$$

The symmetric kernel satisfies $K_N \kappa_j = -\nu_N \kappa_j$, so $\int \kappa_j (I + J_N)v = 0$. The absolute row bounds justify these exchanges of integration. If $\mathcal{A}_{N,0}v = 0$, first pair with each κ_j to show that all normalization integrals vanish, then use the five-dimensional null-space description to obtain $v = 0$. This operator is the identity plus a compact operator, so the Fredholm alternative gives a bounded inverse. One can also reduce this directly to finite-dimensional linear algebra: the compact image of the kernel operator is equicontinuous in $C(D)$, and a finite partition of unity approximates it in operator norm by finite-rank operators. Absorb the small remainder into an invertible identity-plus-small operator. For the remaining identity-plus-finite-rank operator, injectivity is equivalent to surjectivity. The proof applies both to $C(D)$ and to $\mathcal{X}$, since null vectors are automatically continuous; it does not use norm density of $C(D)$ in L^∞ . For microscopic F , every normalization integral of the solution vanishes automatically, and the solution satisfies the unmodified equation. Its energy equals $\int F u$, so $\nu_N^{-1} \in L^1$ gives (3.23). Estimating all four terms separately yields

$$N(k) \int N_1 N_2 N_3 |\Delta(u/N)| \, d\Gamma_k \leq N(k) \left(\nu_N(k) |u(k)| + \int K_{a,N}(k, p) |u(p)| \, dp \right),$$

which proves (3.24). Uniformity in the parameter will be established in the next proof. $\square$

Proposition 3.10 (A uniform resolvent bound). *For $s \geq 1$, the collision operator satisfies*

$$\|(I + sL_N)^{-1}F\|_{\mathcal{X}} + s\|(I + sL_N)^{-1}F\|_{\mathcal{Y}} \leq C\|F\|_{\mathcal{X}}, \quad F \in Q_N \mathcal{X}. \quad (3.25)$$

Continuous inputs give continuous outputs.

Proof. For $0 \leq z \leq 1$, define

$$\mathcal{A}_{N,z}v = v + K_N \frac{v}{\nu_N + z} + \sum_j \kappa_j \int \kappa_j \frac{v}{\nu_N + z} \, dk.$$

Each is the identity plus a compact operator. Operator-norm continuity at $z = 0$ follows from (3.22): first remove corner inputs uniformly, after which $z/(\nu_N + z)$ tends to zero uniformly on the remaining domain. For the finite-rank part, use $\nu_N^{-1} \in L^1$. Parameter derivatives of each fixed order satisfy $|\partial_\theta^\alpha \nu_N| \leq C_\alpha \nu_*$ and $|\partial_\theta^\alpha K_N| \leq C_\alpha K_{a,n*}$, which also gives operator-norm continuity in the parameter.

For $z > 0$, if $\mathcal{A}_{N,z}v = 0$, pairing with κ_j gives $(1 + z) \int \kappa_j v / (\nu_N + z) = 0$. Consequently $u = v/(\nu_N + z)$ satisfies $(z + L_N)u = 0$. It belongs to L^2 , and nonnegativity forces $u = 0$. The null space at $z = 0$ was already excluded in Proposition 3.9. The inverse norms are therefore uniformly bounded on the compact set $\Theta \times [0, 1]$. This also proves uniformity in (3.23).

For microscopic F , solve $\mathcal{A}_{N,z}v = F$. The same pairings make the finite-rank term vanish. Thus $(z + L_N)^{-1}F = v/(\nu_N + z)$, with unweighted norm at most $Cz^{-1}\|F\|_\infty$ and weighted norm at most $C\|F\|_\infty$. Set $z = 1/s$ and multiply by z to obtain the assertion. Continuity follows from the same inverse construction on $C(D)$; there is no need to treat the unbounded multiplier ν_N^{-1} as a bounded similarity transformation. $\square$

Under parameter differentiation, the space $Q_N\mathcal{X}$ moves with N . The following change of coordinates identifies all five-dimensional null spaces with a fixed reference null space, so that subsequent derivatives can be taken on fixed Banach spaces.

Lemma 3.11 (Parameter coordinates with a fixed five-dimensional null space). *There is a real unitary operator U_N on L^2 satisfying $U_N P_* = P_N U_N$, where $P_* = P_{n_*}$ and $Q_* = I - P_*$. The operator $U_N - I$ has finite rank, and U_N, U_N^* and their parameter derivatives of every fixed order are uniformly bounded on $\mathcal{X}, \mathcal{Y}, H_\nu$. Set $\tilde{L}_N = U_N^* L_N U_N$. Its null space is fixed, and all parameter derivatives of fixed order map $\mathcal{Y}$ boundedly into $\mathcal{X}$.*

Proof. The overlap Gram matrix of the two null spaces is $\int N n_* \Psi \Psi^T$, which is strictly positive definite. Their largest principal angle is therefore strictly less than $\pi/2$, and on the compact parameter set $\|P_N - P_*\| \leq r < 1$. Let $E = P_N - P_*$ and choose

$$U_N = [(I - P_N)(I - P_*) + P_N P_*](I - E^2)^{-1/2}.$$

Projection algebra gives $T^*T = TT^* = I - E^2$ for the first factor T , as well as $TP_* = P_N T$; moreover, E^2 commutes with both projections. The displayed operator is therefore unitary and has the required intertwining property. For uniform bounds on the different spaces, express P_N by its five-dimensional Gram formula. The integral kernels of its parameter derivatives are finite sums of tensor products of uniformly bounded functions. Since $\nu_*^{-1} \in L^1$, for $\mathcal{E} = \mathcal{X}, \mathcal{Y}, H_\nu$ and every fixed multi-index α , the maps

$$\partial_\theta^\alpha E : \mathcal{E} \longrightarrow L^2, \quad \partial_\theta^\alpha E : L^2 \longrightarrow \mathcal{X}$$

are uniformly bounded. These estimates require no inversion of the combined generating family $\{N\Psi, n_*\Psi\}$. Set

$$h(z) = \frac{(1 - z)^{-1/2} - 1}{z}, \quad h(0) = \frac{1}{2}.$$

On L^2 , $\|E^2\| \leq r^2 < 1$, so the power series for $h(E^2)$ and all its parameter derivatives of fixed order converge uniformly in operator norm. The identity

$$(I - E^2)^{-1/2} - I = E h(E^2) E$$

together with the two endpoint mappings above shows that the nonidentity part of the inverse square root and its parameter derivatives are uniformly bounded from each $\mathcal{E}$ to $\mathcal{X}$. Since $\mathcal{X}$ embeds continuously into all three spaces and $T - I$ has the same kind of finite-rank integral kernel, U_N, U_N^* and all the specified parameter derivatives have the asserted bounds. Their nonidentity parts have finite rank, and the estimates remain valid as N approaches n_* . Finally, L_N and its parameter derivatives are bounded from $\mathcal{Y}$ to $\mathcal{X}$: for the diagonal term use $|\partial^\alpha \nu_N| \leq C\nu_*$, and for the remaining terms use (3.20). The Leibniz rule gives the assertion for the conjugated operator. $\square$

Lemma 3.12 (Two-norm estimates for the cubic remainder). *Write*

$$N^{-1}\mathcal{C}(N(1 + q)) = -L_N q + \mathcal{R}_N(q).$$

Then $P_N \mathcal{R}_N(q) = 0$. If $\|q_i\|_{\mathcal{X}} \leq \kappa \leq 1$,

$$\begin{aligned} \|\mathcal{R}_N(q)\|_{\mathcal{X}} &\leq C\kappa\|q\|_{\mathcal{Y}}, \\ \|\mathcal{R}_N(q_1) - \mathcal{R}_N(q_2)\|_{\mathcal{X}} &\leq C\kappa\|q_1 - q_2\|_{\mathcal{Y}}. \end{aligned} \quad (3.26)$$

In each term of the second and third state derivatives and of parameter derivatives of any fixed order, exactly one input may be placed in $\mathcal{Y}$, with the others in $\mathcal{X}$, with uniform constants.

Proof. Each remainder term is a product of two or three function values on the same quadruple. Place only one input in the weighted space $\mathcal{Y}$ and bound the others in the unweighted supremum norm. The remaining integral is controlled precisely by the diagonal frequency or by one of the two absolute row bounds. Let $(s_0, s_1, s_2, s_3) = (1, 1, -1, -1)$. Expand the four cubic monomials term by term. The constant terms cancel because $\sum_i s_i/N_i = 0$, and the linear terms satisfy

$$\sum_i \frac{s_i}{N_i} \sum_{j \neq i} q_j = -\Delta(q/N).$$

Thus the linear part is exactly $-L_N q$. All terms in the remainder can be written as

$$\mathcal{R}_N(q)_0 = \int N_1 N_2 N_3 \sum_{i=0}^3 \frac{s_i}{N_i} \left(\sum_{\substack{j < l \\ j, l \neq i}} q_j q_l + \prod_{j \neq i} q_j \right) d\Gamma_{k_0}.$$

There are twelve quadratic terms and four cubic terms, all retaining the four-leg indicators. In any term, designate one input v_j and bound the other inputs in the unweighted supremum norm. Since N_j/N_i is uniformly bounded, the remaining absolute integral is at most a constant multiple of

$$\begin{cases} \nu_N(k_0)|v_0(k_0)|, & j = 0, \\ \int K_{1,N}(k_0, p)|v_1(p)| dp, & j = 1, \\ \int K_{2,N}(k_0, p)|v_j(p)| dp, & j = 2, 3. \end{cases}$$

This gives the bound from a $\mathcal{Y}$ input to a $\mathcal{X}$ output for each position in the product. Expand a difference by changing one factor at a time. Exactly one factor is the difference of the two states; designate it as v_j . At least one remaining input contributes a factor κ . The two stated estimates follow from (3.20). Parameter derivatives act only on the weights N and the output factor N^{-1} , not on the cutoff indicators, so the same proof applies at every fixed order. Finally, five-moment conservation gives the microscopic range. No product bound $\mathcal{Y} \times \mathcal{Y} \rightarrow \mathcal{X}$ is used. $\square$

3.5 Linearization and spatial products in a thin corner layer

We now allow the distribution to differ from N by a finite amplitude inside the corner layer, requiring closeness to N only outside it. To control the sign of the linearization, we first estimate, for a fixed output, the conditional mass of the other legs entering a corner layer. Opposite and adjacent legs play different geometric roles and must be treated separately.

Lemma 3.13 (Conditional corner-layer mass by leg position). *There exist $\rho_0, C > 0$, depending only on R , such that, for $0 < \eta \leq \rho_0$ and all $k \in D$,*

$$\int \mathbf{1}_{k_1 \in E_\eta} d\Gamma_k \leq C\eta M(k). \quad (3.27)$$

For each corner ξ , if $k \in E_{\rho_0}(\xi)$, then, for $i = 2, 3$,

$$\int \mathbf{1}_{k_i \in E_{\eta} \setminus E_{\eta}(\xi)} d\Gamma_k \leq C\eta M(k). \quad (3.28)$$

No small conditional-mass assertion is made when an adjacent leg and the output lie in the same corner layer.

Proof. Choose ρ_0 sufficiently small that the corresponding dimensionless corner depth lies in the range of Proposition 3.3, and that an input at a different corner has one coordinate uniformly separated from zero. This neighborhood is fixed before η is chosen. Use the scaling in (3.13)–(3.15), and denote the output corner depth by $d = \max d_j$. On resonance, each $|X^{(j)}Y^{(j)}| \leq 2d$. First observe that if the product density of any one coordinate is bounded by B in this range, while the other two densities are unchanged, the sum of all six contributions is at most $CBd^2[1 + \log(1/d)]^2$. If the designated coordinate is the sole negative one, use directly $B \int_0^{2d} s[1 + \log(1/s)]^2 ds$. If it is positive, first integrate the other positive coordinate to obtain $Cs[1 + \log(1/s)]$, then integrate against the negative density $\log(d/s)$. For the two-negative patterns, divide both negative variables by d ; the resulting endpoint terms contain only integrable logarithms. This proves the uniform bounded-density replacement estimate.

If an adjacent input lies at a different corner, one coordinate X or Y lies in a far-end strip of width η and has absolute value bounded below by a fixed positive constant. The product density of this strip is bounded by $\int_{\text{strip}} dX/|X| \leq C\eta$; the third indicator can only shrink the domain. Divide the replacement estimate by $M(k) \geq cd^2[1 + \log(1/d)]^2$ to obtain (3.28).

For the opposite leg, when the output has distance at least η from every corner, the spherical kernel is uniformly bounded. The input corner layer has volume $O(\eta^3)$ and $M(k) \geq c\eta^2$, giving $C\eta$. If the output corner depth satisfies $d \leq \eta$, distinguish the input corners. An opposite leg at the same corner forces both adjacent legs, by momentum conservation and all the indicators, into a cube of side 2η at that corner. Its linear scale is 2η , and the dimensionless output corner depth within it is $d/(2\eta)$. The six-dimensional volume and the quadratic-energy δ distribution together scale by $(2\eta)^4$. Apply the unweighted version of Proposition 3.3, absorbing the upper bound on compact sets away from corners into the constant. Its upper bound then applies throughout $0 < d/(2\eta) \leq 1/2$ and bounds the frequency of this small cube by $C\eta^2 d^2[1 + \log(\eta/d)]^2$. Dividing by the full frequency gives $O(\eta^2)$. If the opposite leg is at a different corner, one coordinate $X + Y$ lies in a far-end strip of width η . In the quadrant where both coordinates are negative, set $s = -X - Y, t = XY$. Then $s \geq c > 0, |t| \leq 2d$, and the root separation $\sqrt{s^2 - 4t}$ is uniformly nonzero. In a mixed-sign quadrant, the root separation is $\sqrt{s^2 + 4|t|}$. The product density is therefore at most $C\eta$, and the replacement estimate applies again. At the corners themselves, every row vanishes, so no division by zero frequency is needed. $\square$

Proposition 3.14 (Linearized coercivity under two-sided bounds in the corner layer). *Fix $0 < a \leq 1 \leq b < \infty$. There exist $\kappa_0, \eta_0, \gamma > 0$ such that, if*

$$a \leq f/N \leq b, \quad |f/N - 1| \leq \kappa \quad \text{on } D \setminus E_{\eta}, \quad \kappa \leq \kappa_0, \quad \eta \leq \eta_0,$$

then, for every $h \in L^2(D)$ with $P_N h = 0$,

$$\operatorname{Re} \langle h, -N^{-1}DC(f)[Nh] \rangle \geq \gamma \|h\|_{H_{\nu}}^2. \quad (3.29)$$

Moreover, the full operator can be written as $\nu_f + K_f$, with

$$c\nu_N \leq \nu_f \leq C\nu_N \quad (D \setminus Z), \quad |K_f(k, p)| \leq CK_{a,N}(k, p), \quad \int |K_f(k, p)|^2 dp \leq C\sqrt{\nu_*(k)}.$$

This quadratic form also extends continuously to H_ν , and the inequality remains valid there. Its diagonal term is the integral of $\nu_f |h|^2$, and its off-diagonal term is defined by the bounded form of the weighted kernel. Equivalently, it is the limit obtained by approximating with bounded functions in H_ν ; a function belonging only to H_ν is not treated directly as both entries of an unweighted L^2 inner product. The corner rows remain zero. All bounds also hold after averaging DC between positive states satisfying the same conditions. The statement retains the full gain and loss terms and does not require the right null space for general f to remain $N\Psi$.

Proof. There are three steps. Conditional-mass estimates first preserve a positive lower bound for the diagonal frequency. The off-diagonal part is then treated as a small perturbation of the compact equilibrium kernel. Finally, weak convergence separates the mass of the limit from mass that may concentrate at the corners. The diagonal coefficient is

$$\nu_f(k_0) = \int a_0 d\Gamma_{k_0}, \quad a_0 = f_1 f_3 + f_1 f_2 - f_2 f_3 = f_1 f_3 + f_2(f_1 - f_3).$$

An individual a_0 may change sign. Set $r = \sqrt{\eta}$ and take η sufficiently small that $4r \leq \rho_0$, with ρ_0 from Lemma 3.13. When the output is at least distance r from the corners, the spherical and planar kernels, together with $M(k) \geq cr^2$, bound the conditional mass of each corner-layer input by $C\eta^3/r^3$. Consequently $\nu_f/\nu_N = 1 + O(\kappa + \eta^3/r^3)$. When the output is within distance r of ξ , mark the part where $k_1 \in E_{4r}$ or an adjacent input lies at another corner. Lemma 3.13 bounds its conditional mass by $O(r + \eta)$. On the complement there is at most one bad nonoutput leg, and it can only be an adjacent leg at the same corner: two such adjacent bad legs would force the opposite leg into E_{4r} . If the bad leg is k_2 , the other two lie outside the layer, and $|k_1 - k_3| = |k_2 - k_0| \leq C(r + \eta)$. Thus $|f_1 - f_3| \leq C(\kappa + r)$ and $a_0 \geq N_-^2 - C_b(\kappa + r)$. Exchange the roles if the bad leg is k_3 ; when there is no bad leg, use the RJ identity. On the marked part, use only the fixed absolute upper bound. This gives

$$0 < c\nu_N \leq \nu_f \leq C\nu_N \quad \text{on } D \setminus Z. \quad (3.30)$$

Moreover, as $\kappa, \eta \rightarrow 0$, $\nu_f/\nu_N \rightarrow 1$ almost everywhere.

The other three coefficients are

$$a_1 = f_0 f_3 + f_0 f_2 - f_2 f_3, \quad a_2 = f_1 f_3 + f_0 f_3 - f_0 f_1, \quad a_3 = f_1 f_2 + f_0 f_2 - f_0 f_1,$$

and enter the kernel with signs $+a_1, -a_2, -a_3$ and the corresponding factors N_i/N_0 . The fixed positive two-sided bounds control their absolute values by $K_{a,N}$, and the squared-row bound follows from Proposition 3.8. Denote the resulting off-diagonal kernel by K_f . Subtract the equilibrium value of each coefficient one factor at a time. Terms outside the corner layer give $C\kappa$ times the weighted absolute kernel. If the bad factor is a fixed input or output, the resulting operator contains $\chi_{E_\eta} \tilde{K}_{i,N}$ or $\tilde{K}_{i,N} \chi_{E_\eta}$, whose Hilbert–Schmidt norm tends to zero. If the bad factor is on an unfixed leg, retain its indicator inside the spherical or planar fiber. For almost every pair of fixed legs, the fiber area of this indicator tends to zero; the exceptions are only degenerate diagonals or coordinate lines. The weighted Hilbert–Schmidt kernel provides a majorant, so another application of dominated convergence gives

$$\|\nu_N^{-1/2}(K_f - K_N)\nu_N^{-1/2}\|_{2 \rightarrow 2} \leq C\kappa + o_{\eta \rightarrow 0}(1).$$

All errors are uniform on compact parameter sets.

It remains to exclude escape of mass into the corners. If the conclusion failed, choose $\kappa_j, \eta_j \rightarrow 0$ and unit vectors $v_j = \sqrt{\nu_{N_j}} h_j$, with $P_{N_j} h_j = 0$, for which the left-hand side tends to zero or is negative. Pass to a subsequence with $N_j \rightarrow N$ and $v_j \rightharpoonup v$. The dual null-space vectors $\kappa_{N_j}/\sqrt{\nu_{N_j}}$ converge strongly in L^2 , so v still satisfies the weighted microscopic constraint. Compactness gives convergence of the kernel

quadratic forms. Set $\alpha_j = \nu_{f_j}/\nu_{N_j}$. Since $\alpha_j \geq a_* > 0$, the sequence is uniformly bounded, and $\alpha_j \rightarrow 1$ almost everywhere,

$$\liminf \int \alpha_j |v_j|^2 \geq \|v\|_2^2 + a_*(1 - \|v\|_2^2).$$

This follows by expanding $v_j = v + (v_j - v)$ and applying dominated convergence and weak convergence. Proposition 3.6 therefore bounds the limiting energy below by $c\|v\|_2^2 + a_*(1 - \|v\|_2^2) > 0$, a contradiction. $\square$

Finally, consider the products arising when the collision operator is differentiated in space. Apply spatial interpolation to each fixed momentum quadruple before integrating over the resonance measure. The same measure retains the correlations between the legs; the required momentum estimates reduce to its single-leg marginals and two-leg conditional averages.

Lemma 3.15 (Third-order products on the same four legs). *Set $d\mu_* = \prod_i n_*(k_i) d\Xi_D$ and $dm_*(k) = n_*^2 \nu_* dk$. Every leg has marginal m_* . For $i \neq j$, the conditional averaging operator P_{ij} is determined by*

$$\int_D g(k)(P_{ij}h)(k) dm_*(k) = \int g(k_i)h(k_j) d\mu_*$$

for all bounded g, h . It is a positive contraction on each $L^p(m_*)$; set

$$\omega_\eta = \max_{i \neq j} \|\chi_{E_\eta} P_{ij} \chi_{E_\eta}\|_{2 \rightarrow 2} \rightarrow 0.$$

Suppose that a real function e on $\mathbb{T}_X^3 \times D$ satisfies $|e| \leq K_0$ everywhere and $|e| \leq \kappa \leq 1$ outside the layer. For $l = 2, 3$, if the total spatial derivative order is l and at least two distinct e legs carry derivatives of positive order, then every product of two or three factors satisfies

$$\left\| \prod_j \partial_X^{\beta_j} e(X, k_{i_j}) \right\|_{L_X^2 L^2(\mu_*)} \leq C_{K_0} (\kappa^{1/3} + \omega_\eta^{1/3}) \|e\|_{H_X^l H_\nu}. \quad (3.31)$$

No statistical independence of the legs is assumed. General products of total order $l \leq 3$ satisfy the bound $C_{K_0} \|e\|_{H_X^l H_\nu}$ without a small coefficient. If the total order is at most three and at least one factor is a derivative of a parameter error $a(X)$, then, in the range $\|a\|_{H_X^3} \leq A_0$, the corresponding product of scalar and Hilbert-valued functions is bounded by $C_{A_0} \|a\|_{H_X^3} \|e\|_{H_X^3 H_\nu}$. The remaining coefficients are bounded derivatives of fixed finite order of prescribed functions, or bounded zeroth-order factors e .

Proof. Equality of the four marginals gives the positive contraction property. Its unitary representation on $L^2(m_*)$ is exactly $\nu_*^{-1/2} K_i \nu_*^{-1/2}$, so Lemma 3.5 implies that the corner-restricted norm tends to zero. Interpolation with the L^1, L^∞ contractions gives

$$\|\chi_{E_\eta} P_{ij} \chi_{E_\eta}\|_{p \rightarrow p} \leq \omega_\eta^{2 \min(1/p, 1-1/p)}.$$

For a periodic function at fixed momentum, integration by parts and Hölder's inequality give

$$\|\nabla e\|_4^2 \leq C \|e\|_\infty \|D^2 e\|_2, \quad \|\nabla e\|_6^2 \leq C \|e\|_\infty \|D^2 e\|_3, \quad \|D^2 e\|_3^2 \leq C \|\nabla e\|_6 \|D^3 e\|_2.$$

Eliminating between the last two inequalities yields $\|\partial^j e\|_{6/j} \leq C \|e\|_\infty^{1-j/3} \|e\|_{H^3}^{j/3}$ for $j = 1, 2$; the second-order estimate is similar. At zeros of the Hessian norm, one may first regularize the square root and then pass to the limit.

Apply these estimates in X for a fixed momentum quadruple, and then apply Hölder's inequality on the same measure μ_* . For the third-order distribution $2 + 1$, the squared integral leaves $F_i^{2/3} F_j^{1/3}$, where

$F_i = \|e(\cdot, k_i)\|_{H^3}^2$. If a differentiated leg lies outside the layer, its small amplitude supplies at least $\kappa^{1/3}$. If both legs lie inside the layer, the conditional-average bound at $p = 3$ or $3/2$ supplies $\omega_\eta^{2/3}$ in the squared integral. For the third-order distribution $1 + 1 + 1$, write $\chi_i = \mathbf{1}_{E_\eta}(k_i)$ and use

$$\int \chi_i \chi_j \chi_k F_i^{1/3} F_j^{1/3} F_k^{1/3} d\mu_* \leq \left(\int \chi_i \chi_j \sqrt{F_i F_j} d\mu_* \right)^{2/3} \left(\int \chi_k F_k d\mu_* \right)^{1/3},$$

followed by the L^2 corner-restricted bound for the two-leg conditional average. Taking square roots gives the asserted exponent $1/3$ in both cases. The second-order distribution $1 + 1$ gives the stronger exponent $1/2$; zeroth-order legs are bounded by K_0 . Without corner restrictions, the same calculation using only contraction gives the general Moser bound.

Finally, abbreviate spatial derivatives of total order j by a_j, e_j ; these subscripts denote derivative order, not collision legs. The highest-order distributions are $a_3 e_0, a_2 e_1, a_1 e_2$. Use, respectively, $L_X^2 \times L_X^\infty H_\nu$, $L_X^6 \times L_X^3 H_\nu$, and $L_X^\infty \times L_X^2 H_\nu$. If an additional e leg occurs, first use its zeroth-order bound or the same lower-order four-leg Moser estimate; for example, $a_1 e_1 e_1$ is controlled by $a_1 \in L^\infty$ and the second-order product bound. Multiple parameter derivatives are treated by the same distributions of total order at most three. The entire proof uses only $H_X^3 H_\nu$ and the joint zeroth-order amplitude bound, not $\sup_k H_X^3$ or $H_X^3 L_k^\infty$. $\square$

4 The macroscopic limit for positive Rayleigh–Jeans data on a smooth Euler interval

We consider the four-wave collision operator $\mathcal{C}$ defined in the preceding section, with spatial domain $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ and velocity $v(k) = 2k$. Writing $\varepsilon = 4\pi/c$ for the equivalent small parameter, set

$$\Psi(k) = (1, k^{(1)}, k^{(2)}, k^{(3)}, |k|^2)^\top, \quad \mathcal{T} = \partial_t + v(k) \cdot \nabla_X, \quad c = \frac{4\pi}{\varepsilon}.$$

All momentum integrals are over $D = [-R, R]^3$. We prove that, for positive local Rayleigh–Jeans initial data, the strong-collision equation has a solution throughout any prescribed interval of smooth positive Euler existence, and we obtain convergence rates for both the macroscopic parameters and the distribution. The collision frequency vanishes at the corners, so the proof treats both the usual dissipative region and the corner layers that retain a free-transport trace. Here $v_j(k) = 2k^{(j)}$; the subscript in k_i is reserved for the i th collision leg. The proof is organized around the five moments of the same evolving distribution. The exact entropy decomposition first identifies the moment-matched Rayleigh–Jeans equilibrium and the microscopic deviation; a third-order energy with a nonlinear corrector and a corner comparison then control the original initial-value problem. The main result is Theorem 4.3: the distribution error is $O(c^{-3/4})$, whereas the second-order Sobolev error of the macroscopic parameters is $O(c^{-1})$. We first construct a local solution and the nonlinear corrector, establish the joint energy and corner positivity estimates, and then use a first-exit argument to reach the prescribed Euler interval.

4.1 Five-moment parameters and the symmetric hyperbolic system

Define the positive parameter domain and its associated distributions by

$$\mathcal{O} = \{\theta \in \mathbb{R}^5 : \min_{k \in D} \theta \cdot \Psi(k) > 0\}, \quad N_\theta(k) = \frac{1}{\theta \cdot \Psi(k)}. \quad (4.1)$$

The set $\mathcal{O}$ is open and convex. Set

$$U(\theta) = \int_D \Psi N_\theta \, dk, \quad \mathcal{F}_j(\theta) = \int_D v_j \Psi N_\theta \, dk, \quad (4.2)$$

$$M(\theta) = \int_D N_\theta^2 \Psi \Psi^\top \, dk, \quad J_j(\theta) = \int_D v_j N_\theta^2 \Psi \Psi^\top \, dk. \quad (4.3)$$

Direct differentiation gives $DU = -M$ and $D\mathcal{F}_j = -J_j$. Every nonzero coefficient vector defines a nonzero polynomial $b \cdot \Psi$, so M is positive definite; each J_j is real symmetric. For $\theta, \vartheta \in \mathcal{O}$,

$$(\theta - \vartheta) \cdot [U(\theta) - U(\vartheta)] = - \int_D \frac{[(\theta - \vartheta) \cdot \Psi]^2}{(\theta \cdot \Psi)(\vartheta \cdot \Psi)} \, dk. \quad (4.4)$$

Thus U is injective, and the inverse function theorem makes it a smooth diffeomorphism from $\mathcal{O}$ onto its open image $U(\mathcal{O})$. Parameters will be recovered from moments only within this image.

Proposition 4.1 (Exact entropy decomposition under five-moment matching). *Fix a spatial point. Let $f > 0$ almost everywhere, $f \in L^1(D)$, $N = N_\theta$, and $\bar{N} = N_{\bar{\theta}}$, where $\theta, \bar{\theta} \in \mathcal{O}$. Write $U_f = \int_D \Psi f \, dk$ and define the relative entropy, allowing the value $+\infty$, by*

$$\mathcal{H}(f \mid g) = \int_D \left(\frac{f}{g} - 1 - \log \frac{f}{g} \right) \, dk.$$

Then the three-point identity is

$$\mathcal{H}(f \mid \bar{N}) = \mathcal{H}(f \mid N) + \mathcal{H}(N \mid \bar{N}) + (\bar{\theta} - \theta) \cdot [U_f - U(\theta)]. \quad (4.5)$$

In particular, if $U_f = U(\theta)$, then

$$\mathcal{H}(f \mid \bar{N}) = \mathcal{H}(f \mid N) + \mathcal{H}(N \mid \bar{N}). \quad (4.6)$$

Among positive distributions with these five moments, N uniquely minimizes both $\mathcal{H}(f \mid \bar{N})$ and $-\int_D \log f \, dk$, up to almost-everywhere equality.

The parameter potential $\mathcal{P}(\theta) = -\int_D \log(\theta \cdot \Psi) \, dk$ and the entropy in moment coordinates $\mathcal{E}(U(\theta)) = -\int_D \log N_\theta \, dk$ satisfy

$$D_\theta \mathcal{P} = -U, \quad D_{\bar{\theta}}^2 \mathcal{P} = M, \quad \nabla_U \mathcal{E} = -\theta, \quad D_U^2 \mathcal{E} = M^{-1}. \quad (4.7)$$

On every fixed compact convex set $\Theta \subseteq \mathcal{O}$, all $\theta, \bar{\theta} \in \Theta$ satisfy

$$\mathcal{H}(N_\theta \mid N_{\bar{\theta}}) \asymp_\Theta |\theta - \bar{\theta}|^2 \asymp_\Theta |U(\theta) - U(\bar{\theta})|^2. \quad (4.8)$$

Proof. Expanding the three entropy densities pointwise leaves $(f - N)(\bar{N}^{-1} - N^{-1})$. Both reciprocals are linear combinations of Ψ , so integration gives (4.5). The cross term is absolutely integrable. The two relative entropies involving f are finite if and only if $\log^- f \in L^1(D)$: the positive logarithmic part is controlled by $\log^+ f \leq f$, and both RJ states have positive upper and lower bounds. If the negative logarithmic part is not integrable, both sides of (4.5) equal $+\infty$; the additive identity involves no subtraction $\infty - \infty$. The solutions with positive two-sided bounds considered here satisfy the finiteness condition. Moment matching makes the cross term vanish. The inequality $x - 1 - \log x \geq 0$, with equality only at $x = 1$, proves the first unique-minimizer assertion. Moreover, $\int_D (f/N - 1) \, dk = \theta \cdot [U_f - U(\theta)] = 0$, so $-\int \log f + \int \log N = \mathcal{H}(f \mid N)$, proving the second.

Direct differentiation of $\mathcal{P}$ gives the first two derivative formulas; $DU = -M$ and $M\theta = U$ give the other two. With $\delta\theta = \bar{\theta} - \theta$,

$$\begin{aligned} \mathcal{H}(N_\theta \mid N_{\bar{\theta}}) &= \mathcal{P}(\bar{\theta}) - \mathcal{P}(\theta) - D\mathcal{P}(\theta) \cdot \delta\theta \\ &= \int_0^1 (1-s) \delta\theta^\top M(\theta + s\delta\theta) \delta\theta \, ds. \end{aligned}$$

Uniform positive definiteness of M on Θ gives the quadratic entropy comparison. Integrating $DU = -M$ along the same segment and pairing with $\delta\theta$ gives both bounds for the moment difference. $\square$

A simple matching example is $N = 1$. Choose a real ζ with $|\zeta|R^2 < 1$ and set

$$f_\zeta(k) = 1 + \zeta k^{(1)} k^{(2)}.$$

This distribution is strictly positive on D , and

$$\int_D \Psi(f_\zeta - 1) dk = \zeta \int_D \Psi k^{(1)} k^{(2)} dk = 0.$$

Each component of the last integrand is odd in at least one coordinate. Its moment-matched equilibrium is therefore exactly $N = 1$, whereas $\mathcal{H}(f_\zeta | 1) > 0$ when $\zeta \neq 0$: equality of all five moments still permits a nonzero microscopic deviation.

The preceding identity is the concrete three-point formula for the convex entropy under five-moment constraints. It holds at each (t, X) for the same $f, N, \bar{N}$, hence also after spatial integration; the collision term and its four-leg dissipation continue to be computed from the original distribution f . Its infinitesimal form explains the orthogonal energy used below. At $f = N$, suppose that bounded variations δf and $\delta N = -N^2 \Psi \cdot \delta\theta$ have matching five moments. Set $h = (\delta f - \delta N)/N$ and $\kappa_N = N\Psi$; then $\int_D \kappa_N h dk = 0$. Since $(-\log)''(N) = N^{-2}$, the second variation of the entropy is

$$\left\| \frac{\delta f}{N} \right\|_2^2 = \|h\|_2^2 + \delta\theta^\top M \delta\theta.$$

Thus the zeroth-order orthogonality is compatible with the entropy Hessian. The third-order differentiated energy with a corrector used below is not a spatial derivative of the total entropy: its control requires the full collision and corner-layer arguments. In particular, zero initial microscopic entropy need not imply zero initial error relative to the corrected approximation.

The five-moment conservation laws are equivalent to

$$M(\theta)\theta_t + \sum_{j=1}^3 J_j(\theta)\partial_j\theta = 0, \quad \theta_t + \sum_j A_j(\theta)\partial_j\theta = 0, \quad A_j = M^{-1}J_j. \quad (4.9)$$

The positive matrix M is a symmetrizer. Write $J(\xi) = \sum_j \xi_j J_j$. For every $\xi \in \mathbb{R}^3$, $M^{-1/2}J(\xi)M^{-1/2}$ is real symmetric; repeated eigenvalues cause no difficulty in the following construction. This local construction belongs to the theory of quasilinear symmetric hyperbolic systems [9]; we give the energy and approximation arguments needed here.

Proposition 4.2 (Local positive Euler solutions). *Let $s \geq 4$ be an integer, and suppose $\theta_0 \in H^s(\mathbb{T}^3; \mathbb{R}^5)$ has range compactly contained in $\mathcal{O}$. There exist $T_0 > 0$ and a unique solution*

$$\theta \in C([0, T_0]; H^s) \cap C^1([0, T_0]; H^{s-1})$$

of (4.9) with initial value θ_0 , whose range remains compactly contained in $\mathcal{O}$. Smooth initial data yield a smooth solution. The existence time is independent of c .

Proof. Fix a compact neighborhood $K \Subset \mathcal{O}$ of the range of the initial data. If $q_- \leq \theta \cdot \Psi \leq q_+$, then, with $G_D = \int_D \Psi \Psi^\top dk$,

$$q_+^{-2} G_D \leq M(\theta) \leq q_-^{-2} G_D.$$

Consequently, $E_s = \sum_{|\alpha| \leq s} \int (\partial^\alpha \theta)^\top M(\theta) \partial^\alpha \theta \, dX$ is equivalent to $\|\theta\|_{H^s}^2$. The multivariate Leibniz formula, the embedding $H^s \hookrightarrow W^{1,\infty}$, and interpolation give

$$\sum_{|\alpha| \leq s} \|[\partial^\alpha, A_j(\theta)] \partial_j z\|_2 \leq C_K (\|\nabla \theta\|_\infty \|z\|_{H^s} + \|\theta\|_{H^s} \|\nabla z\|_\infty). \quad (4.10)$$

Symmetry of $MA_j = J_j$ permits integration by parts in the leading transport term, so smooth solutions satisfy $|E'_s| \leq C_K \|\nabla \theta\|_\infty E_s$.

To construct a solution, let S_δ be convolution with an even periodic mollifier and solve the Banach-space ordinary differential equation

$$(\theta_\delta)_t = - \sum_j S_\delta \{A_j(\theta_\delta) \partial_j S_\delta \theta_\delta\}, \quad \theta_\delta(0) = \theta_0.$$

For fixed δ , the right-hand side is locally Lipschitz on the positive parameter domain in H^s . For $z = \partial^\alpha \theta_\delta$, split the principal term as

$$\langle Mz, S_\delta(A_j \partial_j S_\delta z) \rangle = \langle MS_\delta z, A_j \partial_j S_\delta z \rangle + \langle [S_\delta, M]z, A_j \partial_j S_\delta z \rangle.$$

Use symmetry for the first term and $\|[S_\delta, M]z\|_2 \leq C\delta \|\nabla M\|_\infty \|z\|_2$ together with $\|\partial S_\delta z\|_2 \leq C\delta^{-1} \|z\|_2$ for the second. The same energy estimate therefore holds for the mollified equation, uniformly in δ . The equation also gives $\|\theta_{\delta,t}\|_\infty \leq C_K \|\nabla \theta_\delta\|_\infty$. Choose a short time on which both the energy and the positive distance to the boundary of K are controlled. This gives a common existence interval.

On this interval, θ_δ is bounded in $L^\infty H^s$ and its time derivative in $L^\infty H^{s-1}$. Compact embedding and equicontinuity yield a subsequence converging strongly in $C_t H^{s-1}$, and the limit satisfies (4.9). Finally, mollify the differentiated equation for the limit once more. Integration by parts in the convolution kernel gives

$$\|[S_\delta, A] \partial z\|_2 \leq C \|\nabla A\|_\infty \|z\|_2.$$

Together with (4.10), this gives a uniform integrable bound for the time derivative of the mollified energy. Letting $\delta \rightarrow 0$ proves continuity of $E_s(t)$; weak H^s continuity and continuity of the positive energy norm give strong $C_t H^s$ continuity. For the difference of two solutions, use the first solution's M in an L^2 energy. Local Lipschitz continuity of the coefficients and the first derivatives of the other solution control the right-hand side; Gronwall's inequality proves uniqueness. Higher-order energies preserve smoothness successively. $\square$

From now on, fix a smooth positive Euler solution $\bar{\theta}$ on a finite interval $[0, T]$, with range compactly contained in $\mathcal{O}$, and write $\bar{N} = N_{\bar{\theta}}$. This may be any interval of smooth positive continuation of the solution in Proposition 4.2. Constants may depend on its positivity margin and finitely many derivative bounds. More precisely, fix a compact convex parameter neighborhood $\Theta \Subset \mathcal{O}$ whose interior contains $\bar{\theta}([0, T] \times \mathbb{T}^3)$, and set

$$K_5 = 1 + \max_{j+|\alpha| \leq 5} \|\partial_t^j \partial_X^\alpha \bar{\theta}\|_{L^\infty([0, T] \times \mathbb{T}^3)}.$$

The fixed data in the uniform estimates below are R, T, Θ, K_5 ; the temporary two-sided ratio bounds will also be determined by these data. We use C_K for a constant depending only on them, independent of Λ, η, c . Dependence on an auxiliary parameter already chosen will be indicated explicitly in the subscript. Higher derivatives of the smooth initial data are used only for classical regularity at each fixed c ; they do not enter the uniform threshold c_* or the constants in the following three convergence rates.

Theorem 4.3 (Strong-collision limit on a prescribed positive Euler interval). *Suppose $\theta_0 \in C^\infty(\mathbb{T}^3; \mathbb{R}^5)$ has range compactly contained in $\mathcal{O}$, $\bar{\theta}(0) = \theta_0$, and $\bar{\theta}$ is the smooth positive Euler solution on $[0, T]$ specified above. There exist $c_*, C_T, m_T, M_T > 0$ such that, for every $c \geq c_*$, the initial-value problem*

$$\mathcal{T}f_c = c\mathcal{C}(f_c), \quad f_c(0, X, k) = N_{\theta_0(X)}(k) \quad (4.11)$$

has a unique positive solution on $[0, T]$. For every finite m it belongs to $C_t C_X^m C_k(D)$; for $m \geq 1$ it also belongs to $C_t^1 C_X^{m-1} C_k(D)$, and $m_T \leq f_c \leq M_T$. Its five moments lie in $U(\mathcal{O})$. Defining $N_c = N_{\theta_c}$ by these moments, we have

$$\|f_c - \bar{N}\|_{L_t^\infty H_X^3 L_k^2} + \|\theta_c - \bar{\theta}\|_{L_t^\infty H_X^3} \leq C_T c^{-3/4}, \quad (4.12)$$

$$\|\theta_c - \bar{\theta}\|_{L_t^\infty H_X^2} \leq C_T c^{-1}, \quad (4.13)$$

$$\|f_c - N_c\|_{L_t^2 H_X^3 H_\nu} \leq C_T c^{-1}. \quad (4.14)$$

Here $H_\nu = L^2(D, \nu_* dk)$, where ν_* is the collision frequency of any fixed positive Rayleigh–Jeans distribution. All constants are independent of c .

We distinguish five objects used in the proof. The prescribed Euler solution determines $\bar{N}$, while the five moments of the evolving solution f_c determine N_c . The microscopic deviation is further split into a corrector and a remainder:

$$\begin{aligned} \bar{N} &= N_{\bar{\theta}}, & N_c &= N_{\theta_c}, & U(\theta_c) &= \int_D \Psi f_c dk, \\ f_c - N_c &= G_c + e_c, & \int_D \Psi G_c dk &= \int_D \Psi e_c dk = 0. \end{aligned}$$

The corrector G_c will be constructed from $\bar{N}$ in Subsection 4.3; e_c is the distribution error controlled by the joint energy. The difference between the two macroscopic states is measured by $\theta_c - \bar{\theta}$.

A spatially homogeneous equilibrium gives the simplest instance of this decomposition. If θ_0 is constant, then $\bar{\theta}(t, X) = \theta_0$, and $f_c(t, X, k) = N_{\theta_0}(k)$ is an exact solution for every c : both transport and collision terms vanish. The forcing in the corrector equation below also vanishes, its small solution is $G_c = 0$, and hence $e_c = 0$.

The corners explain the choice of convergence topology. The collision term vanishes at every corner ξ , so

$$f_c(t, X, \xi) = N_{\theta_0(X-2\xi t)}(\xi).$$

This free-transport trace generally differs from $\bar{N}(t, X, \xi)$. The momentum L^2 and weighted norms in the theorem allow this corner memory; the corner comparison will provide positive two-sided bounds in its neighborhood.

4.2 Local construction and continuation of the transport equation

Lemma 4.4 (Amplitude continuation criterion). *Let $B_m = C_X^m C_k(D)$. For every fixed $c > 0$ and strictly positive $f_0 \in B_m$, equation (4.11) has a unique maximal positive mild solution. If its maximal existence time $T_{\max}$ is finite, then*

$$\sup_{t < T_{\max}} \|f(t)\|_\infty = \infty.$$

For smooth spatial initial data, every finite B_m norm remains finite on each finite interval of existence on which the amplitude is bounded.

Proof. Lemma 3.1 gives local Lipschitz continuity of $\mathcal{C} : B_0 \rightarrow B_0$ and $\|\mathcal{C}(f)\|_\infty \leq C_R \|f\|_\infty^3$. The operators $S(t)f(X, k) = f(X - 2kt, k)$ form a strongly continuous isometric group on B_0 . The contraction mapping theorem applied to

$$f(t) = S(t)f_0 + c \int_0^t S(t-s)\mathcal{C}(f(s)) \, ds$$

on a bounded ball of continuous paths gives a unique local solution, with existence time depending only on c and the amplitude bound of the ball. Reapplying the same local construction near a finite maximal endpoint extends the solution, and uniqueness identifies the extension with the original solution. This proves the amplitude criterion.

Positivity follows from the decomposition of the original polynomial

$$\begin{aligned} \mathcal{C}(f)_0 &= \mathcal{G}(f)_0 + \mathcal{B}(f)_0 f_0, \\ \mathcal{G}(f)_0 &= \int f_1 f_2 f_3 \, d\Gamma_{k_0} \geq 0, \\ \mathcal{B}(f)_0 &= \int (f_2 f_3 - f_1 f_2 - f_1 f_3) \, d\Gamma_{k_0}. \end{aligned}$$

Along transport characteristics, the equivalent integral equation obtained with the integrating factor for $c\mathcal{B}(f)$ is also a contraction on the closed nonnegative ball. Its solution therefore agrees with the one above. On an interval where the amplitude is at most M , $f(t) \geq (\min f_0) \exp(-cC_R M^2 t) > 0$.

The group $S(t)$ commutes with ∂_X . In the full Leibniz expansion of each product of three factors, an m th-order derivative can fall entirely on only one factor. Hence

$$\|f(t)\|_{B_m} \leq \|f_0\|_{B_m} + cC_R M^2 \int_0^t \|f(s)\|_{B_m} \, ds + c \int_0^t P_m(\|f(s)\|_{B_{m-1}}) \, ds,$$

where P_m is a finite polynomial. Induction and Gronwall's inequality finish the proof. These higher-order norms at fixed c justify differentiation; uniform energy bounds are proved separately below. $\square$

4.3 The nonlinear corrector in two norms

The local equilibrium satisfies $\mathcal{C}(\bar{N}) = 0$, but its transport residual $\mathcal{T}\bar{N}$ generally does not vanish. We construct a corrector G_c with zero five moments so that $c\mathcal{C}(\bar{N} + G_c)$ cancels the leading part of this residual. Degeneracy of the frequency at the corners calls for two distinct controls: a small amplitude throughout momentum space and a stronger c^{-1} bound after multiplication by the frequency. Both norms will be retained below.

We first construct and differentiate the equation in the Banach spaces from the preceding section,

$$\mathcal{X} = L^\infty(D), \quad \mathcal{Y} = \{h : \nu_* h \in L^\infty(D)\}, \quad \|h\|_{\mathcal{Y}} = \|\nu_* h\|_\infty,$$

with functions identified almost everywhere. Write $X_0 = C(D)$ for the closed subspace of continuous functions. We will subsequently show that continuous inputs give outputs and parameter derivatives in X_0 . For a general $\mathcal{Y}$ input, the diagonal term $\nu_N h$ is only known to be bounded; continuous representatives will be recovered after the parameter estimates. For $s \geq 1$, set

$$\|h\|_{\mathcal{B}_s} = \|h\|_\infty + s\|\nu_* h\|_\infty.$$

For fixed s , the spaces $\mathcal{B}_s$ and $\mathcal{X}$ have the same underlying set and equivalent norms. Its continuous subspace is again denoted by X_0 , equipped with the same $\mathcal{B}_s$ norm. Write

$$\tilde{\mathcal{C}}_N(q) = N^{-1}\mathcal{C}(N(1+q)) = -L_N q + \mathcal{R}_N(q). \quad (4.15)$$

The remainder $\mathcal{R}_N$ consists exactly of the quadratic and cubic terms. Conservation of the five moments gives $P_N \mathcal{R}_N(q) = 0$.

Lemma 4.5 (The nonlinear cell problem and parameter derivatives). *There is $\delta_0 > 0$, depending on Θ , such that for real $F \in Q_N X_0$ with $\|F\|_\infty \leq \delta_0$, the equation*

$$q - s\tilde{\mathcal{C}}_N(q) = F, \quad P_N q = 0, \quad (4.16)$$

has a unique real solution in a fixed small $\mathcal{B}_s$ ball, and $\|q\|_{\mathcal{B}_s} \leq C_\Theta \|F\|_\infty$. Suppose $N = N(y)$ and $F = F(y)$ depend smoothly on a finite-dimensional parameter y , the specified finitely many derivatives of N are bounded, and $\max_{|\alpha| \leq j} \|\partial_y^\alpha F\|_\infty \leq K_j \delta$. Then

$$\max_{|\alpha| \leq j} (\|\partial_y^\alpha q\|_\infty + s \|\nu_* \partial_y^\alpha q\|_\infty) \leq C_j \delta. \quad (4.17)$$

The constants are uniform for $s \geq 1$.

Proof. By Proposition 3.10, $R_{N,s} = (I + sL_N)^{-1} Q_N : \mathcal{X} \rightarrow \mathcal{B}_s$ is uniformly bounded. For $\|q_i\|_\infty \leq \kappa$, Lemma 3.12 gives

$$\|\mathcal{R}_N(q_1) - \mathcal{R}_N(q_2)\|_\infty \leq C\kappa \|\nu_*(q_1 - q_2)\|_\infty.$$

Thus $q \mapsto R_{N,s}[F + s\mathcal{R}_N(q)]$ is a contraction on the ball of radius $2C\|F\|_\infty$. First ensure $C\kappa < 1/2$, then choose δ_0 . Iteration from zero remains in the same range of Q_N . The original cubic operator and the resolvent preserve continuous functions, so iteration with continuous input remains in X_0 . The supremum and essential supremum of a continuous function on the closed cube agree; hence $X_0 \hookrightarrow \mathcal{X}$ is a closed isometric embedding, and the limit is continuous. Iteration also preserves real values. Taking the ball radius below $1/2$ gives $1 + q \geq 1/2$.

Parameter differentiation requires a fixed projected space. Retain the reference state n_* and write $P_* = P_{n_*}$, $Q_* = I - P_*$. Use the operator U_N from Lemma 3.11, which satisfies $U_N P_* = P_N U_N$; the operators U_N, U_N^* and their specified parameter derivatives are uniformly bounded on $\mathcal{X}, \mathcal{Y}$. Set $z = U_N^* q$, $\tilde{L}_N = U_N^* L_N U_N$, and define

$$\tilde{R}_{N,s} = U_N^* R_{N,s} U_N = (I + s\tilde{L}_N)^{-1} Q_*, \quad \tilde{\mathcal{R}}_N(z) = U_N^* \mathcal{R}_N(U_N z).$$

On the fixed space $Q_* \mathcal{X}$, the resolvent satisfies

$$\partial_\theta (I + s\tilde{L}_N)^{-1} = -s(I + s\tilde{L}_N)^{-1} (\partial_\theta \tilde{L}_N) (I + s\tilde{L}_N)^{-1}.$$

Reading the right-hand side from right to left gives an s^{-1} gain from $\mathcal{X}$ to $\mathcal{Y}$, a bounded parameter derivative from $\mathcal{Y}$ to $\mathcal{X}$, and a uniform bound from $\mathcal{X}$ to $\mathcal{B}_s$. Every higher parameter derivative is controlled by the same pairing of factors s and s^{-1} .

In the nonlinear equation on the fixed space, the operator acting on the highest y derivative is $I - s\tilde{R}_{N,s} D \tilde{\mathcal{R}}_N(z)$. Its inverse is uniformly controlled by the contraction constant above. Every remaining term in the finite-order chain rule contains parameter derivatives or lower state derivatives. Lemma 3.12 permits exactly one input of each multilinear term to be estimated in $\mathcal{Y}$, with the others in $\mathcal{X}$. Each coefficient s is therefore canceled by the s^{-1} gain of a lower derivative. Induction gives (4.17). Derivatives of U_N^* are already included in this fixed-space equation; no separate membership of F_y in the range of the original projection Q_N is required.

Finally, recover continuous representatives of the parameter derivatives. For fixed s , the argument above gives the required smoothness of $q(y)$ in $\mathcal{X}$. Its first difference quotients are continuous functions and converge in the $\mathcal{X}$ norm. The closed embedding gives a unique continuous representative of the limit, and the convergence is in fact uniform. Induction on higher difference quotients shows that all the stated derivatives belong to X_0 and depend continuously on the parameters. This recovery uses closedness only at each fixed s ; the estimates uniform in s have already been proved in $\mathcal{X}, \mathcal{Y}$. $\square$

Apply this result with $F = -\mathcal{T} \log \bar{N}$. The Euler equation gives

$$\int_D \bar{N} \Psi F \, dk = - \int_D \Psi \mathcal{T} \bar{N} \, dk = 0.$$

The forcing F may be large on the prescribed interval. Choose a fixed $\Lambda \geq 1$ and apply Lemma 4.5 with $s = c/\Lambda$ and F/Λ to obtain

$$\Lambda q_c - c\tilde{\mathcal{C}}_{\bar{N}}(q_c) = -\mathcal{T} \log \bar{N}, \quad P_{\bar{N}} q_c = 0, \quad G_c = \bar{N} q_c. \quad (4.18)$$

Thus Λ reduces the forcing in the corrector equation while the positive parameter range of the initial data is unchanged. From now on, Λ depends on the prescribed interval, but not on c . For $c \geq \Lambda \geq \Lambda_0$, all required time and space derivatives of total order $|\alpha| \leq 4$ satisfy

$$\|\partial^\alpha G_c\|_\infty \leq C_K \Lambda^{-1}, \quad c\|\nu_* \partial^\alpha G_c\|_\infty \leq C_K. \quad (4.19)$$

The forcing $F = -\mathcal{T} \log \bar{N}$ uses one more derivative than $\bar{\theta}$. Thus its total space-time derivatives through order four, and the corresponding derivatives of G_c , are precisely controlled by K_5 . Taking the small parameter in the lemma to be C_K/Λ and $s = c/\Lambda$ gives both the Λ^{-1} bound in the uniform norm and the c^{-1} bound in the weighted norm. The nonlinear products in the parameter derivatives use estimates with one $\mathcal{Y}$ input. The corner-frequency estimate in Lemma 3.3 gives $\nu_*^{-3/2} \in L^1(D)$. Since $\min(A, B)^2 \leq A^{1/2} B^{3/2}$,

$$\|\partial^\alpha G_c\|_2 \leq C_K \Lambda^{-1/4} c^{-3/4}, \quad \|\partial^\alpha G_c\|_{H_\nu} + \|\partial^\alpha G_c\|_1 \leq C_K c^{-1}. \quad (4.20)$$

The exponent $3/4$ comes from the critical integrability of the corner frequency: squaring and integrating the pointwise bound $|\partial^\alpha G_c| \leq C_K \min\{\Lambda^{-1}, (c\nu_*)^{-1}\}$ uses exactly $\nu_*^{-3/2} \in L^1$. Let $f^{\text{app}} = \bar{N} + G_c$. Its residual and initial error are

$$\mathcal{E}^{\text{app}} = \mathcal{T} f^{\text{app}} - c\mathcal{C}(f^{\text{app}}) = \bar{N}\{\mathcal{T} q_c + (\mathcal{T} \log \bar{N}) q_c - \Lambda q_c\}, \quad (4.21)$$

$$f_c(0) - f^{\text{app}}(0) = -G_c(0). \quad (4.22)$$

In particular,

$$\begin{aligned} \|G_c\|_{L_t^\infty H_X^3 L_k^2} &\leq C_K \Lambda^{-1/4} c^{-3/4}, \\ \|\mathcal{E}^{\text{app}}\|_{L_t^\infty H_X^3 L_k^2} &\leq C_K \Lambda^{3/4} c^{-3/4}, \quad \|\mathcal{E}^{\text{app}}/\bar{N}\|_\infty \leq C_K. \end{aligned} \quad (4.23)$$

The projection condition gives $\int \Psi G_c = 0$, whereas

$$\int_D \Psi \mathcal{E}^{\text{app}} \, dk = \sum_j \partial_j \int_D v_j \Psi G_c \, dk. \quad (4.24)$$

This vector is generally nonzero; the macroscopic equation below retains it.

4.4 Orthogonal decomposition and differentiated equations through order three

Work for now on an interval where the original solution exists. Use $U_c = \int \Psi f_c$ to define $N = N_{\theta_c}$, and abbreviate $G = G_c$, $f = f_c$, and $\theta = \theta_c$. Throughout this subsection and the energy proof that follows, N denotes the actual moment-matched equilibrium; the prescribed Euler distribution remains $\bar{N}$. Set

$$e = f - N - G, \quad a = \theta - \bar{\theta}, \quad \kappa_N = N\Psi, \quad \eta_\alpha = \frac{\partial_X^\alpha e}{N}, \quad a_\alpha = \partial_X^\alpha a, \quad H_\alpha = \eta_\alpha - \kappa_N \cdot a_\alpha.$$

These adapted derivative variables are obtained by first differentiating the distribution error e in space and then dividing by N . In particular,

$$\eta_j = \partial_j(e/N) + (\partial_j \log N)(e/N).$$

This choice preserves exact five-moment orthogonality at each order; variation of the denominator appears in the connection terms below. The macroscopic and microscopic components thus use the same transport velocity, and the highest-order cross terms cancel in the joint energy. For every $0 \leq |\alpha| \leq 3$,

$$\int_D \kappa_N \eta_\alpha \, dk = \int_D \Psi \partial^\alpha e \, dk = 0. \quad (4.25)$$

Define

$$E_l = \frac{1}{2} \sum_{|\alpha|=l} \int_{\mathbb{T}^3} (\|\eta_\alpha\|_2^2 + a_\alpha^\top M a_\alpha) \, dX = \frac{1}{2} \sum_{|\alpha|=l} \|H_\alpha\|_{L_{X,k}^2}^2, \quad (4.26)$$

$$D_l = \sum_{|\alpha|=l} \int_{\mathbb{T}^3} \mathfrak{a}_N(\eta_\alpha, \eta_\alpha) \, dX, \quad E = \sum_{l=0}^3 \omega_l E_l, \quad D_e = \sum_{l=0}^3 \omega_l D_l. \quad (4.27)$$

The fixed decreasing weights $1 = \omega_0 > \omega_1 > \omega_2 > \omega_3 > 0$ will be chosen in the energy estimate. In the parameter neighborhood where $E \leq e_*$, Proposition 3.6 and (4.25) give

$$E \asymp \|e\|_{H_X^3 L_k^2}^2 + \|a\|_{H_X^3}^2, \quad \|e\|_{H_X^l H_\nu}^2 \leq C \sum_{p=0}^l D_p. \quad (4.28)$$

We now derive the differentiated identities, including order zero. Set

$$B = \mathcal{T} \log N, \quad C_\alpha = \frac{\partial^\alpha N}{N} + \kappa_N \cdot \partial^\alpha \theta, \quad s_\alpha = \kappa_N \cdot \partial^\alpha \theta.$$

Here $\partial_j = \partial/\partial X_j$. In C_{ij} and s_{ij} the indices indicate the successive spatial derivatives $\partial_i \partial_j$, and repeated indices are allowed. Sums in the energy are still indexed by multiindices α . Direct differentiation gives

$$\begin{aligned} C_0 &= 2, & C_i &= 0, & C_{ij} &= 2s_i s_j, \\ C_{ijk} &= 2(s_i s_{jk} + s_j s_{ik} + s_k s_{ij}) - 6s_i s_j s_k. \end{aligned} \quad (4.29)$$

Consequently,

$$H_\alpha = \frac{\partial^\alpha (f - G)}{N} + \kappa_N \cdot \partial^\alpha \bar{\theta} - C_\alpha. \quad (4.30)$$

Set

$$\begin{aligned} S_\alpha &= \kappa_N \cdot \mathcal{T} \partial^\alpha \bar{\theta} + 2B \kappa_N \cdot \partial^\alpha \bar{\theta} - (\mathcal{T} + B) C_\alpha, \\ \mathcal{R}_\alpha^{\text{src}} &= S_\alpha + \frac{\partial^\alpha \mathcal{T} \bar{N}}{N}. \end{aligned} \quad (4.31)$$

Differentiating (4.30) and using (4.21) yields

$$\mathcal{T} H_\alpha = \frac{c}{N} \partial^\alpha [\mathcal{C}(f) - \mathcal{C}(f^{\text{app}})] - B H_\alpha + \mathcal{R}_\alpha^{\text{src}} - \frac{\partial^\alpha \mathcal{E}^{\text{app}}}{N}. \quad (4.32)$$

Its collision term is orthogonal to κ_N . In particular, $C_0 = 2$ gives

$$\mathcal{R}_0^{\text{src}} = (1 - \rho^2) \kappa_N \cdot \mathcal{T} \bar{\theta} + 2B(\kappa_N \cdot \bar{\theta} - 1), \quad \rho = \bar{N}/N. \quad (4.33)$$

The actual five-moment equation is

$$\theta_t = - \sum_j A_j(\theta) \partial_j \theta + M^{-1} \sum_j \partial_j Z_j, \quad Z_j = \int_D v_j \Psi(e + G) \, dk. \quad (4.34)$$

It is obtained by taking moments of the conservation law for the same distribution; in particular, it contains no collision coefficient c . Split the flux into the distribution-error and known-corrector parts:

$$Z_j^e = \int_D v_j \Psi e \, dk, \quad Z_j^G = \int_D v_j \Psi G \, dk, \quad Z_j = Z_j^e + Z_j^G.$$

Spatial derivatives commute with this moment integral, so

$$\|Z\|_{H_X^3} \leq C\|e\|_{H_X^3 L_k^2} + C\|G\|_{H_X^3 L_k^1} \leq C_K(\sqrt{E} + c^{-1}).$$

Subtract the Euler equation from (4.34) and use the product rule in H_X^2 to obtain the first estimate below. Three spatial derivatives suffice for $\operatorname{div} Z$; third-order control of θ_t is not required. By (4.20), (4.28), and $H_X^3 \hookrightarrow C_X^1$,

$$\begin{aligned} \|\theta_t - \bar{\theta}_t\|_{H_X^2} + \|B - \bar{B}\|_{H_X^2 L_k^\infty} &\leq C_K(\sqrt{E} + c^{-1}), \\ \|B\|_\infty &\leq C_K, \quad \sum_{|\alpha| \leq 3} \|\mathcal{R}_\alpha^{\text{src}}\|_{L_X^2 L_k^\infty} \leq C_K(\sqrt{E} + c^{-1}), \end{aligned} \quad (4.35)$$

where $\bar{B} = \mathcal{T} \log \bar{N}$. To explain the highest-order case of the last estimate, note that for $1 \leq |\beta| \leq 2$,

$$\mathcal{T}s_\beta = Bs_\beta + \kappa_N \cdot \left(\partial^\beta \theta_t + \sum_j v_j \partial_j \partial^\beta \theta \right).$$

Substituting this into the third-order formula in (4.29), the highest time-derivative products are $\theta_{tij}\theta_i$ and $\theta_{ti}\theta_{jk}$; the highest spatial terms are $\theta_{ijk}\theta_i$ and products of two second derivatives. Estimate $\theta_{tij}\theta_i$ and $\theta_{ijk}\theta_i$ in $L^2 \times L^\infty$, and the remaining products in $L^6 \times L^3$. All these norms are controlled by $\theta_t \in H^2$ and $\theta \in H^3$. At $\theta = \bar{\theta}$, $S_\alpha = -\partial^\alpha \mathcal{T} \bar{N} / \bar{N}$; hence every term after subtraction contains either a parameter error or a microscopic moment from (4.34). Equation (4.33) verifies the zeroth-order endpoint of the same argument.

4.5 Third-order energy under two-sided corner-layer bounds

Let E_η be the union of the η -neighborhoods of the eight corners, with distance measured in the ℓ^∞ norm. On a temporary interval of existence, assume

$$0 < d_- \leq f/N \leq d_+ < \infty, \quad \sup_{k \notin E_\eta} |f/N - 1| \leq \kappa_b, \quad E \leq e_*. \quad (4.36)$$

The constants $d_\pm$ are fixed, and deviations within the corner layers need not be small. Write $v_\eta = \eta^2[1 + \log(R/\eta)]^2$. If

$$cv_\eta \geq C_K/\kappa_b, \quad (4.37)$$

then (4.19) and frequency comparison give $|e| \leq C\kappa_b$ on $D \setminus E_\eta$, while $|e| \leq K_0$ everywhere, where K_0 depends only on the fixed positive two-sided bounds. This converts the assumptions on the distribution f into assumptions on the error e used in the differentiated estimates.

Proposition 4.6 (Joint energy estimate). *On the interval above, there are fixed decreasing order weights and constants $\gamma, C_K > 0$ such that, for sufficiently small κ_b, η, e_* and sufficiently large Λ , provided (4.37) holds,*

$$E' + \gamma c D_e \leq C_K E + C_K \Lambda^{3/2} c^{-3/2}. \quad (4.38)$$

The smallness threshold can be written as

$$C_K(\kappa_b^{1/3} + \omega_\eta^{1/3} + \Lambda^{-1} + \sqrt{e_*}) \leq \gamma_0, \quad (4.39)$$

where $\omega_\eta \rightarrow 0$ is the corner-restricted two-leg operator norm in Lemma 3.15. Constants may depend on $d_\pm$, the positive parameter domain, and finitely many derivatives of the prescribed Euler solution, but not on c, η, Λ or any uncontrolled higher-order norm of the actual solution.

The estimates play distinct roles. The highest collision derivatives produce D_l ; parameter variations and the corrector produce terms controlled by lower-order dissipation and the known residual. The actual five-moment equation handles the transport connection terms. Fixed decreasing order weights then absorb the lower-triangular dissipation terms successively. The four small quantities in (4.39) correspond respectively to the error outside the corner layers, two-leg interactions in the corner layers, the corrector amplitude, and the unknown macroscopic parameter error.

Proof. The proof separates collision differentiation, parameter variation, and transport. Let $F = N + G$ and write

$$\mathcal{C}(f) - \mathcal{C}(f^{\text{app}}) = \underbrace{\mathcal{C}(F + e) - \mathcal{C}(F)}_{T(e; N, G)} + \underbrace{\mathcal{C}(N + G) - \mathcal{C}(\bar{N} + G)}_{V(a; G)}. \quad (4.40)$$

First consider T . The distributions $F + se$, $0 \leq s \leq 1$, have the same five moments and the same moment-matched equilibrium N . They are positive, have uniform two-sided bounds, and are close to N outside the corner layers. Proposition 3.14 and (4.25) give

$$\left\langle \eta_0, \frac{T}{N} \right\rangle = \int_0^1 \left\langle \eta_0, \frac{DC(F + se)[N\eta_0]}{N} \right\rangle ds \leq -\gamma_1 D_0. \quad (4.41)$$

This supplies the negative term in the zeroth-order energy directly.

For positive order $|\alpha| = l$, the chain rule for the cubic polynomial gives

$$\partial^\alpha T = DC(f)[\partial^\alpha e] + [DC(f) - DC(F)]\partial^\alpha F + \mathcal{L}_\alpha. \quad (4.42)$$

The coefficients of the first term retain all values of f in the corner layers, including those not close to equilibrium. Nevertheless, (4.25) and Proposition 3.14 yield

$$\left\langle \eta_\alpha, \frac{DC(f)[N\eta_\alpha]}{N} \right\rangle \leq -\gamma_1 \mathbf{a}_N(\eta_\alpha, \eta_\alpha). \quad (4.43)$$

We estimate all remaining terms by type. The four signs and leg-index sets in the collision polynomial are

$$(+, 123), \quad (+, 023), \quad (-, 013), \quad (-, 012).$$

For each such index set I , the full derivative formula is

$$\partial^\alpha \prod_{i \in I} f_i = \sum_{\sum_{i \in I} \beta_i = \alpha} \frac{\alpha!}{\prod_{i \in I} \beta_i!} \prod_{i \in I} \partial^{\beta_i} f_i. \quad (4.44)$$

Substitute $f = N + G + e$ and $F = N + G$ separately and subtract. Every remaining term contains at least one factor e . Equation (4.42) separates the terms in which a single factor receives the entire multiindex α ; $\mathcal{L}_\alpha$ is the difference of the remaining multiindex terms.

Fix a positive n_* and use the common four-leg measure $d\mu_* = \prod_i n_*(k_i) d\Xi_D$. Each one-leg marginal is $n_*^2 \nu_* dk$. First compare the varying N weights with this fixed measure. Lemma 3.15 applies directly to the distribution error e : momentum indicators commute with every X derivative, and (4.37) gives the required amplitude bound outside the corner layers. Write $R_l = \|e\|_{H_X^l H_\nu}$. For total order $l = 2, 3$, terms in which at least two positive-order derivatives act on distinct e factors satisfy

$$\|\text{these products}\|_{L_X^2 L^2(\mu_*)} \leq C_{K_0} (\kappa_b^{1/3} + \omega_\eta^{1/3}) R_l. \quad (4.45)$$

Any zeroth-order factor in a product of three factors is estimated only by K_0 . If the total derivative order on e is at most $l - 1$, the ordinary product estimate on the same measure gives $C_{K_0} R_{l-1}$.

To specify how these estimates apply here, divide the remainder into three types. The first contains no G , and all l derivatives fall on e . Apart from the leading term already separated, (4.45) applies exactly. The second contains no G , but at least one derivative falls on N . Terms containing only derivatives of the prescribed reference parameters contribute $C_K R_{l-1}$. Subtract the remaining smooth parameter coefficients along $\bar{\theta} + sa$; their differences contain at least one derivative of a . Through total order three,

$$\begin{aligned} \|a_3 e_0\|_{L_X^2 H_\nu} &\leq \|a_3\|_{L_X^2} \|e\|_{L_X^\infty H_\nu} \leq C \|a\|_{H_X^3} R_3, \\ \|a_2 e_1\|_{L_X^2 H_\nu} &\leq \|a_2\|_{L_X^6} \|e_1\|_{L_X^3 H_\nu} \leq C \|a\|_{H_X^3} R_3, \\ \|a_1 e_2\|_{L_X^2 H_\nu} &\leq C \|a\|_{H_X^3} R_3. \end{aligned}$$

An additional e factor is handled by its amplitude bound or a product estimate on the same measure. All parameter coefficients are expanded according to total derivative order. These three cases and their lower-order versions therefore exhaust products of highest parameter derivatives with microscopic errors, giving $C_K \sqrt{E} R_3$. In particular, the second term in (4.42) obeys the same estimate: indeed,

$$\partial^\alpha N = -N \kappa_N \cdot \partial^\alpha \theta + N C_\alpha,$$

and every coefficient of $DC(f) - DC(F)$ contains an e factor. A prescribed highest derivative $\partial^\alpha \bar{\theta}$ times a lower-order e contributes $C_K R_{l-1}$; an unknown highest derivative $\partial^\alpha a$ contributes $C_K \sqrt{E} R_3$ by the estimates above.

The third type contains one or more factors G . Every required derivative of G is bounded in the uniform norm by C_K/Λ through (4.19). The remaining terms still contain e , so the ordinary bounded-amplitude product estimate on the same measure gives $C_K \Lambda^{-1} R_3$; terms containing a parameter error add $C_K \sqrt{E} R_3$. This classification retains every sign and multiplicity in (4.44), including products involving the output leg.

More explicitly, among the remaining terms without G , let r_N be the total derivative order on the parameter coefficients. If $r_N = 0$, the only nonleading allocations of third-order microscopic derivatives are $2+1$ and $1+1+1$, and the only second-order allocation is $1+1$. If $r_N \geq 1$, the microscopic derivative order is at most $l - 1$. After subtracting the prescribed reference values of the parameter coefficients, the highest unknown allocations are precisely $a_3 e_0, a_2 e_1, a_1 e_2$ and their lower-order or multifactor variants. This includes a zeroth-order e factor times a highest parameter derivative; no additional c^{-1} factor is required on that highest parameter derivative.

The one-leg marginal norm of the test function $\eta_\alpha(k_0)$ is controlled by $D_l^{1/2}$. For any product P above, after removing uniformly bounded weights depending only on the positive parameter domain, the estimate used is

$$\left| \int_{\mathbb{T}^3} \int \eta_\alpha(k_0) P \, d\mu_* \, dX \right| \leq \|\eta_\alpha(k_0)\|_{L_X^2 L^2(\mu_*)} \|P\|_{L_X^2 L^2(\mu_*)}.$$

The marginal measure in the first factor is exactly $n_*^2 \nu_* \, dk_0$. Thus all three classes of product bounds can be paired with $D_l^{1/2}$, without integrating the four legs separately or assuming independence. Applying Young's inequality to the known lower-order terms gives

$$\begin{aligned} \sum_{|\alpha|=l} \left\langle \eta_\alpha, \frac{\partial^\alpha T}{N} \right\rangle &\leq -\gamma_2 D_l + C_l \sum_{p < l} D_p \\ &\quad + C_K (\kappa_b^{1/3} + \omega_\eta^{1/3} + \Lambda^{-1} + \sqrt{E}) \sum_{p=0}^3 D_p. \end{aligned} \tag{4.46}$$

For $l = 0$, the lower-order sum is empty and (4.41) applies directly. To choose the order weights, first fix these constants on the ball $\|a\|_{H^3} \leq 1$, retaining all unknown parameter errors in the $\sqrt{E}$ term. Set $\omega_0 = 1$ and choose successively $0 < \omega_l < \omega_{l-1}/2$ so that

$$\omega_l C_l \leq \frac{\gamma_2}{8} \min_{0 \leq p < l} \omega_p, \quad l = 1, 2, 3.$$

Each D_p receives at most three such lower-order contributions, which are absorbed by its negative term. Decrease the threshold in (4.39) so that the common remainder is at most $\gamma_2 \sum_p \omega_p D_p/4$. Finally, choose e_* so that (4.28) implies $\|a\|_{H^3} \leq 1$ and the smallness condition holds. Thus the order weights are fixed before the energy threshold, and neither depends on time or c . The contribution of T is at most $-\gamma c D_e$.

For V , vary $\theta_s = \bar{\theta} + sa$ while keeping the distribution corrector G fixed. Since $\mathcal{C}(N_{\theta_s}) = 0$, every term obtained after any fixed number of parameter derivatives still contains at least one G factor. If that factor lies on the output leg, the collision integral gives $C\nu_N G$; if it lies on an internal leg, the bound $\int K_{a,N}(k, p)/\nu_*(p) dp \leq C$ from Lemma 3.8 gives the same estimate. In each term, estimate exactly one G in $\mathcal{Y}$ and the others in the uniform norm. Since $c\|\nu_* \partial^\beta G\|_\infty \leq C_K$, the multivariate chain rule yields

$$c\|\mathcal{C}(N + G) - \mathcal{C}(\bar{N} + G)\|_{H_X^3 L_k^2} \leq C_K \|a\|_{H_X^3}. \quad (4.47)$$

The relation between parameter differentiation and normalization can be displayed explicitly. Let $N_s = N_{\theta_s}$, $q_s = G/N_s$, and $b_s = \partial_s \log N_s = -N_s(a \cdot \Psi)$. Write D_N for the Fréchet derivative with respect to the positive distribution N while q is held fixed, namely

$$D_N \tilde{\mathcal{C}}_N(q)[h] = -\frac{h}{N} \tilde{\mathcal{C}}_N(q) + \frac{1}{N} D\mathcal{C}(N(1+q))[h(1+q)].$$

Since $\partial_s N_s = N_s b_s$ and $\partial_s q_s = -b_s q_s$, the chain rule gives

$$\frac{\partial_s \mathcal{C}(N_s + G)}{N_s} = b_s \tilde{\mathcal{C}}_{N_s}(q_s) + D_N \tilde{\mathcal{C}}_{N_s}(q_s)[N_s b_s] - D_q \tilde{\mathcal{C}}_{N_s}(q_s)[b_s q_s].$$

All three terms arise from differentiation with the same distribution corrector G held fixed. Pairing (4.47) with the unweighted energy contributes only $C_K E$.

Finally, integrate (4.32) over the periodic spatial domain. The collision term has zero five moments, so its pairing with H_α can be replaced by the pairing with η_α . Thus

$$\begin{aligned} E' &= c \sum_{l, |\alpha|=l} \omega_l \left\langle \eta_\alpha, \frac{\partial^\alpha [\mathcal{C}(f) - \mathcal{C}(f^{\text{app}})]}{N} \right\rangle \\ &\quad - \sum_{l, |\alpha|=l} \omega_l \int B |H_\alpha|^2 dX dk \\ &\quad + \sum_{l, |\alpha|=l} \omega_l \left\langle H_\alpha, \mathcal{R}_\alpha^{\text{src}} - \frac{\partial^\alpha \mathcal{E}^{\text{app}}}{N} \right\rangle. \end{aligned} \quad (4.48)$$

By (4.35) and the ordinary L^2 Young inequality, the last two lines are bounded by $C_K E + C_K c^{-2} + C_K \Lambda^{3/2} c^{-3/2}$. Combining this with the collision estimates proves the result. The residual term Λq_c and the zeroth-order initial layer have both been retained throughout. $\square$

The initial data $f(0) = N(0) = \bar{N}(0)$ give $a(0) = 0$ and $e(0) = -G(0)$. Proposition 4.6 and (4.20) therefore yield

$$\sup_{s \leq t} E(s) + c \int_0^t D_e(s) ds \leq C_K e^{C_K T} (\Lambda^{-1/2} + T \Lambda^{3/2}) c^{-3/2}, \quad t \leq T, \quad (4.49)$$

whenever the hypotheses hold on $[0, t]$. The right-hand side uses the prescribed T and is uniform on every shorter interval; it contains no negative power of that interval's length.

4.6 Positive exchange in corner layers and amplitude bounds outside them

Lemma 4.7 (Corner-layer comparison). *Assume (4.36) and (4.49) hold on $[0, t]$. For each corner ξ , set*

$$F_\xi(s, X) = N_{\theta_0(X-2\xi s)}(\xi), \quad F_- \leq F_\xi \leq F_+, \\ A_* = \min(1, N_-/(2F_+)), \quad B_* = \max(1, 2N_+/F_-),$$

where $N_- \leq N_\theta \leq N_+$ uniformly on the parameter domain in use. Then

$$A_* - r_c^{\text{cor}} \leq \frac{f(s, X, k)}{F_\xi(s, X)} \leq B_* + r_c^{\text{cor}}, \quad k \in E_\eta(\xi), \quad s \leq t, \quad (4.50)$$

and

$$r_c^{\text{cor}} \leq C_{K, \Lambda, T, d_\pm} [\eta + c\eta v_\eta + \psi(\eta) + c^{-1/4} v_\eta^{-1/4}], \quad \psi(\eta) \longrightarrow 0. \quad (4.51)$$

Proof. Fix $k_0 \in E_\eta(\xi)$ and partition the original integration variables into

$$\begin{aligned} \mathcal{B} &= \{k_1 \in E_{4\eta}\} \cup \{k_2 \text{ or } k_3 \text{ lies in } E_\eta \text{ at another corner}\}, \\ \mathcal{A}_2 &= \mathcal{B}^c \cap \{k_2 \in E_\eta(\xi)\}, \\ \mathcal{A}_3 &= \mathcal{B}^c \cap \{k_3 \in E_\eta(\xi)\}, \\ \mathcal{A}_0 &= \mathcal{B}^c \cap \{k_2, k_3 \notin E_\eta\}. \end{aligned}$$

If both k_2, k_3 belong to $E_\eta(\xi)$, momentum conservation forces $k_1 \in E_{3\eta}$, so $\mathcal{A}_2$ and $\mathcal{A}_3$ are disjoint. Lemma 3.13 gives $\int_{\mathcal{B}} d\Gamma_{k_0} \leq C\eta\nu_*(k_0)$. On $\mathcal{A}_2$, the sum of the four cubic terms is

$$f_1 f_3 (f_2 - f_0) + f_0 f_2 (f_3 - f_1),$$

and the corresponding expression on $\mathcal{A}_3$ interchanges indices 2, 3. The first term defines an exchange generator with nonnegative rates:

$$\begin{aligned} (\mathcal{J}_f h)(k_0) &= \int_{\mathcal{A}_2} f_1 f_3 [h(k_2) - h(k_0)] d\Gamma_{k_0} \\ &\quad + \int_{\mathcal{A}_3} f_1 f_2 [h(k_3) - h(k_0)] d\Gamma_{k_0}. \end{aligned}$$

It exchanges values only at the same X and within the same corner layer, and $\mathcal{J}_f 1 = 0$.

On $\mathcal{A}_0$, all three internal values are close to N . Therefore, for sufficiently small κ_b ,

$$a_0(f) = f_1 f_3 + f_1 f_2 - f_2 f_3 > 0, \quad \frac{f_1 f_2 f_3}{a_0(f)} \in [N_-/2, 2N_+].$$

Set $\nu_b = \int_{\mathcal{A}_0} a_0(f) d\Gamma_{k_0}$ and $\nu_b F_b = \int_{\mathcal{A}_0} f_1 f_2 f_3 d\Gamma_{k_0}$. When $\nu_b = 0$, choose any F_b in the indicated interval. The original equation is then equivalent to

$$\mathcal{T}f = c\mathcal{J}_f f + c\nu_b(F_b - f) + c(S_N + S_G + S_e + S_{\mathcal{B}}). \quad (4.52)$$

Here $S_{\mathcal{B}}$ is the entire original polynomial integrated over the exceptional region. The other three source terms come from the decomposition $f_3 - f_1 = (N_3 - N_1) + (G_3 - G_1) + (e_3 - e_1)$ on $\mathcal{A}_2$ and the corresponding expression on $\mathcal{A}_3$.

With the positive two-sided bounds fixed, $c\|S_{\mathcal{B}}\|_{L_s^1 L_{X,k}^\infty(E_\eta)} \leq C_T c\eta v_\eta$. On $\mathcal{A}_2$, $k_1 - k_3 = k_2 - k_0 = O(\eta)$, and the momentum Lipschitz constant of N_θ is uniform on the parameter domain. Thus S_N obeys the same bound. Lemma 3.8 gives

$$\sup_{k \in E_\eta, \theta \in \Theta} \int_D \frac{K_{a,\theta}(k, p)}{\nu_*(p)} dp =: \psi(\eta) \longrightarrow 0.$$

Together with $c\|\nu_*G\|_\infty \leq C_K$, this yields $c\|S_G\|_{L_s^1 L_{X,k}^\infty(E_\eta)} \leq C_T \psi(\eta)$.

In S_e , the factors e_1, e_3 lie in $D \setminus E_\eta$. The square-row estimate of the same lemma gives

$$\int_D K_{a,\theta}(k,p)^2 dp \leq C\sqrt{\nu_*(k)}.$$

Cauchy's inequality and $H_X^3(H_\nu) \hookrightarrow L_X^\infty(H_\nu)$ therefore imply

$$\begin{aligned} \left| \int_{D \setminus E_\eta} K_{a,\theta}(k,p) e(s, X, p) dp \right| &\leq C\nu_*(k)^{1/4} v_\eta^{-1/2} \|e(s, X)\|_{H_\nu} \\ &\leq C v_\eta^{-1/4} \|e(s, X)\|_{H_\nu}, \quad k \in E_\eta. \end{aligned}$$

The first line uses $\nu_*(p) \geq c_0 v_\eta$ on the input region and the square-row estimate; the second uses $\nu_*(k) \leq C_0 v_\eta$ on the output region. Consequently,

$$\sup_{X,k \in E_\eta} |S_e(s, X, k)| \leq C v_\eta^{-1/4} \|e(s)\|_{H_X^3 H_\nu}.$$

Take this joint supremum before integrating in time. The bound $\int_0^t \|e(s)\|_{H_X^3 H_\nu}^2 ds \leq C_{K,\Lambda,T} c^{-5/2}$ in (4.49), followed by Cauchy's inequality in time, gives

$$c\|S_e\|_{L_s^1 L_{X,k}^\infty(E_\eta)} \leq C_{K,\Lambda,T} c^{-1/4} v_\eta^{-1/4}. \quad (4.53)$$

The collision row vanishes at a corner, so $f(s, X, \xi) = F_\xi(s, X)$. For nearby momenta, $\mathcal{T} \log F_\xi = 2(k - \xi) \cdot \nabla_X \log F_\xi = O(\eta)$. Set $y = f/F_\xi$. Since exchange occurs at the same X and within the same corner layer, (4.52) becomes

$$\mathcal{T}y = c\mathcal{J}_f y + c\nu_b(F_b/F_\xi - y) + c(S_N + S_G + S_e + S_B)/F_\xi - y\mathcal{T} \log F_\xi.$$

Fix the coefficients along this solution. Transport together with exchange preserves positivity and constants; after adding the nonnegative killing rate $c\nu_b$, the evolution is contractive in L^∞ . Explicitly, incorporate the loss rate into an integrating factor along characteristics and iterate the positive gain term to obtain a positive evolution. Comparison with constants shows that its norm is at most one. The relaxation target F_b/F_ξ lies in $[A_*, B_*]$. Initially $y(0) = 1 + O(\eta)$, and the last term is $O(\eta)$ by the temporary amplitude bound. The preceding $L_s^1 L^\infty$ estimates now give (4.50)–(4.51). $\square$

Lemma 4.8 (Amplitude bound outside the corner layers). *On the same interval as in Lemma 4.7, set $h = (f - f^{\text{app}})/\bar{N}$. If the actual parameter error is sufficiently small, then*

$$\sup_{s \leq t, X, k \notin E_\eta} |h(s, X, k)| \leq C_{K,\Lambda,T} \{(c\nu_\eta)^{-1} + c^{-3/4} v_\eta^{-3/4}\}. \quad (4.54)$$

Proof. Apply Proposition 3.14 to the linear interpolation from f^{app} to f . Relative to $\bar{N}$, these distributions have fixed positive two-sided bounds, and their deviations outside the corner layers are controlled by $C\kappa_b + C_K/\Lambda + C\sqrt{E}$. The resulting equation is

$$\mathcal{T}h + \bar{B}h + c\nu_{\text{eff}}h = -cK_{\text{eff}}h - \mathcal{E}^{\text{app}}/\bar{N}, \quad h(0) = -q_c(0),$$

where $\nu_{\text{eff}} \asymp \nu_*$ and all three signed off-diagonal kernels are dominated by $CK_{a,\bar{N}}$. Outside the corner layers, when $c\nu_\eta \rightarrow \infty$, $c\nu_{\text{eff}} + \bar{B} \geq ac\nu_*$ for a fixed $a > 0$. Along the transport characteristic of the output momentum, the integrating factor bounds the source contribution by $C/(c\nu_*)$. Boundedness of $c\nu_*q_c(0)$ gives the same bound for the initial contribution. For the gain term, the square-row estimate and the spatial embedding for $H_X^3 L_k^2$ give

$$|K_{\text{eff}}h(s, X, k)| \leq C\nu_*(k)^{1/4} \|h(s)\|_{H_X^3 L_k^2}.$$

Integrating this against the exponential loss gives $C\nu_*(k)^{-3/4} \sup_s \|h(s)\|_{H_X^3 L_k^2}$. Equation (4.49) bounds the final norm by $O(c^{-3/4})$, proving the claim. $\square$

4.7 Continuation on the prescribed interval and convergence rates

Proof of Theorem 4.3. Choose all parameters in the following order. First fix the prescribed Euler interval and its compact positive parameter neighborhood. Let $N_{\pm}, F_{\pm}, A_*, B_*$ denote the known bounds in Lemma 4.7. Set

$$d_- = \min\{1/4, F_- A_*/(4N_+)\}, \quad d_+ = \max\{4, 2F_+(B_* + 1)/N_-\}. \quad (4.55)$$

These constants are fixed before Λ, κ_b, c . With $d_{\pm}$ fixed, choose the order weights and geometric thresholds in Proposition 4.6. Next choose Λ large and κ_b, e_* small enough that $C_K(\kappa_b^{1/3} + \Lambda^{-1} + \sqrt{e_*})$ is below half the threshold in (4.39), and that the parameter C^1 error keeps the parameters within the positive neighborhood. For the interpolation family relative to $\bar{N}$ in Lemma 4.8, take the uniform thresholds corresponding to the slightly wider bounds $d_-/2, 2d_+$, and also require $C\kappa_b + C_K/\Lambda + C\sqrt{e_*}$ to lie below its smallness threshold outside the corner layers. These finitely many requirements can be met simultaneously.

Fix

$$\frac{1}{3} < \beta < \frac{1}{2}, \quad \eta = c^{-\beta}. \quad (4.56)$$

For each c , this layer is kept fixed throughout the time interval. Finally, take c sufficiently large that $c \geq \Lambda$, the geometric thresholds hold, ω_{η} is sufficiently small, and $cv_{\eta} \geq C_K/\kappa_b$. This is possible because $cv_{\eta} \asymp c^{1-2\beta}(\log c)^2 \rightarrow \infty$. The full order of choices is therefore

$$\begin{aligned} (R, T, \Theta, K_5) &\longrightarrow d_{\pm} \longrightarrow (\omega_l)_{l=0}^3 \\ &\longrightarrow (\Lambda, \kappa_b, e_*) \longrightarrow \beta \longrightarrow c \geq c_*. \end{aligned}$$

In particular, $\eta = c^{-\beta}$ is determined last. Both Λ and the factor $\Lambda^{3/2}$ in the energy estimate remain fixed; the prescribed T is not shortened to remove this factor.

Lemma 4.4 constructs the same maximal solution from the original initial data. Initially $N_c = \bar{N}$, and the inverse moment map is defined by (4.4) and the nonsingular Jacobian $-M$. Let t_* be the first time this solution violates (4.36) or the parameter-neighborhood condition, or reaches its maximal existence endpoint. The initial distribution ratio is one, and $E(0) = O(\Lambda^{-1/2}c^{-3/2})$; hence all these conditions hold strictly at the initial time for large c .

For every $t < t_*$, estimate (4.49) holds with constants depending only on the prescribed T . By (4.56),

$$\begin{aligned} c\eta v_{\eta} &= O(c^{1-3\beta}(\log c)^2) \longrightarrow 0, \\ c^{-1/4}v_{\eta}^{-1/4} &= O(c^{-1/4+\beta/2}(\log c)^{-1/2}) \longrightarrow 0. \end{aligned}$$

Lemma 4.7 therefore gives $r_c^{\text{cor}} \leq \min(A_*/2, 1)$. By (4.55), $2d_- \leq f_c/N_c \leq d_+/2$ on the corner layers. Outside those layers, use

$$f_c/N_c - 1 = (\bar{N}/N_c)(q_c + h) + (\bar{N}/N_c - 1).$$

Lemma 4.8, the bound $|q_c| \leq C/(cv_{\eta})$, and the parameter energy give

$$\sup_{k \notin E_{\eta}} |f_c/N_c - 1| \leq C_{K,\Lambda,T} \{(cv_{\eta})^{-1} + c^{-3/4}v_{\eta}^{-3/4} + c^{-3/4}\} \longrightarrow 0. \quad (4.57)$$

The limit uses $-3/4 + 3\beta/2 < 0$. Increasing c further makes the right-hand side less than $\kappa_b/2$, the energy less than $e_*/2$, and the C^1 norm of the parameter difference strictly less than the margin of the chosen positive neighborhood.

Thus $t_* < T$ cannot be the first time a temporary inequality becomes an equality. The continuity used here is $C_t B_3$ continuity at each fixed c and continuity of the local inverse moment map, both supplied

by the local construction. If the parameter branch were to reach the boundary first, the error estimate and $H^3 \hookrightarrow C^1$ would keep its range in a fixed compact subset of the interior of Θ . The original inverse function theorem would then continue the same branch at that endpoint, excluding this type of exit as well. If t_* is the maximal existence endpoint, the bounds just obtained give

$$2d_- N_- \leq f_c \leq (d_+/2) N_+.$$

Lemma 4.4 then continues the same solution and its finite spatial derivatives. The parameters remain in a compact positive set, so the inverse moment map extends simultaneously along the same branch. This also excludes the maximal endpoint and proves existence through T . Every comparison function and residual integral is defined from the original initial time.

With Λ, T now fixed, (4.49) gives $\|e\|_{L_t^\infty H_X^3 L_k^2} + \|a\|_{L_t^\infty H_X^3} \leq C_T c^{-3/4}$ and $\|e\|_{L_t^2 H_X^3 H_\nu} \leq C_T c^{-5/4}$. Combining these with (4.20) proves (4.12) and (4.14).

It remains to establish the sharper macroscopic H^2 rate. The integrability $\nu_*^{-1} \in L^1$ and the moment integrals defining the remainder flux give

$$\|Z^e\|_{L_t^1 H_X^3} \leq C_T c^{-5/4}, \quad \|Z^G\|_{L_t^1 H_X^3} \leq C_T c^{-1}. \quad (4.58)$$

Subtracting the Euler equation yields

$$a_t + \sum_j A_j(\theta) \partial_j a = - \sum_j [A_j(\theta) - A_j(\bar{\theta})] \partial_j \bar{\theta} + M^{-1} \sum_j \partial_j Z_j.$$

Use the second-order energy with weight $M(\theta)$, $\mathcal{M}_2 = \sum_{|\alpha| \leq 2} \int (\partial^\alpha a)^\top M \partial^\alpha a \, dX$. Uniform bounds for M_t and $J_{j,X}$ require only $\theta_t \in H^2$ and $\theta \in H^3$. The worst commutator is $\partial^2 A \partial a$, estimated in $L^6 \times L^3$. The H^2 norm of $M^{-1} \operatorname{div} Z$ is controlled by $C\|Z\|_{H^3}$. Hence

$$\mathcal{M}_2' \leq C \mathcal{M}_2 + C \sqrt{\mathcal{M}_2} \|Z\|_{H^3}.$$

Integrate the inequality for $\sqrt{\mathcal{M}_2 + \delta^2}$ and let $\delta \downarrow 0$. Using $a(0) = 0$ and (4.58), we obtain $\|a\|_{L_t^\infty H^2} \leq C_T/c$, which is (4.13). $\square$

Corollary 4.9 (Entropy and dissipation). *The solution in Theorem 4.3 satisfies*

$$\sup_{t \leq T} \int_{\mathbb{T}^3 \times D} [f_c/\bar{N} - 1 - \log(f_c/\bar{N})] \, dX \, dk \leq C_T c^{-3/2}, \quad (4.59)$$

$$\int_0^T \frac{1}{4} \int_{\mathbb{T}^3} \int \prod_{i=0}^3 f_{c,i} [\Delta(1/f_c)]^2 \, d\Xi_D \, dX \, dt \leq C_T c^{-2}. \quad (4.60)$$

Proof. Uniform positive two-sided bounds make the entropy density comparable to $|f_c - \bar{N}|^2$, so the first estimate follows from (4.12). Set $r = (f_c - N_c)/N_c$ and $w = r/(1+r)$. Since $\Delta(1/N_c) = 0$, $\Delta(1/f_c) = -\Delta(w/N_c)$. The fixed two-sided comparison of $\prod f_{c,i}$ with $\prod N_{c,i}$, boundedness of $\mathfrak{a}_N$, and $|w| \leq C|r|$ give

$$\frac{1}{4} \int \prod f_{c,i} [\Delta(1/f_c)]^2 \, d\Xi_D \leq C \|f_c - N_c\|_{H_\nu}^2.$$

The second estimate follows from (4.14). $\square$

Remark 4.10. The convergence topology in the theorem allows the corner trace to persist:

$$f_c(t, X, \xi) = N_{\theta_0(X-2\xi t)}(\xi).$$

This trace generally differs from $\bar{N}(t, X, \xi)$. The corner-layer estimates provide positive two-sided comparison bounds and do not require the trace to vary with c . The time interval of the macroscopic result is determined by the smooth positive lifespan of the prescribed Euler solution.

5 The Onsager tensor and its exact degeneracy

Moment matching separates the macroscopic parameters from the microscopic deviation. To determine the first-order contribution of this deviation to the flux, we project the Euler transport forcing off the null space, solve for its response using the weighted collision form, and take the five moment fluxes. This defines the Onsager tensor for the cube-truncated model. Write $\theta = (\alpha, \beta, \gamma) \in \Theta_+ = \mathcal{O}$, with $\beta \in \mathbb{R}^3$, and $N = (\alpha + \beta \cdot k + \gamma|k|^2)^{-1}$. All uniform statements are made on a fixed compact set $\Theta \Subset \Theta_+$. Reciprocity is a basic feature of Onsager theory [8]; here its definition, null directions and transport properties follow from the four-leg dissipation form of this model.

5.1 The weighted cell problem and transport tensor

Set $\kappa_a = N\Psi_a$, $a = 0, \dots, 4$, and retain the projections P_N, Q_N from Proposition 3.6. These κ_a are the natural generators. With the Gram matrix M in (4.3), the projection is

$$P_N F = (N\Psi)^T M^{-1} \int_D N\Psi F \, dk.$$

This is the same projection as that obtained from the orthonormal basis in Section 3. For $F \in Q_N L^\infty(D)$, write $L_{N,Q}^{-1} F$ for the normalized solution of

$$u \in Q_N H_\nu, \quad \mathfrak{a}_N(u, h) = \langle F, h \rangle \quad (h \in H_\nu). \quad (5.1)$$

The pairing on the right is the unweighted integral pairing.

Lemma 5.1 (Weighted inverse and smooth parameter dependence). *Equation (5.1) has a unique solution, and*

$$\|u\|_{H_\nu} + \|\nu_* u\|_\infty \leq C_\Theta \|F\|_\infty. \quad (5.2)$$

If $F = F_\theta \in Q_N L^\infty$ depends smoothly on θ , then u_θ depends smoothly on θ with values in $\mathcal{Y} = \{u : \nu_* u \in L^\infty\}$. Each derivative of fixed order is uniformly bounded on compact parameter sets. For $F, H \in Q_N L^\infty(D)$, the following pairing is well defined:

$$\langle F, L_{N,Q}^{-1} H \rangle = \mathfrak{a}_N(L_{N,Q}^{-1} F, L_{N,Q}^{-1} H). \quad (5.3)$$

Proof. Proposition 3.9 gives the solution and (5.2). For every $h \in H_\nu$,

$$|\langle F, h \rangle| \leq \|F / \sqrt{\nu_*}\|_2 \|h\|_{H_\nu} \leq C \|F\|_\infty \|h\|_{H_\nu},$$

since $\nu_*^{-1} \in L^1$. The equation initially obtained for bounded test functions therefore extends continuously to all of H_ν . Testing with another weighted solution proves (5.3).

For parameter dependence, use the reference state n_* , set $Q_* = I - P_{n_*}$, and fix the null space with U_N from Lemma 3.11. The operators $\tilde{L}_N = U_N^* L_N U_N$ form a smooth family of bounded isomorphisms between the fixed spaces $Q_* \mathcal{Y}$ and $Q_* \mathcal{X}$. Their inverse norms are uniformly controlled by (5.2). The identity

$$\partial_\theta \tilde{L}_N^{-1} = -\tilde{L}_N^{-1} (\partial_\theta \tilde{L}_N) \tilde{L}_N^{-1}$$

and its successive derivatives give the claim. Conjugating back by U_N returns to the original spaces. The solutions remain in weighted spaces throughout; an unweighted spectral gap is not needed. $\square$

Define the fifteen forcing terms, their responses, and the transport tensor by

$$\begin{aligned} \mathcal{G}_j^a &= Q_N(v_j \kappa_a), & u_j^a &= L_{N,Q}^{-1} \mathcal{G}_j^a, \\ \mathbb{K}_{ij}^{ab}(N) &= \langle \mathcal{G}_i^a, u_j^b \rangle, & 1 \leq i, j \leq 3, & \quad 0 \leq a, b \leq 4. \end{aligned} \quad (5.4)$$

The index j specifies a spatial direction, and a specifies mass, one of the three momentum components, or energy. The fifteen forcing terms thus correspond to variations of five parameters in three spatial directions. The projection removes the invariant component, the weighted inverse gives the microscopic response, and the final pairing extracts its contribution to flux (i, a) . The forcing terms are real and bounded. The five-dimensional Gram formula for the projection and Lemma 5.1 show that the tensor is smooth in θ .

Proposition 5.2 (Reciprocity and nonnegativity). *The tensor (5.4) is a real symmetric positive semidefinite 15×15 matrix:*

$$\mathsf{K}_{ij}^{ab} = \mathsf{K}_{ji}^{ba}, \quad \sum_{i,j,a,b} \xi_i^a \mathsf{K}_{ij}^{ab} \xi_j^b \geq 0 \quad (\xi \in \mathbb{R}^{3 \times 5}). \quad (5.5)$$

Each parameter derivative of fixed order is uniformly bounded on compact positive parameter sets. For a real gradient array ξ , set $F_\xi = \sum_{j,a} \mathcal{G}_j^a \xi_j^a$. Then

$$\xi^\top \mathsf{K}(N) \xi = \max_{h \in Q_N H_\nu} \left\{ 2 \operatorname{Re} \int_D F_\xi \bar{h} \, dk - \mathsf{a}_N(h, h) \right\}. \quad (5.6)$$

The unique maximizer is $L_{N,Q}^{-1} F_\xi$.

Proof. Equation (5.3) gives $\mathsf{K}_{ij}^{ab} = \mathsf{a}_N(u_i^a, u_j^b)$, and hence reciprocity. With $F_\xi = \sum_{j,a} \mathcal{G}_j^a \xi_j^a$ and $u_\xi = L_{N,Q}^{-1} F_\xi$,

$$\sum_{i,j,a,b} \xi_i^a \mathsf{K}_{ij}^{ab} \xi_j^b = \mathsf{a}_N(u_\xi, u_\xi) \geq 0. \quad (5.7)$$

Pairings of parameter derivatives are controlled by (5.2) and $\nu_*^{-1} \in L^1$. Completing the square gives

$$2 \operatorname{Re} \int_D F_\xi \bar{h} \, dk - \mathsf{a}_N(h, h) = \mathsf{a}_N(u_\xi, u_\xi) - \mathsf{a}_N(h - u_\xi, h - u_\xi).$$

The last term is nonnegative and vanishes on $Q_N H_\nu$ exactly when $h = u_\xi$. $\square$

To relate this tensor to Euler transport, let $N = N_{\theta(t,X)}$ be a smooth positive Euler solution and set $p = 1/N = \theta \cdot \Psi$. The five-moment equations give

$$F_E = -\mathcal{T} \log N = N \mathcal{T} p, \quad P_N F_E = 0, \quad F_E = \sum_{j,a} \mathcal{G}_j^a \partial_j \theta_a. \quad (5.8)$$

Indeed, $N \Psi \cdot \theta_t$ belongs to the null space, while $\int \kappa_a F_E = -\int \Psi_a \mathcal{T} N = 0$. Thus, for $z = L_{N,Q}^{-1} F_E$,

$$\int_D v_i \Psi_a N z \, dk = \langle \mathcal{G}_i^a, z \rangle = \sum_{j,b} \mathsf{K}_{ij}^{ab}(N) \partial_j \theta_b. \quad (5.9)$$

This identifies the coefficients and signs of all five first-order fluxes.

5.2 Null directions in the gradient space

Write a gradient array as $\xi = (g, B, h)$, with $g, h \in \mathbb{R}^3$ and $B \in \mathbb{R}^{3 \times 3}$, according to

$$g_j = \xi_j^0, \quad B_{jl} = \xi_j^l \quad (1 \leq l \leq 3), \quad h_j = \xi_j^4.$$

When $\xi = \nabla_X \theta$, these are $g = \nabla_X \alpha$, $B_{jl} = \partial_j \beta_l$ and $h = \nabla_X \gamma$. Set

$$B^\circ = \frac{B + B^T}{2} - \frac{\operatorname{tr} B}{3} I.$$

Theorem 5.3 (Kernel and rank of the transport tensor). *For every positive parameter θ ,*

$$\ker \mathbf{K}(N) = \{(g, B, 0) : g \in \mathbb{R}^3, (B + B^T)/2 = bI \text{ for some } b \in \mathbb{R}\}. \quad (5.10)$$

The kernel has dimension 7 and the tensor has rank 8. Uniformly for $\theta \in \Theta$,

$$c_\Theta(|B^\circ|^2 + |h|^2) \leq \sum_{i,j,a,b} \xi_i^a \mathbf{K}_{ij}^{ab}(N) \xi_j^b \leq C_\Theta(|B^\circ|^2 + |h|^2). \quad (5.11)$$

Proof. By (5.7) and weighted coercivity, the quadratic form vanishes exactly when u_ξ belongs to the null space. The five normalization conditions then give $u_\xi = 0$, equivalently $F_\xi = 0$. Dividing by the positive function N , this condition becomes

$$2g \cdot k + 2k^T B k + 2(h \cdot k)|k|^2 \in \text{span}\{1, k^{(1)}, k^{(2)}, k^{(3)}, |k|^2\}. \quad (5.12)$$

Since the cube has nonempty interior, an almost-everywhere polynomial identity is an identity of coefficients. The cubic terms give $h = 0$, and the quadratic terms give $B^\circ = 0$. The skew-symmetric part does not enter $k^T B k$, and the linear terms are already collision invariants. This proves (5.10) and gives dimension $3 + 3 + 1 = 7$.

For the uniform comparison, F_ξ depends only on the eight effective components (B°, h) . On the compact set $|B^\circ|^2 + |h|^2 = 1$, $\theta \in \Theta$, the five-dimensional projection is continuous in the parameters, and (5.12) excludes zero. Hence $\|F_\xi\|_2 \geq c_\Theta > 0$. Testing the cell equation with the bounded function F_ξ and using Cauchy's inequality for the form yields

$$\|F_\xi\|_2^4 \leq \mathbf{a}_N(u_\xi, u_\xi) \mathbf{a}_N(F_\xi, F_\xi) \leq C_\Theta \mathbf{a}_N(u_\xi, u_\xi) \|F_\xi\|_2^2.$$

This proves the lower bound. Microscopic coercivity and weighted duality give the upper bound:

$$\mathbf{a}_N(u_\xi, u_\xi) \leq C_\Theta \|F_\xi / \sqrt{\nu_*}\|_2^2 \leq C_\Theta(|B^\circ|^2 + |h|^2).$$

All integrals use the cubic domain and its four-leg resonance measure. $\square$

Some pointwise gradient arrays make the null directions explicit. Take $g = h = 0$. If $B = \lambda I$, then $2k^T B k = 2\lambda|k|^2$ is a collision invariant; if $B^T = -B$, then $k^T B k = 0$. Both give zero dissipation. In contrast, $B = \text{diag}(1, -1, 0)$ has a nonzero traceless symmetric part, so (5.11) gives positive dissipation. These are gradient arrays at a single spatial point, not global affine parameter fields on the torus.

The degeneracy concerns gradients of the dual parameter θ . The mass component also satisfies an identity for every individual state. If $f \in L^1(D)$ and N have the same five moments, then

$$\int_D v_i(f - N) dk = 2 \int_D k^{(i)}(f - N) dk = 0, \quad \mathcal{G}_i^0 = 0, \quad \mathbf{K}_{ij}^{0b} = \mathbf{K}_{ij}^{a0} = 0. \quad (5.13)$$

The mass equation remains part of the five-moment system; momentum matching makes its residual flux vanish.

5.3 The spatial Fourier symbol and transport coupling

For $\ell \in \mathbb{R}^3$, define

$$\mathbf{D}_N(\ell)_{ab} = \sum_{i,j} \ell_i \mathbf{K}_{ij}^{ab}(N) \ell_j. \quad (5.14)$$

This matrix is real symmetric and nonnegative. A Fourier amplitude $a = (a_\alpha, a_\beta, a_\gamma)$ gives the array $\xi_j^b = \ell_j a_b$; the common factor i from spatial differentiation cancels in the quadratic form. Thus the fifteen gradient components of a single mode cannot be chosen independently: they must have the form $\ell \otimes a$. Substituting this constraint into (5.10) reduces the seven null directions to the single mass direction.

Proposition 5.4 (The unique null direction at nonzero frequency). *If $\ell \neq 0$, then*

$$\ker D_N(\ell) = \text{span}\{e_{\text{mass}}\}, \quad \text{rank } D_N(\ell) = 4, \quad (5.15)$$

where $e_{\text{mass}} = (1, 0, 0, 0, 0)^T$. *The uniform comparison*

$$c_\Theta |\ell|^2 (|a_\beta|^2 + |a_\gamma|^2) \leq a^* D_N(\ell) a \leq C_\Theta |\ell|^2 (|a_\beta|^2 + |a_\gamma|^2) \quad (5.16)$$

holds also for complex amplitudes.

Proof. Theorem 5.3 gives $a_\gamma = 0$ and $\text{sym}(\ell \otimes a_\beta) = bI$. Contracting with a nonzero $z \perp \ell$ on both sides gives $b = 0$. Contracting with ℓ gives $\ell \cdot a_\beta = 0$, and applying the matrix to ℓ then yields $a_\beta = 0$. The component a_α is arbitrary, proving the kernel statement. For $|\ell| = 1$, take the minimum on the unit sphere of the remaining four-dimensional amplitude space. Equation (5.11) and homogeneity give the uniform comparison. The complex case follows by separating real and imaginary parts. $\square$

Use M, J_j from (4.3). For a unit direction ω , write $J(\omega) = \sum_j \omega_j J_j$ and $e_{\beta \cdot \omega} = (0, \omega_1, \omega_2, \omega_3, 0)^T$. The Gram definitions give

$$J(\omega) e_{\text{mass}} = 2M e_{\beta \cdot \omega}. \quad (5.17)$$

Thus no eigenvector of $M^{-1}J(\omega)$ lies in the null direction of (5.15). This is the Shizuta–Kawashima coupling condition for the five-moment system [10, Theorem 1.1]. We give a direct decay proof.

Our Fourier convention is

$$\widehat{\vartheta}(\ell) = (2\pi)^{-3} \int_{\mathbb{T}^3} \vartheta(X) e^{-i\ell \cdot X} dX, \quad \ell \in \mathbb{Z}^3.$$

For a fixed positive matrix M , define

$$\|\vartheta\|_{H^s, M}^2 = (2\pi)^3 \sum_{\ell \in \mathbb{Z}^3} (1 + |\ell|^2)^s \widehat{\vartheta}(\ell)^* M \widehat{\vartheta}(\ell).$$

For $s = 0$, Parseval's identity identifies this with $\int_{\mathbb{T}^3} \vartheta^* M \vartheta dX$. For general s it is equivalent to the usual H^s norm, uniformly on the compact parameter set under consideration.

Proposition 5.5 (Fourier decay of the constant-parameter system). *Fix $\theta \in \Theta$ and consider on $\mathbb{T}^3$ the constant-coefficient system*

$$M \partial_t \vartheta + \sum_i J_i \partial_i \vartheta = c^{-1} \sum_{i,j} K_{ij} \partial_i \partial_j \vartheta, \quad c > 0. \quad (5.18)$$

For every $\ell \in \mathbb{Z}^3 \setminus \{0\}$,

$$\|\widehat{\vartheta}(t, \ell)\|_M \leq C_\Theta \exp\left(-a_\Theta \frac{c|\ell|^2}{c^2 + |\ell|^2} t\right) \|\widehat{\vartheta}(0, \ell)\|_M, \quad \|z\|_M^2 = z^* M z. \quad (5.19)$$

All five zero-frequency components are constant. For $c \geq 1$ and mean-zero data, the system also satisfies, on every $H^s(\mathbb{T}^3; \mathbb{R}^5)$,

$$\|\vartheta(t)\|_{H^s, M} \leq C_\Theta e^{-a_\Theta t/c} \|\vartheta(0)\|_{H^s, M}. \quad (5.20)$$

Proof. Set $r = |\ell| > 0$, $\omega = \ell/r$, and

$$A = M^{-1/2} J(\omega) M^{-1/2}, \quad B = M^{-1/2} D_N(\omega) M^{-1/2}, \quad e = \frac{M^{1/2} e_{\text{mass}}}{\sqrt{M_{\alpha\alpha}}}, \quad b = (I - ee^T) A e.$$

Both matrices are real symmetric, $Be = 0$, and $B \geq a_0(I - ee^T)$, $\|A\| + \|B\| \leq C_0$. By (5.17),

$$|b|^2 = \frac{4}{M_{\alpha\alpha}} \left(e_{\beta \cdot \omega}^T M e_{\beta \cdot \omega} - \frac{|e_{\text{mass}}^T M e_{\beta \cdot \omega}|^2}{M_{\alpha\alpha}} \right) \geq a_0 > 0.$$

The expression in parentheses is a Schur complement of a positive Gram matrix, so the lower bound is uniform on $\Theta \times \mathbb{S}^2$.

Let $S = be^T - eb^T$. This is a real skew-symmetric matrix and $e^T[S, A]e = -2|b|^2$. For a fixed sufficiently small $\eta > 0$, set

$$d = r/c, \quad \alpha_d = \frac{\eta d}{1 + d^2}, \quad H = I + i\alpha_d S.$$

Then H is Hermitian and $I/2 \leq H \leq 3I/2$. For $y = M^{1/2}\hat{\vartheta}$, the equation reads $y' = \mathcal{G}y$, with $\mathcal{G} = -irA - c^{-1}r^2B$. Direct multiplication gives

$$H\mathcal{G} + \mathcal{G}^*H = -2c^{-1}r^2B + \alpha_d r[S, A] - i\alpha_d c^{-1}r^2(SB + BS). \quad (5.21)$$

Write $y = ze + q$, $q \perp e$. Since $Be = 0$, the matrix $SB + BS$ has only cross blocks between e and $e^\perp$. Hence $E = y^*Hy$ satisfies

$$E' \leq -a_1 c^{-1} r^2 |q|^2 - a_1 \alpha_d r |z|^2 + C \alpha_d r |q|^2 + C \alpha_d r (1 + d) |z| |q|.$$

Young's inequality absorbs the mixed term using half of $\alpha_d r |z|^2$, at the cost of at most

$$C \alpha_d r (1 + d^2) |q|^2 = C \eta c^{-1} r^2 |q|^2.$$

Choosing η sufficiently small yields

$$E' \leq -a_2 \{c^{-1} r^2 |q|^2 + \alpha_d r |z|^2\} \leq -a_3 \frac{cr^2}{c^2 + r^2} E.$$

Integration and the comparison of H with I give (5.19). At $\ell = 0$, the matrix equation is $M\partial_t \hat{\vartheta} = 0$. If $c \geq 1$ and $|\ell| \geq 1$, then $c|\ell|^2/(c^2 + |\ell|^2) \geq 1/(2c)$. Summing over Fourier modes gives (5.20). Density of finite Fourier sums and continuity of each mode also give a strongly continuous semigroup on H^s . $\square$

The tensor in this section uses the c normalization in (4.11). With $\varepsilon = 4\pi/c$, the tensor multiplying the first-order parameter ε is $\mathbf{K}/(4\pi)$:

$$c^{-1}\mathbf{K} = \varepsilon \frac{\mathbf{K}}{4\pi}.$$

6 Strong resonance currents and first-order transport of all five moments

The preceding section defined the transport coefficients through a weighted cell problem. We now identify them for the solution family of Theorem 4.3. The key quantity is the microscopic entropy isolated in Proposition 4.1. Its exact time balance relates collision dissipation to the residual flux and determines the norm of the resonance current after its weak limit has been found. We retain the same initial data and prescribed Euler interval throughout. To fix the limiting measure, this section uses the local abbreviation $N = \bar{N} = N_{\hat{\theta}}$; the actual moment-matched equilibrium is always denoted by $N_c = N_{\theta_c}$. Set

$$d_c = \frac{f_c}{N_c}, \quad w_c = 1 - d_c^{-1}, \quad z_c = cw_c, \quad F_E = -\mathcal{T} \log N. \quad (6.1)$$

The variable w_c measures the relative deviation of the reciprocal distribution, since $1/f_c = (1 - w_c)/N_c$. It gives $\Delta(1/f_c) = -\Delta(w_c/N_c)$, while the factor c extracts the first-order microscopic response. By (5.8), $F_E \in Q_N L^\infty(D)$. Use the weighted inverse of Lemma 5.1 to define

$$L_N u = F_E, \quad \int_D N \Psi u \, dk = 0. \quad (6.2)$$

This cell problem depends on (t, X) , and all five normalization integrals are well defined. The tensor K below has the c normalization in (5.4). Write

$$\mathcal{H} = L^2((0, T) \times \mathbb{T}^3; H_\nu), \quad \mathcal{B} = L^2((0, T) \times \mathbb{T}^3; L^1(D)).$$

Theorem 6.1 (Strong limits of resonance currents and moment fluxes). *Under the assumptions of Theorem 4.3, the following limits hold on the entire prescribed interval $(0, T)$:*

$$z_c \longrightarrow u \quad \text{strongly in } \mathcal{H}, \quad (6.3)$$

$$c(f_c - N_c) \longrightarrow Nu \quad \text{strongly in } \mathcal{H} \cap \mathcal{B}. \quad (6.4)$$

Define the signed four-leg resonance current

$$\mathcal{J}_c = \frac{c}{2} \sqrt{\prod_{i=0}^3 f_{c,i}} \Delta(1/f_c).$$

Then

$$\mathcal{J}_c \longrightarrow -\frac{1}{2} \sqrt{\prod_{i=0}^3 N_i} \Delta(u/N) \quad \text{strongly in } L^2(dt \, dX \, d\Xi_D), \quad (6.5)$$

and the physical collision term satisfies

$$c\mathcal{C}(f_c) \longrightarrow \mathcal{T}N \quad \text{in } L^2((0, T) \times \mathbb{T}^3 \times D). \quad (6.6)$$

For $i = 1, 2, 3$ and $a = 0, \dots, 4$, set

$$Z_{c,i}^a = \int_D v_i \Psi_a(f_c - N_c) \, dk.$$

All fifteen components satisfy

$$cZ_{c,i}^a \longrightarrow \sum_{j=1}^3 \sum_{b=0}^4 K_{ij}^{ab}(N) \partial_j \bar{\theta}_b \quad \text{strongly in } L^2((0, T) \times \mathbb{T}^3). \quad (6.7)$$

Consequently, the actual five-moment parameter obeys the first-order law

$$\begin{aligned} M(\theta_c) \partial_t \theta_c + \sum_i J_i(\theta_c) \partial_i \theta_c \\ = \frac{1}{c} \sum_{i,j} \partial_i [K_{ij}(N_c) \partial_j \theta_c] + o(c^{-1}). \end{aligned} \quad (6.8)$$

The remainder belongs to $L^2(0, T; H^{-1}(\mathbb{T}^3; \mathbb{R}^5))$, with norm $o(c^{-1})$.

The proof has five steps. Theorem 4.3 first gives weighted microscopic bounds. The kinetic equation then identifies the weak limit on a fixed resonance measure. The entropy balance determines the current norm, so weak convergence becomes strong convergence. An exact Gram equation from moment matching recovers the five-dimensional null component left undetermined by dissipation. Finally, the adjoint of the collision-difference operator returns the physical collision term, and integration gives all five moment fluxes. Ratios in the corner layer need only fixed positive upper and lower bounds.

6.1 Actual moment matching and the entropy identity

Theorem 4.3 and the decomposition $f_c - N_c = e_c + G_c$ in its proof give positive constants d_-, d_+ , independent of c , such that

$$d_- \leq d_c \leq d_+, \quad \|\theta_c - \bar{\theta}\|_{L_t^\infty H_X^2} \leq Cc^{-1}, \quad (6.9)$$

$$\|f_c - N_c\|_{L_t^\infty L_{X,k}^2} \leq Cc^{-3/4}, \quad \|f_c - N_c\|_{L_{t,X}^2 H_\nu} \leq Cc^{-1}. \quad (6.10)$$

The three-dimensional embedding $H_X^2 \hookrightarrow C_X^0$ yields uniform convergence $N_c/N \rightarrow 1$, and $\nabla_X \theta_c \rightarrow \nabla_X \bar{\theta}$ strongly in $L_{t,X}^2$. All parameters remain in a common compact subset of the positive domain.

Define

$$H_c(t) = \int_{\mathbb{T}^3} \mathcal{H}(f_c | N_c) dX = \int_{\mathbb{T}^3 \times D} \Phi(d_c) dX dk, \quad \Phi(d) = d - 1 - \log d, \quad (6.11)$$

$$D_c(t) = \frac{1}{4} \int_{\mathbb{T}^3} \int \prod_i f_{c,i} [\Delta(1/f_c)]^2 d\Xi_D dX. \quad (6.12)$$

The ratio bounds and (6.10) imply

$$\sup_{t \leq T} H_c(t) \leq Cc^{-3/2}, \quad \|z_c\|_{L_{t,X}^2 H_\nu} \leq C, \quad \sup_{t \leq T} c \int_{X,k} |w_c|^2 \leq Cc^{-1/2}. \quad (6.13)$$

Applying (4.6) to the actual moment match gives

$$\int_{\mathbb{T}^3} \mathcal{H}(f_c | N) dX = H_c(t) + \int_{\mathbb{T}^3} \mathcal{H}(N_c | N) dX. \quad (6.14)$$

By (4.8) and (6.9), the second term is uniformly $O(c^{-2})$. We use the balance for H_c itself to retain the exact relation between microscopic dissipation and the actual moment fluxes. In particular, w_c is uniformly bounded and tends to zero in volume measure. The difference between the two microscopic coordinates satisfies

$$c[(d_c - 1) - w_c] = cd_c w_c^2 \longrightarrow 0 \quad \text{in } L_t^\infty L_{X,k}^1. \quad (6.15)$$

All entropy balances below are integrated from the given initial time. Moment matching gives $\int_D \Psi(f_c - N_c) dk = 0$, and hence

$$H_c = \int_{X,k} -\log f_c - \int_{X,k} -\log N_c.$$

The four-leg symmetry of (3.4) and periodic transport give $-cD_c$ for the time derivative of the first term. For the second, set $\mathcal{E}(U(\theta)) = \int_D -\log N_\theta dk$. Equation (4.7) gives $\nabla_U \mathcal{E} = -\theta$. The moment equation for the same solution is

$$\partial_t U(\theta_c) + \sum_i \partial_i \mathcal{F}_i(\theta_c) = - \sum_i \partial_i Z_{c,i}. \quad (6.16)$$

The equilibrium entropy flux is $\int_D -v_i \log N_\theta dk$, whose divergence integrates to zero on the torus. Therefore

$$\frac{d}{dt} \int_X \mathcal{E}(U(\theta_c)) = - \sum_i \int_X \partial_i \theta_c \cdot Z_{c,i}.$$

Combining the two derivatives gives

$$H'_c + cD_c = \sum_i \int_X \partial_i \theta_c \cdot Z_{c,i}, \quad H_c(0) = 0. \quad (6.17)$$

The positive classical solution of Theorem 4.3 justifies every differentiation. Passing this identity to the limit will use only the parameter and spatial-gradient convergence in (6.9); time derivatives of the actual parameters have already been eliminated by the exact calculation.

Lemma 6.2 (Strong convergence of bounded multipliers). *Let (Ω, λ) be a finite measure space. Suppose $b_n \rightarrow b$ in λ -measure and $|b_n|, |b| \leq C$. If $\sigma \ll \lambda$ is a finite positive measure, then for every fixed $V \in L^2(\sigma)$,*

$$\|(b_n - b)V\|_{L^2(\sigma)} \rightarrow 0.$$

Proof. For $\delta > 0$, let $A_n = \{|b_n - b| > \delta\}$. The finite measure $|V|^2 \sigma$ is absolutely continuous with respect to λ , and $\lambda(A_n) \rightarrow 0$. Thus

$$\|(b_n - b)V\|_{L^2(\sigma)}^2 \leq \delta^2 \|V\|_{L^2(\sigma)}^2 + 4C^2 \int_{A_n} |V|^2 d\sigma.$$

Let $n \rightarrow \infty$ and then $\delta \downarrow 0$. □

6.2 A common resonance measure and the limiting weak equation

On the Hilbert space $\mathcal{H}$ defined above, fix the measure and operator

$$d\mu = \prod_{i=0}^3 N_i dt dX d\Xi_D, \quad R_N h = \frac{1}{2} \Delta(h/N). \quad (6.18)$$

Here N is the fixed limiting distribution, which still varies with (t, X) . The common measure places every resonance current in the same Hilbert space. Changes in the actual distributions will appear as explicit multipliers. Four-leg symmetry gives the exact marginal identity

$$\int |b(t, X, k_i)|^2 d\mu = \int_{t, X, k} N^2 \nu_N |b|^2 dt dX dk, \quad i = 0, 1, 2, 3. \quad (6.19)$$

Consequently $R_N : \mathcal{H} \rightarrow L^2(\mu)$ is bounded. The identity also shows that changing a momentum function on a volume null set does not alter the μ -equivalence class of any leg. Thus the operator is defined on the whole weighted Hilbert space.

Set

$$y_c = \frac{N}{N_c} z_c, \quad m_c = \prod_{i=0}^3 \frac{f_{c,i}}{N_i}. \quad (6.20)$$

The variable y_c replaces the matched denominator by the reference denominator, so $y_c/N = z_c/N_c$. The multiplier m_c records the full change of the four-leg product relative to the reference measure. Each $f_c/N - 1$ tends to zero in volume L^2 and is uniformly bounded. Applying (6.19) and subtracting the four factors in turn gives

$$m_c \rightarrow 1, \quad \sqrt{m_c} \rightarrow 1 \quad \text{in } L^2(\mu), \quad 0 < m_- \leq m_c \leq m_+ < \infty. \quad (6.21)$$

All four legs come from one distribution at the same (t, X) . Equation (6.19) transfers volume convergence to the resonance measure. Moreover, for every fixed $V \in L^2(\mu)$,

$$(m_c - 1)V \rightarrow 0, \quad (\sqrt{m_c} - 1)V \rightarrow 0 \quad \text{in } L^2(\mu). \quad (6.22)$$

This is Lemma 6.2 with $\lambda = \sigma = \mu$. It places the multiplier on the fixed test function, allowing the microscopic vector on the other side to converge only weakly.

By (6.13), extract a subsequence $z_c \rightharpoonup u_0$ in $\mathcal{H}$. Uniform macroscopic convergence also gives $y_c \rightharpoonup u_0$. Since $\nu_*^{-1} \in L^1(D)$, there is a continuous embedding

$$\mathcal{H} \hookrightarrow \mathcal{B}. \quad (6.23)$$

Apply (6.15) to the identity $\int N_c \Psi d_c z_c dk = 0$ and let $c \rightarrow \infty$. This yields

$$\int_D N \Psi u_0 dk = 0 \quad \text{for almost every } (t, X). \quad (6.24)$$

More explicitly, first pair with an arbitrary smooth space-time function. Then use uniform convergence $N_c \rightarrow N$ and the admissible weighted dual test $N \Psi / \sqrt{\nu_*}$.

Take a real test function $\psi(t, X, k) = \chi(t, X) b(k)$, where χ is smooth with time support in $(0, T)$ and b is real and bounded. Test the kinetic equation with the fixed function ψ/N . Since $\Delta(1/N_c) = 0$, its four-leg weak form gives exactly

$$\int c \mathcal{C}(f_c) \frac{\psi}{N} dt dX dk = - \int m_c R_N y_c R_N \psi d\mu. \quad (6.25)$$

On the left, substitute $\mathcal{T} f_c = c \mathcal{C}(f_c)$ and integrate by parts. Strong volume L^2 convergence of f_c gives the limit $\int (\mathcal{T} N) \psi / N$; only χ and the fixed N are differentiated. By (6.22) and weak convergence, the right-hand side tends to $-\int R_N u_0 R_N \psi d\mu$. Using a countable dense family of bounded functions b gives the almost-everywhere variational equation

$$\mathbf{a}_N(u_0, b) = \int_D F_E b dk.$$

As $F_E / \sqrt{\nu_*} \in L^2$, continuity of the form extends the equation to every real test in H_ν , and then to complex tests by the sesquilinear convention. Equation (6.24) and weighted coercivity imply $u_0 = u$, with u defined by (6.2). Every weakly convergent subsequence therefore has the same limit.

6.3 Norm convergence and recovery of the five-dimensional kernel

By (6.12) and $\Delta(w_c/N_c) = -\Delta(1/f_c)$,

$$c^2 \int_0^T D_c(t) dt = \|\sqrt{m_c} R_N y_c\|_{L^2(\mu)}^2. \quad (6.26)$$

Multiply (6.17) by c and integrate from the initial time to T :

$$c H_c(T) + c^2 \int_0^T D_c = \int_{t,X,k} z_c F_c, \quad F_c = N_c d_c(v \cdot \nabla_X p_c), \quad p_c = N_c^{-1} = \theta_c \cdot \Psi. \quad (6.27)$$

The first term tends to zero by (6.13). Set $p = N^{-1} = \bar{\theta} \cdot \Psi$ and $F = N(v \cdot \nabla_X p)$. Then

$$\begin{aligned} F_c - F &= N_c d_c v \cdot \nabla_X (p_c - p) + (N_c - N) d_c (v \cdot \nabla_X p) \\ &\quad + N(d_c - 1)(v \cdot \nabla_X p). \end{aligned} \quad (6.28)$$

After division by $\sqrt{\nu_*}$, the first term is controlled by $\|\nabla_X \theta_c - \nabla_X \bar{\theta}\|_{L_{t,X}^2}$ and the second by uniform macroscopic convergence. In the third term, $d_c - 1$ is uniformly bounded and tends to zero in volume measure, while the fixed function $|N(v \cdot \nabla_X p)|^2 / \nu_*$ is integrable. Lemma 6.2, applied with volume measure and this density, gives

$$F_c / \sqrt{\nu_*} \longrightarrow F / \sqrt{\nu_*} \quad \text{strongly in } L_{t,X,k}^2. \quad (6.29)$$

The right-hand side of (6.27) therefore tends to $\int u F$. The five moment normalizations allow us to add $N \partial_t p$, so

$$\int u F = \int u F_E = \int_0^T \int_X \mathbf{a}_N(u, u) = \|R_N u\|_{L^2(\mu)}^2.$$

On the other hand, (6.22) gives $\sqrt{m_c}R_N y_c \rightharpoonup R_N u$. Equations (6.26)–(6.27) give convergence of the norms. Hence

$$\sqrt{m_c}R_N y_c \longrightarrow R_N u \quad \text{strongly in } L^2(\mu). \quad (6.30)$$

Removing the density in (6.18) gives (6.5). Its sign follows from $\Delta(w_c/N_c) = -\Delta(1/f_c)$.

Strong convergence of the resonance difference controls only the complement of the null space: adding $N\Psi \cdot \zeta^{\text{ker}}(t, X)$ to y_c leaves the difference unchanged. The null component must therefore be recovered from actual moment matching. The positive lower bound on m_c and (6.22), applied to the fixed vector $R_N u$, first yield $R_N(y_c - u) \rightarrow 0$ strongly in $L^2(\mu)$. Write $\kappa = N\Psi$ and $M_N = \int_D \kappa \kappa^T dk$. The extension of the unweighted projection is

$$P_N h = \kappa^T M_N^{-1} \int_D \kappa h dk.$$

It is bounded on H_ν because $\nu_*^{-1} \in L^1$. Decompose

$$y_c - u = g_c + \kappa^T \zeta_c^{\text{ker}}, \quad g_c = Q_N(y_c - u).$$

The identity $\mathfrak{a}_N(Q_N h, Q_N h) = \mathfrak{a}_N(h, h)$ and (3.17) imply $g_c \rightarrow 0$ strongly in $\mathcal{H}$. The projection here is the extension of the unweighted one.

Actual moment matching gives

$$\int_D \kappa b_c y_c dk = 0, \quad b_c = d_c(N_c/N)^2, \quad (6.31)$$

and consequently

$$\left(\int_D b_c \kappa \kappa^T dk \right) \zeta_c^{\text{ker}} = - \int_D \kappa b_c g_c dk - \int_D \kappa (b_c - 1) u dk. \quad (6.32)$$

The matrix on the left is uniformly positive, since $\int b_c \kappa \kappa^T \geq (\inf b_c) M_N$. The uniform L^2 bound on $\kappa/\sqrt{\nu_*}$ gives, at each (t, X) ,

$$|\zeta_c^{\text{ker}}| \leq C(\|g_c\|_{H_\nu} + \|(b_c - 1)u\|_{H_\nu}). \quad (6.33)$$

Since b_c is uniformly bounded and tends to one in volume measure, Lemma 6.2 gives $\|(b_c - 1)u\|_{\mathcal{H}} \rightarrow 0$. Thus $\zeta_c^{\text{ker}} \rightarrow 0$ in $L^2_{t,X}$ and $y_c \rightarrow u$ strongly in $\mathcal{H}$. Finally,

$$z_c - u = (N_c/N)(y_c - u) + (N_c/N - 1)u$$

proves (6.3). Decomposing $c(f_c - N_c) = N_c d_c z_c$ into a strongly convergent difference and a bounded multiplier acting on the fixed u gives strong convergence in $\mathcal{H}$. The embedding (6.23) then gives (6.4).

6.4 The physical collision term and the first-order macroscopic equation

On the physical space $L^2_{t,X,k}$, define $S_N \varphi = \frac{1}{2} \Delta \varphi$ with values in $L^2(\mu)$. Equation (6.19) gives

$$\|S_N \varphi\|_{L^2(\mu)}^2 \leq 4 \int_{t,X,k} N^2 \nu_N |\varphi|^2 \leq C \|\varphi\|_{L^2_{t,X,k}}^2. \quad (6.34)$$

Its Hilbert adjoint $S_N^* : L^2(\mu) \rightarrow L^2_{t,X,k}$ is therefore bounded. The four-leg weak identity is

$$c\mathcal{C}(f_c) = -S_N^*[m_c R_N y_c]. \quad (6.35)$$

Equation (6.30) and multiplier convergence on the fixed limit give $m_c R_N y_c \rightarrow R_N u$ strongly in $L^2(\mu)$. Hence

$$c\mathcal{C}(f_c) \longrightarrow -S_N^* R_N u = -N L_N u = \mathcal{T}N,$$

which proves (6.6).

Every $v_i \Psi_a$ is bounded on D , so (6.4) gives the strong limits of all five moment fluxes simultaneously. By (5.9),

$$\int_D v_i \Psi_a N u \, dk = \sum_{j,b} K_{ij}^{ab}(N) \partial_j \bar{\theta}_b,$$

proving (6.7). Smoothness of the tensor on compact positive parameter sets and (6.9) also give

$$K(N_c) \nabla_X \theta_c - K(N) \nabla_X \bar{\theta} \longrightarrow 0 \quad \text{strongly in } L_{t,X}^2. \quad (6.36)$$

The bound is the uniform parameter difference times the norm of the fixed $\nabla_X \bar{\theta}$, plus $C \|\nabla_X \theta_c - \nabla_X \bar{\theta}\|_2$. Write the flux difference as

$$Z_{c,i} = c^{-1} \sum_j K_{ij}(N_c) \partial_j \theta_c + \mathfrak{R}_{c,i}^{\text{flux}}, \quad \|\mathfrak{R}_c^{\text{flux}}\|_{L_{t,X}^2} = o(c^{-1}).$$

Denote its spatial divergence by $\mathfrak{R}_c^{\text{mac}} = \sum_i \partial_i \mathfrak{R}_{c,i}^{\text{flux}}$. Then $\|\mathfrak{R}_c^{\text{mac}}\|_{L_t^2 H_X^{-1}} = o(c^{-1})$. Substitute into (6.16) and use $DU = -M$, $D\mathcal{F}_i = -J_i$ to obtain (6.8); spatial divergence maps L_X^2 continuously into H_X^{-1} . In conserved variables, the same law reads

$$\partial_t U(\theta_c) + \sum_i \partial_i \mathcal{F}_i(\theta_c) = -c^{-1} \sum_{i,j} \partial_i [K_{ij}(N_c) \partial_j \theta_c] - \mathfrak{R}_c^{\text{mac}} \quad \text{in } L_t^2 H_X^{-1}.$$

The mass row also satisfies $Z_{c,i}^0 = 2 \int k^{(i)} (f_c - N_c) \, dk = 0$, in agreement with the zero mass row of the tensor. This completes the proof of Theorem 6.1.

Corollary 6.3 (Macroscopic entropy and first-order dissipation). *Define the equilibrium entropy and entropy flux by*

$$\mathcal{E}(U(\theta)) = - \int_D \log N_\theta \, dk, \quad \mathcal{Q}_i(\theta) = - \int_D v_i \log N_\theta \, dk.$$

Under the assumptions of Theorem 6.1,

$$c^2 D_c(t) \longrightarrow \int_{\mathbb{T}^3} \sum_{i,j} (\partial_i \bar{\theta})^\top K_{ij}(N) \partial_j \bar{\theta} \, dX \quad \text{in } L^1(0, T). \quad (6.37)$$

The actual parameter also satisfies the local entropy law

$$\begin{aligned} \partial_t \mathcal{E}(U(\theta_c)) + \sum_i \partial_i [\mathcal{Q}_i(\theta_c) - \theta_c \cdot Z_{c,i}] \\ = -\frac{1}{c} \sum_{i,j} (\partial_i \theta_c)^\top K_{ij}(N_c) \partial_j \theta_c + r_c^\mathcal{E}, \quad \|r_c^\mathcal{E}\|_{L_{t,X}^1} = o(c^{-1}). \end{aligned} \quad (6.38)$$

Proof. Set $J_c^\mu = \sqrt{m_c} R_N y_c$ and $J^\mu = R_N u$. Strong convergence of the resonance current and Cauchy's inequality give

$$\| |J_c^\mu|^2 - |J^\mu|^2 \|_{L^1(\mu)} \leq (\|J_c^\mu\|_2 + \|J^\mu\|_2) \|J_c^\mu - J^\mu\|_2 \longrightarrow 0.$$

Integrate over (X, k_0, k_1, k_2, k_3) and use the fixed-time version of (6.26), together with (5.7) and (5.8), to obtain (6.37).

Since $\nabla_U \mathcal{E} = -\theta$ and $D_\theta Q_i = \mathcal{F}_i^\top$, the equilibrium flux satisfies $\theta \cdot \partial_i \mathcal{F}_i = -\partial_i Q_i$. Multiply (6.16) by $-\theta_c$ and apply the chain rule:

$$\partial_i \mathcal{E}(U(\theta_c)) + \sum_i \partial_i [Q_i(\theta_c) - \theta_c \cdot Z_{c,i}] = - \sum_i \partial_i \theta_c \cdot Z_{c,i}.$$

Substitute the strong flux approximation already proved. The remainder satisfies

$$\|r_c^\mathcal{E}\|_{L_{t,X}^1} \leq \|\nabla_X \theta_c\|_{L_{t,X}^2} \|Z_c - c^{-1} K(N_c) \nabla_X \theta_c\|_{L_{t,X}^2} = o(c^{-1}).$$

This proves (6.38). Proposition 5.2 gives nonnegativity of the leading dissipation. $\square$

Remark 6.4 (Topology, initial layer and normalization). The time domain of the strong convergence is $(0, T)$, and its norms cover the entire prescribed interval. At the initial time $z_c(0) = 0$, whereas the solution of (6.2) can be nonzero there. This is why the conclusions use time-integrated topologies. The momentum norm retains the degenerate weight in H_ν ; the cell solution need not belong to unweighted $L^2(D)$. The free-transport corner traces of Theorem 4.3 are preserved. Equation (6.8) describes the first-order flux of the actual solution. A limit for $c(\theta_c - \bar{\theta})$ requires further analysis. Under $c = 4\pi/\varepsilon$, the tensor coefficient becomes $\varepsilon K/(4\pi)$. No rate for strong convergence of the resonance current is asserted here.

7 Structural relations and further questions

Both models begin with the full collision difference, but resonance compatibility enters at different stages. In the pinned chain, it relates values at different points, gives regularity of measurable solutions, and determines all invariants through periodic matching. At critical resonances, the same geometry has another consequence: the energy mismatch and the function difference share a vanishing factor, allowing the singular measure and the squared difference to be treated together. Kernel classification, compact resonance averaging and construction of the closed form thus form a connected argument.

In the three-dimensional model, collision invariants determine both the equilibrium family and the error decomposition. Actual five-moment matching cancels the cross term in relative entropy. Collisions control deviations that leave these moments unchanged, while transport makes the moments vary in space. The residual flux joins these effects in the microscopic entropy balance (6.17). That balance determines the strong resonance-current limit and then the first-order moment fluxes. The Onsager tensor defined by the weighted cell problem therefore also appears in the limit of microscopic entropy production and in the local macroscopic entropy law.

Corner memory specifies the topology of this macroscopic description. The free-transport trace (1.14) persists throughout the existence interval, while the distribution approaches the Euler equilibrium in momentum-integrated norms. The estimates control the contribution of the corner layer to total dissipation and flux; pointwise relaxation at the corners does not hold. The first-order microscopic quantities and resonance currents converge in weighted, time-integrated spaces. Their strong traces at the initial time require an initial-layer analysis.

Several further questions arise. For the pinned chain, a spatial diffusion limit requires simultaneous rescaling of transport and collisions, together with flux and nonlinear stability estimates for the two conserved densities. The nondegeneracy constants of the resonance charts change near the acoustic endpoint $d = 1/2$ and the zero-coupling endpoint $d = 0$; uniform estimates there need separate analysis.

For the three-dimensional model, the first-order law describes the five-moment parameters of the actual kinetic solution. Comparing them with an independently solved nonlinear viscous macroscopic system requires stability of that system and initial-layer estimates. Enlarging the momentum domain

requires control of the equilibrium, collision frequency and transport coefficients as the cutoff radius R grows. Finally, connecting this strong-collision limit to inhomogeneous Schrödinger dynamics requires a kinetic derivation matched to local RJ initial data, with uniform control of the weak nonlinearity, large-volume and strong-collision parameters.

Companion code. The manuscript and Lean sources are available in the companion repository:

<https://github.com/veridiscoversLab/resonance-kinetic-lean>.

The repository documents the theorem statements, their formal entry points and the verification procedure. The pinned-chain classification, spectral gap and closed realization, the Onsager kernel and rank, and the constant-parameter five-moment decay have completed end-to-end formalizations. Formalization of the nonlinear limit over the prescribed Euler interval and the final strong-current theorem remains in progress; their paper proofs are given above.

Notation index

This index collects objects defined in the text. The one-dimensional model uses $x \in \mathbb{T}_{2\pi}$, a fixed parameter $0 < d < 1/2$, and $\omega_d(x) = \sqrt{1 - 2d \cos x}$. In that section, Ξ is the coarea measure on the regular resonance surface and $m(dx) = dx/(2\pi)$ is normalized circle measure. The table below mainly concerns the three-dimensional model.

Notation	Meaning and convention
$k \in D = [-R, R]^3, X \in \mathbb{T}^3$	$R > 0$ is fixed; $k^{(j)}$ is a momentum component and k_i a collision leg. The velocity is $v(k) = 2k$ and $\mathbb{T}^3 = \mathbb{T}_{2\pi}^3$. The measures dX and dk are periodic and ordinary Lebesgue measure, respectively.
$\Delta\varphi, \Xi_D, \Gamma_k$	$\Delta\varphi = \varphi_0 + \varphi_1 - \varphi_2 - \varphi_3$, $\varphi_i = \varphi(k_i)$. The four-leg resonance measure is Ξ_D ; Γ_k is its fibre measure with output leg $k_0 = k$ fixed.
$\mathcal{T}, c = 4\pi/\varepsilon$	$\mathcal{T} = \partial_t + v \cdot \nabla_X$. The kinetic equation is $\mathcal{T}f_c = c\mathcal{C}(f_c)$, and the limit is $c \rightarrow \infty$.
$\Psi, \Theta_+ = \mathcal{O}$	$\Psi = (1, k^{(1)}, k^{(2)}, k^{(3)}, k ^2)^\top$; $\Theta_+ = \{\theta : \min_D \theta \cdot \Psi > 0\}$ is the positive parameter domain.
$N_\theta, \bar{N}, N_c$	$N_\theta = (\theta \cdot \Psi)^{-1}$. The prescribed Euler equilibrium is $\bar{N} = N_{\bar{\theta}}$; $N_c = N_{\theta_c}$ matches all five moments of f_c . Each local abbreviation N is specified where it is used.
$\kappa_N = N\Psi, P_N, Q_N$	P_N is the unweighted L^2 orthogonal projection onto $\text{span}\{N\Psi_a\}_{a=0}^4$, extended to the weighted space; $Q_N = I - P_N$.
$\mathcal{H}(f N), H_c(t)$	Relative entropy is defined in (1.9). Here $H_c = \int_{\mathbb{T}^3} \mathcal{H}(f_c N_c) dX$. Pointwise moment matching gives the exact decomposition (4.6).
ν_*, H_ν	ν_* is the collision frequency of a fixed positive RJ state, and $H_\nu = L^2(D, \nu_* dk)$. On compact positive parameter sets, $\nu_N \asymp \nu_*$.
$Z, r(k), E_\eta(\xi)$	$Z = \{\pm R\}^3$ is the corner set and $r(k) = \text{dist}_\infty(k, Z)$. Set $E_\eta(\xi) = \{k \in D : \ k - \xi\ _\infty \leq \eta\}$ and $E_\eta = \bigcup_{\xi \in Z} E_\eta(\xi)$.
$\mathcal{X}, \mathcal{Y}$	$\mathcal{X} = L^\infty(D)$ and $\mathcal{Y} = \{h : \nu_* h \in L^\infty(D)\}$, with norms $\ h\ _\infty$ and $\ \nu_* h\ _\infty$.
$L_N, \mathfrak{a}_N$	$L_N = -N^{-1}DC(N)N$. The collision form $\mathfrak{a}_N$ is (3.6); $L_{N,Q}^{-1}$ is its weighted inverse with five moment normalizations.
q_c, G_c, e_c, a	q_c solves the nonlinear resolvent equation and $G_c = \bar{N}q_c$. The errors are $e_c = f_c - N_c - G_c$ and $a = \theta_c - \bar{\theta}$. Section 4 abbreviates e_c, G_c as e, G at fixed c . Its adapted derivative variables are $\eta_\alpha = \partial_X^\alpha e / N_c$.
$\xi = (g, B, h), B^\circ$	For $\theta = (\alpha, \beta, \gamma)$, set $g = \nabla_X \alpha$, $B_{jl} = \partial_{X_j} \beta_l$, $h = \nabla_X \gamma$, and $B^\circ = (B + B^\top)/2 - (\text{tr } B)I/3$.
$K_{ij}^{ab}, D_N(\ell)$	The Onsager tensor and spatial Fourier dissipation symbol, defined by (5.4) and (5.14). The first-order coefficient is $c^{-1}K$.
z_c, u, R_N	$z_c = c(1 - N_c/f_c)$. Its limit satisfies $L_{\bar{N}}u = -\mathcal{T} \log \bar{N}$, $P_{\bar{N}}u = 0$. The operator $R_N h = \frac{1}{2} \Delta(h/N)$ takes values in the common resonance space.
$\mathcal{J}_c, Z_{c,i}^a$	$\mathcal{J}_c = \frac{c}{2} \sqrt{\prod_i f_{c,i}} \Delta(1/f_c)$ is the signed resonance current. The residual flux of quantity a in direction i is $Z_{c,i}^a = \int_D v_i \Psi_a (f_c - N_c) dk$.

For an integer $s \geq 0$, the squared $H_X^s H_\nu$ norm is $\sum_{|\alpha| \leq s} \int_{\mathbb{T}^3 \times D} \nu_* |\partial_X^\alpha h|^2 dX dk$. The space $H^{-1}(\mathbb{T}^3)$ is the dual of $H^1(\mathbb{T}^3)$. Time norms use the stated interval; time-integrated convergence does not include strong endpoint traces. Momentum integrals without an explicit domain use the domain of the model under discussion.

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